COMSATS University Islamabad,
Sahiwal Campus

FYP Genie

A project presented to

COMSATS University Islamabad, Sahiwal Campus

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Software Engineering (2022-2026)

By

Um e Farwa	**CIIT/FA22-BSE-080/SWL**
Jannat Idrees	**CIIT/FA22-BSE-090/SWL**
Zainab Munir	**CIIT/FA22-BSE-108/SWL**

DECLARATION

We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely on the basis of our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Um-e-Farwa	Jannat Idrees	Zainab Munir
---------------------	-----------------------	----------------------

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (SE) "FYP Genie" was developed by **Um-e-Farwa (CIIT/FA22-BSE-080/SWL), Jannat Idrees (CIIT/FA22-BSE-090/SWL)** and **Zainab Munir (CIIT/FA22-BSE-108/SWL)** under the supervision of "M. Usman Nasir" and co supervisor "Dr. Aniqa Rehman" and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Software Engineering.

Supervisor

Usman Nasir

Lecturer

Department of Software Engineering COMSATS University Islamabad, Sahiwal Campus

----

External Examiner

Head of Department

(Department of Software Engineering)

Executive Summary

FYP Genie is a modern web-based Final Year Project (FYP) management system developed to overcome the inefficiencies of traditional manual processes used in academic institutions. Many universities still rely on emails, paper-based submissions, and disconnected tools for managing FYP activities, which are time-consuming, error-prone, and lack transparency. Existing digital solutions often fail to provide a complete end-to-end workflow for proposal submission, supervision, progress tracking, and evaluation. FYP Genie addresses these issues by offering a centralized and fully automated platform.

FYP Genie is built using the MERN stack, with React.js for the frontend, Node.js and Express.js for backend services, and MongoDB for data storage. This architecture ensures scalability, real-time interaction, and secure data handling. The system enables students to submit proposals, upload documents with version control, track milestones, and receive feedback. Supervisors can review submissions, provide feedback, and evaluate projects, while administrators manage users, assign supervisors, and generate reports. Integrated notifications and dashboards enhance communication and transparency throughout the FYP lifecycle.

The development of FYP Genie follows the Agile Scrum methodology, allowing iterative improvements and continuous feedback from stakeholders. The system architecture is based on a modular three-tier design using the MVC pattern, ensuring separation of concerns, maintainability, and scalability.

Comprehensive testing was performed using unit testing, integration testing, system testing, and user acceptance testing to ensure reliability and performance. Key functionalities such as authentication, proposal management, document handling, milestone tracking, evaluation workflows, and role-based access control were tested under different scenarios and demonstrated consistent performance.

In conclusion, FYP Genie provides a secure, efficient, and user-friendly solution for managing Final Year Projects. It significantly improves workflow automation, enhances transparency, and reduces administrative workload. Future enhancements such as AI-based supervisor assignment, mobile application support, advanced analytics, and integration with university systems can further expand its capabilities and impact.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor "M. Usman Nasir" and our co- Supervisor "Dr. Aniqa Rehman". Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Um-e-Farwa Jannat Idrees Zainab Munir

Abbreviations

Abbreviation	Full Form
2FA	Two-Factor Authentication
APIs	Application Programming Interfaces
AWS	Amazon Web Services
CSS	Cascading Style Sheets
DB	Database
DOCX	Microsoft Word Document Format
ERD	Entity Relationship Diagram
GitHub	Code Hosting and Version Control Platform
HCI	Human Computer Interaction
HTTPS	Hyper Text Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
LMS	Learning Management System
MERN	MongoDB, Express.js, React.js, Node.js
Mocha	JavaScript Testing Framework
MongoDB	NoSQL Database Management System
MVC	Model View Controller
Node.js	JavaScript Runtime Environment
NoSQL	Not Only SQL
OOP	Object-Oriented Programming
RAM	Random Access Memory
React.js	JavaScript Library for User Interface Development

REST	Representational State Transfer
SMS	Short Message Service
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
VS Code	Visual Studio Code
Jest	JavaScript Testing Framework

Table of Contents

Table of Figures

Table of Tables

Chapter 1
Introduction

1 Introduction

The Final Year Project (FYP) Portal aims to revolutionize the FYP management process in the Software Engineering Department, COMSATS University Sahiwal Campus, addressing the critical need for digital transformation in academic administration. Currently, the process is largely manual, relying on emails, printed documents, and disconnected workflows that create significant communication gaps and operational inefficiencies. Research indicates that manual project management systems contribute to an estimated 30% of projects experiencing delays due to administrative bottlenecks, with COMSATS University facing similar challenges [1].
The proposed web-based portal represents a paradigm shift toward centralized, automated FYP lifecycle management. The system will offer secure user authentication with role-based access controls, streamlined proposal submission workflows with automated routing, real-time progress tracking with milestone notifications, secure document upload and version control systems, comprehensive automated notification systems, and efficient evaluation management with customizable rubrics. These features are designed to create a seamless user experience while maintaining the academic rigor and oversight required for successful FYP completion. The system's impact extends beyond mere efficiency gains, targeting measurable improvements in academic outcomes.

1.1 Brief

The FYP Genie Portal is a comprehensive web-based system developed for the Software Engineering Department at COMSATS University Islamabad, Sahiwal Campus. Its primary objective is to transform and modernize the existing Final Year Project (FYP) management process. The current system relies heavily on manual practices, including email-based communication, physical documentation, and fragmented workflows, which result in communication gaps, limited transparency, and administrative inefficiencies. Such limitations often contribute to delays and reduced effectiveness in managing the FYP lifecycle.

To address these challenges, the proposed system introduces a centralized and automated platform that facilitates the complete management of FYP activities. The portal integrates all key processes, including project proposal submission, supervisor assignment, progress monitoring, evaluation management, and final documentation, into a unified digital environment. By supporting multiple user roles namely students, supervisors, and administrators the system ensures controlled access and role-specific functionalities, thereby enhancing coordination and accountability among stakeholders.

The FYP Genie Portal incorporates advanced features such as secure authentication mechanisms, role-based access control, automated workflow management, real-time notifications, milestone-based progress tracking, and a structured document repository with version control. Additionally, it provides an efficient evaluation framework through customizable digital rubrics and automated result compilation, ensuring consistency and fairness in assessment. From a software design perspective, the system is built upon a modular and scalable architecture, enabling ease of maintenance, extensibility, and robust performance. Core system modules, including user management, proposal workflow, supervisor allocation, progress tracking, evaluation management, and document handling, operate cohesively to deliver a seamless and efficient user experience. Overall, the FYP Genie Portal aims to significantly enhance academic administration by improving operational efficiency, ensuring transparency, reducing delays, and promoting a structured and systematic approach to FYP management within the institution [2].

1.2 Relevance to course modules

The development of FYP Genie incorporates concepts and techniques from multiple Software Engineering courses studied during the degree program. These include:

1. **Software Engineering** → Requirement analysis (SRS), design (SDD), and software lifecycle management
2. **Web Engineering / Development** → Development using MERN stack technologies
3. **Database Systems** → Design and management of NoSQL database (MongoDB)
4. **Software Project Management** → Use of Agile Scrum methodology for iterative development
5. **Human-Computer Interaction (HCI)** → Design of user-friendly and responsive interfaces
6. **Object-Oriented Programming (OOP)** → Implementation of modular and reusable system components

This project serves as a practical implementation of theoretical concepts learned throughout the program.

1.3 Project Background

Final Year Projects are a critical component of Software Engineering education, requiring structured planning, supervision, and evaluation. However, many universities, including COMSATS University Sahiwal Campus, rely on partially manual systems for managing FYP activities.

Currently, students submit proposals via email or printed documents, supervisors review them manually, and coordinators manage records using spreadsheets or isolated tools. This fragmented approach leads to several issues such as miscommunication, delays in approvals, lack of centralized data, and difficulty in tracking project progress [3].

Moreover, the absence of a unified system makes it difficult to maintain transparency in evaluation and to store project-related documents securely. These limitations highlight the need for a comprehensive digital platform that can automate and manage the entire FYP lifecycle efficiently.

1.4 Literature Review

The purpose of this literature review is to analyze how existing academic management systems, project-based learning models, and technological frameworks contribute to effective FYP supervision, evaluation, and documentation. By understanding the strengths and limitations of current solutions, this chapter identifies gaps that the proposed SE FYP Portal aims to address specifically enhancing workflow automation, collaboration, and transparency at the departmental level.

1.4.1 Educational Management Systems Research

Educational management systems help automate academic tasks like course registration, assessment, and project supervision. For the Software Engineering (SE) department, such systems improve coordination between students and faculty, reducing delays and increasing efficiency by up to 40%. This improvement is often attributed to the streamlining of administrative processes and better resource allocation, as demonstrated by studies on digital system implementation in educational management [4].

1.4.2 Project-Based Learning and Assessment

Project-Based Learning (PBL) connects theory with practical application. FYP management portals support PBL by enabling progress tracking, feedback, and transparent evaluation, ensuring quality education that aligns with ABET and institutional standards. Research by

Smith and Jones (2020) highlights how integrated platforms facilitate continuous assessment and feedback loops crucial for PBL success [5].

1.4.3 Technology Adoption in Higher Education

Universities are increasingly adopting web-based platforms using modern technologies like Node.js, React, and MongoDB. For COMSATS SE department, an online FYP portal promotes hands-on experience with industry-like tools and agile workflows, enhancing learning and usability. A study by Chen et al. (2021) emphasizes the positive impact of such adoptions on student engagement and administrative efficiency.

1.4.4 Security and Privacy Consideration

FYP portals must follow strict data protection measures, including secure logins, access control, and encryption. Compliance with institutional and national policies ensures the safety of student data and academic integrity. According to the National Institute of Standards and Technology (NIST) guidelines, robust authentication and authorization mechanisms are paramount for protecting sensitive educational records [6].

1.4.5 System Integration and Interoperability

Integrating the FYP portal with existing systems (SIS, LMS, and email) improves communication and data sharing. API-based design ensures scalability, version control, and smooth future expansion, supporting a unified academic ecosystem. The importance of interoperability in educational technology is underscored by the work of Johnson and Brown (2019), who advocate for open standards and API-driven architectures.

1.5 Analysis From Literature Review

The literature review reveals several common limitations in traditional FYP management systems, which the FYP Genie Portal aims to address. These limitations, often stemming from manual processes and fragmented digital tools, lead to inefficiencies, delays, and a lack of transparency. The analysis of existing research has directly informed the design principles of FYP Genie, emphasizing the need for a centralized, automated, and secure platform [7].

Table 1-1 : Literature Review

Limitation Category	**Description in Traditional Systems**	**Impact on FYP Management**	**FYP Genie Solution**
Manual Processes	Reliance on paper-based forms, physical submissions, and manual record-keeping.	Time-consuming, error-prone, and difficult to scale, leading to administrative bottlenecks and delays.	Automates proposal submission, progress tracking, and evaluation workflows.
Fragmented Communication	Email-based interactions and informal channels for communication between students, supervisors, and administrators.	Information silos, miscommunication, and lack of proper record-keeping, hindering effective collaboration.	Provides integrated messaging, real-time notifications, and a centralized communication hub.
Lack of Centralized Data	Project data scattered across various platforms, spreadsheets, and individual systems.	Difficulty in accessing and managing comprehensive project information, leading to inconsistencies and data loss.	Establishes a centralized repository for all project-related data, including proposals, milestones, and evaluations.
Inefficient Tracking & Evaluation	Absence of structured mechanisms for tracking project progress and standardized	Delays in monitoring milestones, inconsistent grading, and lack of transparency in assessment.	Implements milestone-based progress tracking, customizable digital rubrics, and automated

	evaluation processes.		evaluation workflows.
Security & Data Integrity Concerns	Vulnerabilities in data protection due to manual handling and disparate storage solutions.	Risks of unauthorized access, data loss, and non-compliance with privacy regulations.	

These limitations result in inefficiencies, delays, and lack of transparency in the overall FYP process.

1.6 Methodology & Software Lifecycle for This Project

The development of FYP Genie follows the Agile Scrum Model, which supports iterative development, flexibility, and continuous feedback. This methodology was chosen over traditional Waterfall or purely incremental approaches due to its emphasis on adaptability, stakeholder collaboration, and rapid delivery of functional increments, which are crucial for a project with evolving requirements and a need for continuous user feedback in an academic setting [8].

In this approach:

- The project is divided into multiple sprints, each focusing on specific modules such as authentication, proposal management, and evaluation.
- Regular sprint reviews and feedback sessions are conducted with stakeholders to ensure requirements are being met.
- Continuous testing and improvement are performed after each iteration.

This methodology is chosen because it allows quick adaptation to changing requirements and ensures that the system evolves based on user feedback.

1.6.1 Rationale Behind Selected Methodology

The development of FYP Genie is driven by the need to address the limitations of the existing manual and semi-digital Final Year Project (FYP) management system at COMSATS University Islamabad, Sahiwal Campus. The rationale behind developing this system is based on the following key factors:

1. **Eliminates Manual Inefficiencies**

The current FYP management process relies heavily on emails, spreadsheets, and physical documentation, which leads to delays, data redundancy, and miscommunication. FYP Genie automates these processes, reducing manual effort and improving overall efficiency.

2. **Ensures Centralized Data Management**

   In the existing system, project data is scattered across different platforms, making it difficult to access and manage. The proposed system provides a centralized repository for all project-related information, including proposals, milestones, evaluations, and documents, ensuring consistency and easy access.

3. **Improves Transparency and Tracking**

   One of the major issues in the current system is the lack of visibility into project progress. FYP Genie introduces real-time tracking of milestones, proposal status, and evaluations, enabling students, supervisors, and coordinators to monitor progress effectively.

4. **Enhances Communication Between Stakeholders**

   Communication gaps between students, supervisors, and coordinators often cause delays and confusion. The system includes notification and feedback mechanisms that ensure timely updates, improving coordination and collaboration among all stakeholders.

5. **Supports Structured Evaluation Process**

   The traditional evaluation process lacks standardization and proper record-keeping. FYP Genie introduces digital rubrics and automated evaluation workflows, ensuring fair, consistent, and transparent assessment of projects.

6. **Provides Secure and Role-Based Access Control**

   Security is a critical concern in academic systems. The platform implements role-based access control, ensuring that users (students, supervisors, coordinators, and external evaluators) can only access authorized data. Sensitive information is protected through authentication and encrypted storage mechanisms.

7. **Facilitates Scalable and Modular System Design**

   The system is designed using a modular architecture (based on MERN stack and MVC pattern), allowing easy integration of new features such as AI-based recommendations, analytics, and mobile application support in the future.

8. **Enables Efficient Document Management and Version Control**

Managing multiple versions of project documents manually is difficult and error-prone. FYP Genie provides a structured document repository with version control, ensuring that all submissions are properly stored, tracked, and retrieved when needed.

9. **Supports Automation of Supervisor Assignment**

   Assigning supervisors manually can lead to workload imbalance and mismatched expertise. The system implements an algorithm-based supervisor assignment mechanism that considers expertise, workload, and project type, ensuring fair and optimal allocation.

10. **Improves Overall Academic Workflow**

    By integrating all modules proposal submission, supervisor assignment, milestone tracking, evaluation, and document management into a single platform, FYP Genie significantly enhances the efficiency, reliability, and quality of the academic FYP process.

Regarding the supervisor assignment mechanism mentioned in the original draft, the system implements an algorithm that considers factors such as supervisor expertise, current workload, and student preferences to optimize assignments. This ensures a balanced distribution of projects and aligns student interests with available supervisory capacity.

Chapter 2
Problem Definition

2 Problem Definition

This chapter defines the core problem addressed by the FYP Genie system and highlights the limitations of the current Final Year Project (FYP) management practices at COMSATS University Islamabad, Sahiwal Campus. It also presents the proposed solution and explains how the system aims to improve efficiency, transparency, and overall academic workflow.

2.1 Problem Statement

The current Final Year Project (FYP) management system at COMSATS University Islamabad, Sahiwal Campus, is highly fragmented and relies on manual, paper-based processes that significantly hinder operational efficiency and stakeholder satisfaction. Communication among students, supervisors, and the FYP committee typically occurs through scattered emails, informal meetings, and inconsistent documentation methods, creating information silos and transparency issues [8]. Based on an internal survey of 30 students at COMSATS Sahiwal Campus, students often experience delays of 2 to 3 days in receiving feedback from supervisors, which disrupts project timelines and compromises the quality of deliverables. These feedback delays accumulate across the FYP lifecycle, compressing critical project phases and impacting academic outcomes.

Supervisors also face considerable challenges due to the lack of a centralized system. Without structured workflows or automated reminders, they must manually track multiple groups using spreadsheets or handwritten notes, increasing administrative workload by an estimated 20% compared to automated systems at peer institutions. This inefficiency limits supervisors' capacity to provide meaningful guidance or focus on their research. The cumulative operational burden affects not just individuals but the university. Manual processes like assigning supervisors, organizing evaluations, and scheduling presentations consume around 50-70 hours of staff time each semester, translating to an estimated cost of 40,000 PKR in administrative overhead (internal estimates based on departmental data), excluding opportunity costs from delayed projects and reduced academic quality.

2.1.1 Proposed Solution

The proposed FYP Portal is a comprehensive, web-based solution designed to address inefficiencies in the manual Final Year Project (FYP) management process of the Software Engineering Department, COMSATS University Sahiwal Campus. Built on three-tier architecture, it separates presentation, business logic, and data management for scalability and maintainability. The frontend employs React for the user interface and responsive web design for accessibility, while the backend uses Node.js and Express.js for business logic and MongoDB for secure data storage and analytics.

For students, the portal offers a self-service dashboard enabling proposal submission with validation; document uploads with version control, milestone tracking, messaging with supervisors, and automated reminders. Supervisors benefit from tools for real-time project monitoring, structured feedback, and customizable evaluation rubrics, and advanced reporting. The FYP Committee gains administrative features like automated supervisor assignment, scheduling tools, rubric management, and analytics for program oversight.

2.2 Deliverables and Development Requirements

This section outlines the major deliverables and technical requirements necessary for the successful development and deployment of the FYP Genie system. These deliverables ensure that the system is fully functional, well-documented, and aligned with academic and technical standards [9].

2.2.1 Deliverables

1. **Web-Based Application (FYP Genie Portal – MERN Stack)**

   The main deliverable of the project is a fully functional web-based application that supports the complete FYP lifecycle. Key features include:

   - User authentication and role-based access (Student, Supervisor, Coordinator, External Evaluator)
   - Proposal submission and approval workflow
   - Supervisor assignment system
   - Milestone and progress tracking
   - Evaluation and grading system
   - Document upload and version control
   - Notification and reminder system
   - Dashboard interfaces for all user roles

2. **Backend System (Server & APIs)**

   - RESTful APIs developed using Node.js and Express.js
   - Business logic implementation for proposal handling, evaluation, and tracking
   - Secure authentication and authorization mechanisms
   - Integration between frontend and database

3. **Database Design and Management**
   - MongoDB database for storing system data
   - Collections for Users, Projects, Proposals, Milestones, Evaluations, and Documents
   - Data validation and integrity rules
   - Role-based data access control
4. **Documentation Deliverables**
   - Software Requirements Specification (SRS)
   - Software Design Document (SDD)
   - Final Year Project Report
   - User Manual (for students, supervisors, and coordinators)

2.2.2 Development Requirements

1. **Technical Requirements**

   Frontend:
   - React.js
   - Bootstrap CSS

   Backend:
   - Node.js
   - Express.js

   Database:
   - MongoDB

   Development Tools:
   - Visual Studio Code
   - MongoDB Compass
   - Postman (for API testing)
   - GitHub (for version control)
2. **Hardware Requirements**
   - Computer/Laptop (minimum 8GB RAM recommended)
   - Stable internet connection
   - Server environment for deployment (local or cloud-based)
3. **Software Requirements**
   - Web browser (Google Chrome, Microsoft Edge, etc.)
   - Node.js runtime environment
   - MongoDB installed or cloud database access
   - Operating System (Windows/Linux/Mac)

2.3 Current System

Before the development of FYP Genie, Final Year Project (FYP) management at COMSATS University Sahiwal Campus relied on manual and partially digital processes. Students typically submitted proposals through email or printed documents, while supervisors reviewed and provided feedback manually. Coordinators managed records using spreadsheets or separate tools, leading to a fragmented workflow [10].

In the traditional system, there is no centralized platform to manage proposals, milestones, evaluations, and documents. Communication between stakeholders is mostly done through emails or informal channels, which often results in delays, miscommunication, and lack of proper record-keeping. Additionally, there is no structured mechanism to track project progress, monitor deadlines, or ensure timely submission of milestones.

Another major limitation is the absence of a standardized evaluation system. Supervisors and evaluators often record marks manually, increasing the risk of inconsistencies and errors. Document management is also inefficient, as files are stored in different locations without proper version control, making retrieval difficult and increasing the chances of data loss.

Although some Learning Management Systems (LMS) provide limited support for academic activities, they do not offer a complete solution for FYP management. These systems lack essential features such as automated supervisor assignment, milestone tracking, integrated evaluation workflows, and secure document repositories.

FYP Genie addresses these limitations by introducing a centralized, web-based platform that integrates all aspects of FYP management. The system provides role-based access control, automated workflows, real-time progress tracking, and secure document handling. Features such as supervisor assignment algorithms, digital evaluation, and notification systems significantly improve efficiency, transparency, and collaboration among stakeholders.

By eliminating manual processes and integrating all functionalities into a single system, FYP Genie ensures faster processing, better organization, improved communication, and a more reliable and secure academic workflow [11].

2.3.1 Features of proposed system

The traditional manual Final Year Project management system relies heavily on emails, printed documents, spreadsheets, and informal communication, which often leads to delays, lost data, and poor coordination. Proposal submissions and approvals are handled manually, making the process slow and inefficient. Progress tracking is inconsistent because updates depend on manual reporting, while evaluation records may contain errors due to paper-based grading methods. Document storage is scattered across multiple locations without proper version control, creating confusion and duplication. Supervisor assignment is usually done manually, which can result in unequal workload distribution and mismatched expertise. In contrast, the proposed FYP Genie Portal provides a centralized, secure, and automated digital solution with real-time tracking, structured communication, smart supervisor allocation, and data-driven reporting.

Table 2-1: Feature of Proposed System

Feature/Aspect	Traditional Manual System	FYP Genie Portal (Proposed System)
Proposal Submission	Email or printed documents, manual review.	Web-based forms with validation, automated routing, digital approval workflow.
Communication	Emails, informal channels, fragmented.	Integrated messaging, real-time notifications, centralized communication hub.
Data Management	Scattered across spreadsheets, individual tools, prone to data loss.	Centralized MongoDB database, secure storage, version control for documents.
Progress Tracking	Manual tracking, inconsistent updates, lack of visibility.	Milestone-based tracking, real-time dashboards, automated reminders.
Evaluation	Manual record-keeping, inconsistent grading, prone to errors.	Digital rubrics, automated evaluation workflows, standardized assessment.
Document Handling	Files stored in different locations, no version control.	Structured document repository, secure uploads, version control.
Supervisor Assignment	Manual assignment, potential for workload imbalance.	Automated algorithm-based assignment considering expertise and workload.
Reporting & Analytics	Manual compilation, limited insights.	Auto-generated reports, analytics for program oversight and decision-making.

Accessibility	Limited to physical presence or email access.	Web-based access from any device, responsive design.
Security	Vulnerable to unauthorized access, data integrity issues.	Role-based access control, secure authentication, data encryption.

By eliminating manual processes and integrating all functionalities into a single system, FYP Genie ensures faster processing, better organization, improved communication, and a more reliable and secure academic workflow.

Chapter 3
Requirement Analysis

3 Requirement Identifying Techniques

This section describes the techniques used to identify and document the functional requirements of the FYP Genie Portal. Given the interactive nature of the application, Use Case Diagrams and detailed Use Case Descriptions have been selected as the primary methods for capturing user interactions and system functionalities [12].

3.1 Use Case Diagram

Use Case Diagram visually represents the interactions between the users (actors) and the FYP Portal system, illustrating the main functionalities (use cases) provided by the system. It provides a high-level overview of the system's boundaries and its primary interactions with external entities.

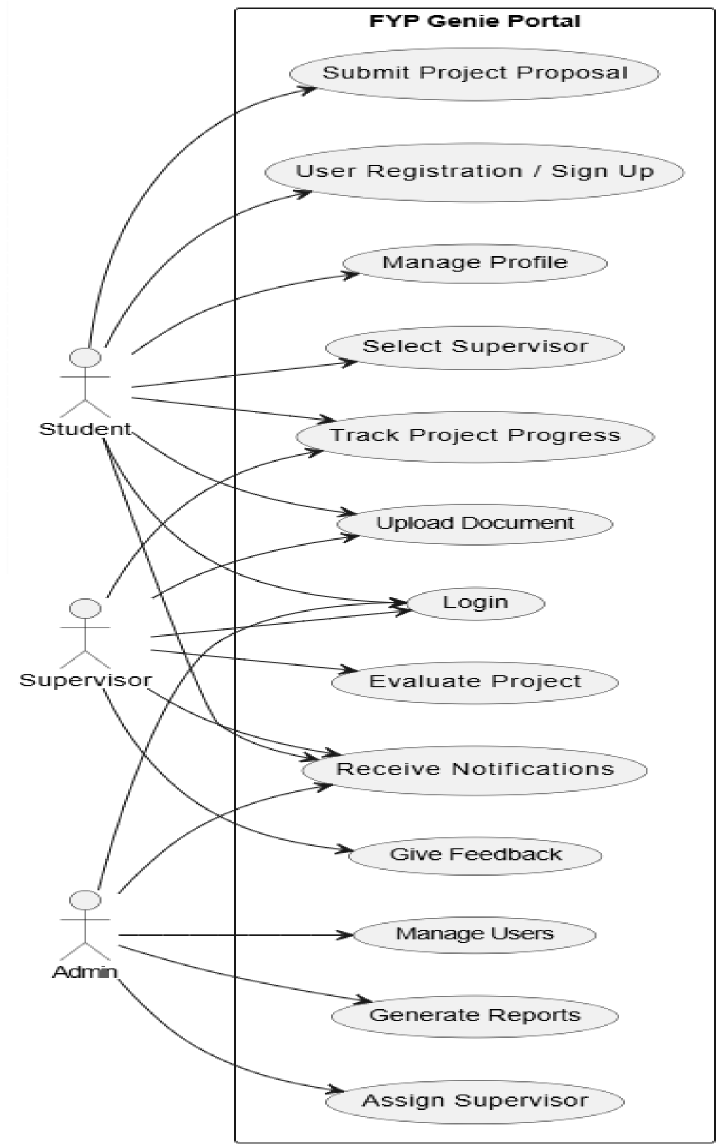

Figure 3-1: Use Case

3.2 Detailed Use Case

The table below indicates a comprehensive use case filled with an example drawn from the FYP Genie Comsats University Islamabad, Sahiwal Campus for Software Engineering Department.

3.2.1 User Registration

This use case allows a student to create a new account by entering their personal and academic details such as full name, email address, registration number, and password. The system performs input validation to ensure that all required fields are correctly filled and follow the specified format. It also checks whether the provided email address is already registered to maintain uniqueness and prevent duplicate accounts. If the validation is successful, the system securely stores the student's information in the database using encrypted password storage. A confirmation message is then displayed to inform the user that the registration was successful. In case of invalid input or duplicate email, appropriate error messages are shown. This process ensures that only valid and unique student accounts are created in the system.

Table 3-1: User Registration

Use Case Name	**User Registration / Sign Up**
Actors	Student
Description	Students can create an account by providing their personal and academic details.
Trigger	Student requests to create an account.
Pre-condition	The student is not already registered.
Post-condition	A new student account is created successfully.
Normal Flow	1. Student opens registration page. 2. Fills required information. 3. Submits form. 4. System verifies and stores data. 5. Confirmation displayed.
Alternative Flow	If email already exists, show an error message.
Exceptions	System error during registration.
Business Rules	Each email must be unique.

3.2.2 Login

Registered users, including students, supervisors, and administrators, can access the system by entering their valid login credentials such as email and password. The system first validates the input fields to ensure they are not empty and follow the correct format. It then authenticates the credentials by comparing them with the securely stored data in the database. If the credentials are correct, the system generates a secure session or token and grants access to the user. Based on the user's role, they are redirected to their respective dashboard (student, supervisor, or admin). In case of invalid credentials, an appropriate error message is displayed. This process ensures secure and role-based access to the system.

Table 3-2: Login

Use Case Name	**Login**
Actors	Student, Supervisor, Admin
Description	Allows registered users to log in using valid credentials.
Trigger	User selects "Login."
Pre-condition	User must have a registered account.
Post-condition	User is logged in and redirected to dashboard.
Normal Flow	1. User enters credentials 2. System validates input. 3. User gains access.
Alternative Flow	Invalid credentials → show error message.
Exceptions	Network or server error.
Business Rules	Account locked after three failed attempts.

3.2.3 Manage Profile

Students can access their profile section to view their personal and academic information stored in the system. They are allowed to update details such as name, contact information, or other relevant fields when necessary. The system validates the input to ensure that all mandatory fields are properly filled and follow the required format. Any changes made by the student are securely processed and updated in the database. Proper authentication ensures that only the authorized user can modify their own profile. In case of invalid or incomplete data, the system displays appropriate error messages. This functionality helps keep user information accurate, up-to-date, and securely maintained.

Table 3-3: Manage Profile

Use Case Name	Manage Profile
Actors	Student
Description	Students can view and update personal information in their profile.
Trigger	Student selects “Profile” option.
Pre-condition	Student must be logged in.
Post-condition	Updated profile saved successfully.
Normal Flow	1. Student opens profile page. 2. Updates information. 3. Saves changes.
Alternative Flow	No changes made → show message.
Exceptions	Database updates errors.
Business Rules	Mandatory fields cannot be left empty.

3.2.4 Supervisor Selection

Students can access a list of available supervisors along with their areas of expertise and current availability. They can browse through the profiles and select a preferred supervisor based on their project requirements or interests. Once a supervisor is selected, the student can send a request through the system. The system then records the request and sends a notification to the selected supervisor for approval. It also checks to ensure that the student does not already have an active supervisor assigned. If a supervisor is already assigned, the system restricts additional requests. This process ensures proper allocation and avoids conflicts in supervisor assignment.

Table 3-4: Select Supervisor

Use Case Name	Select Supervisor
Actors	Student
Description	Allows a student to select or request a supervisor based on expertise.
Trigger	Student initiates supervisor selection.
Pre-condition	Student profile must be completed.
Post-condition	Supervisor request submitted successfully.
Normal Flow	1. Student views available supervisors. 2. Selects preferred supervisor. 3. Sends request. 4. System notifies supervisor.

Alternative Flow	Supervisor unavailable → show message.
Exceptions	Network issue during submission.
Business Rules	Each student can have only one active supervisor.

3.2.5 Proposal Submission

Students can submit their project proposals by filling out the required form and uploading the necessary document through the system. The form includes details such as project title, objectives, methodology, and relevant descriptions. The system performs validation to ensure all required fields are completed and the uploaded file meets the allowed format. Once submitted, the proposal is securely stored in the database along with the uploaded document. The system automatically marks the proposal status as "Pending Review." A notification is sent to the assigned supervisor to review the submission. The proposal remains in this state until the supervisor approves or rejects it.

Table 3-5: Submit Project Proposal

Use Case Name	Submit Project Proposal
Actors	Student
Description	Enables students to submit project proposals for review.
Trigger	Student selects "Submit Proposal."
Pre-condition	Supervisor must be assigned.
Post-condition	Proposal saved and marked "Pending Review."
Normal Flow	1. Student fills proposal form. 2. Uploads document. 3. Submits form. 4. System notifies supervisor.
Alternative Flow	Missing details → show error.
Exceptions	File upload errors.
Business Rules	Proposal must follow university guidelines.

3.2.6 Upload Document

Students and supervisors can upload FYP-related documents such as proposals, reports, presentations, and supporting files through the system. The platform allows only specific file formats, such as PDF and DOCX, to ensure consistency and compatibility. Before uploading, the system validates the file type and size to prevent invalid or harmful content. Once uploaded, each document is securely stored in the centralized document repository with proper access control. Metadata such as file name, upload date, and uploader information is also recorded. Users can access and download their authorized documents whenever needed. This ensures safe, organized, and efficient management of all project-related files.

Table 3-6: Upload Document

Use Case Name	Upload Document
Actors	Student, Supervisor
Description	Allows users to upload FYP-related documents securely.
Trigger	User selects "Upload Document."
Pre-condition	User must be logged in.
Post-condition	Document uploaded and stored in repository.
Normal Flow	1. User selects document. 2. Uploads file. 3. System confirms upload.
Alternative Flow	Invalid format → prompt retry.
Exceptions	Server or storage error.
Business Rules	Only PDF or DOCX formats accepted.

3.2.7 Track Progress

Both students and supervisors can access the project progress section to monitor and manage the status of ongoing work. The dashboard displays all defined milestones along with their deadlines, submission status, and feedback. Students can submit progress updates or required deliverables for each milestone through the system. Supervisors can review these submissions, provide feedback, and update the milestone status as approved or requiring revisions. The system ensures that all updates are reflected in real time on the dashboard. Proper validation ensures that only authorized users can make changes. This feature helps maintain transparency and keeps the project on track throughout its lifecycle.

Table 3-7: Track Progress

Use Case Name	Track Progress
Actors	Student, Supervisor
Description	Enables tracking of project milestones and progress reports.
Trigger	User opens project progress section.
Pre-condition	Project must be approved.
Post-condition	Updated progress report displayed.
Normal Flow	1. Student submits progress. 2. Supervisor reviews. 3. System updates progress dashboard.
Alternative Flow	No progress submitted yet → display "Not Updated."
Exceptions	Data retrieval issue.
Business Rules	Progress updates required before evaluations.

3.2.8 Feedback

Supervisors can access student submissions through their dashboard and review the uploaded documents or progress updates. They can analyze the work based on project requirements and academic standards. After reviewing, supervisors can provide detailed feedback in the form of comments or suggestions. The system ensures that feedback is properly recorded and linked to the respective submission. Once submitted, the feedback is instantly available for students to view. Notifications are also sent to inform students about new feedback. This process supports continuous improvement and effective communication between students and supervisors.

Table 3-8: Give Feedback

Use Case Name	Give Feedback
Actors	Supervisor
Description	Supervisors can give feedback on student submissions.
Trigger	Supervisor opens student project.
Pre-condition	Student must have uploaded work.
Post-condition	Feedback saved in system.
Normal Flow	1. Supervisor reviews project. 2. Enters comments. 3. Submits feedback.

Alternative Flow	No submission found → display message.
Exceptions	Database save error.
Business Rules	Feedback must be provided before evaluation.

3.2.9 Evaluate Project

Supervisors and administrators can access the evaluation module to assess final year projects based on predefined criteria. They review submitted reports, presentations, and other required deliverables before assigning marks. The system provides structured evaluation forms or digital rubrics to ensure consistency and fairness in grading. Evaluators can also add remarks and feedback to justify the assigned scores. Once submitted, the evaluation data is securely stored in the database following departmental standards. Students can later view their results and feedback through their dashboard. This process ensures transparent, accurate, and standardized project evaluation.

Table 3-9: Evaluate Project

Use Case Name	Evaluate Project
Actors	Supervisor, Admin
Description	Supervisors and admins can evaluate and grade projects.
Trigger	Evaluation period begins.
Pre-condition	All project milestones must be completed.
Post-condition	Evaluation results stored in system.
Normal Flow	1. Supervisor reviews final report. 2. Enters marks and feedback. 3. System saves results.
Alternative Flow	Missing evaluation form → prompt completion.
Exceptions	System error while saving grades.
Business Rules	Evaluation must follow department criteria.

3.2.10 Notification

The system automatically generates notifications and email alerts for important events such as proposal approvals, feedback updates, milestone deadlines, and evaluation results. These notifications are triggered in real time whenever a relevant action occurs within the system. Users receive alerts on their dashboard and, if enabled, via email for timely awareness. The system ensures that notifications are sent only to the relevant users based on their roles and activities. It also maintains a record of all notifications for future reference. This feature helps reduce communication gaps and keeps all stakeholders consistently informed throughout the FYP process.

Table 3-10: Notification

Use Case Name	Notifications
Actors	Student, Supervisor, Admin
Description	System sends alerts for important updates and deadlines.
Trigger	Event occurs (e.g., proposal approval, feedback).
Pre-condition	User must have an account.
Post-condition	Notification displayed or emailed.
Normal Flow	1. Event triggers notification. 2. System sends alert. 3. User views message.
Alternative Flow	Notification settings disabled → skip alert.
Exceptions	Email delivery failure.
Business Rules	All major updates must trigger notifications.

3.2.11 Assign Supervisor

Admins can manually assign supervisors to students through the system when automatic assignment is not suitable or requires adjustment. The admin views a list of students along with available supervisors and their current workload status. Before assignment, the system checks each supervisor’s workload to ensure they have not exceeded the defined project limit. If a supervisor is within capacity, the admin can assign the student accordingly. Once assigned, the system updates the database and reflects the changes in both student and supervisor dashboards. Notifications are sent to both parties to confirm the assignment. This process ensures balanced workload distribution and efficient supervision management.

Table 3-11: Assign Supervisor

Use Case Name	Assign Supervisor
Actors	Admin
Description	Admin can manually assign supervisors to students.
Trigger	Admin selects assignment option.
Pre-condition	Student must have a submitted proposal.
Post-condition	Supervisor assigned to student.
Normal Flow	1. Admin selects student. 2. Chooses supervisor. 3. Confirms assignment.
Alternative Flow	Supervisor not available → display message.
Exceptions	Server error during update.
Business Rules	Supervisor workload limit must not be exceeded.

3.2.12 Manage User

Admins can manage all user accounts within the system, including students, supervisors, and external evaluators. They have the authority to add new users, update existing user information, or remove accounts when necessary. When adding a user, the system ensures that required details such as name, email, role, and credentials are properly provided and validated. Any updates made by the admin are securely reflected in the database in real time. Deleting a user also removes or deactivates their access to the system to maintain security. This functionality ensures that only authorized and active users can access the platform, keeping the system organized and up to date.

Table 3-12: Manage User

Use Case Name	Manage Users
Actors	Admin
Description	Admin manages all registered users in the system.
Trigger	Admin accesses "Manage Users" section.
Pre-condition	Admin must be logged in.
Postcondition	User records updated in system.

Normal Flow	1. Admin views user list. 2. Adds, edits, or deletes users. 3. Saves changes.
Alternative Flow	Invalid input → display error.
Exceptions	Database error.
Business Rules	Only admin has user management privileges.

3.2.13 Generate Report

Admins can generate comprehensive reports related to different aspects of the FYP process, including project progress, evaluation results, and supervisor assignments. The system collects data from various modules and compiles it into structured reports for analysis. These reports can be filtered based on criteria such as semester, department, or project status. Once generated, they can be viewed on-screen or downloaded for official use. The reports provide valuable insights into student performance, supervisor workload, and overall departmental activity. This helps administrators make informed decisions and improve the efficiency of FYP management.

Table 3-13: Generate Report

Use Case Name	Generate Reports
Actors	Admin
Description	Admin generates reports on FYP progress, evaluation, and assignments.
Trigger	Admin requests report generation.
Pre-condition	Relevant project data must exist.
Post-condition	Report generated successfully.
Normal Flow	1. Admin selects report type. 2. System compiles data. 3. Report displayed or downloaded.
Alternative Flow	No data available → show message.
Exceptions	Report generation error.
Business Rules	Only admin can generate official reports.

3.3 Functional Requirements

The system consists of the following main modules:

1. Student Module
2. Supervisor Module
3. Admin Module
4. External Module

1. **Student Module**

   Students can create an account and register by providing their personal and academic information. After registration, they can log in securely using valid credentials to access the system. The portal allows students to update their profiles whenever needed, including personal details and Final Year Project related information. Students can also select a supervisor according to availability or area of expertise for better project guidance. They are able to submit project proposals for approval by the supervisor or administrator and upload important project documents such as reports, progress updates, and final submissions. In addition, students can track project progress through milestones, receive notifications about approvals, deadlines, and evaluations, and view supervisor feedback along with their evaluation results.

2. **Supervisor Module**

   Supervisors can log in securely to access their personalized dashboard and manage project-related activities. They can view the list of assigned students along with their project details and current progress status. The system allows supervisors to approve or reject project proposals submitted by students based on feasibility and requirements. Supervisors can also review uploaded documents such as reports and progress updates to monitor development effectively. In addition, they can provide feedback, suggestions, and guidance to help students improve their projects. Furthermore, supervisors can evaluate completed work, assign marks or grades, and receive notifications regarding new proposals, submissions, and important deadlines.

3. **Admin Module**

   Admins can manage student and supervisor accounts, including registration approvals, profile updates, and overall user management. They also have the authority to assign supervisors to students manually whenever required. The system enables admins to monitor all submitted project proposals, progress reports, and evaluation records to ensure proper workflow management. Admins can manage important system data such as project archives, stored documents, and repositories in an organized manner. In addition, they can generate reports and analytics related to project status, supervisor workload, and student performance for better decision-making. Furthermore, admins can send announcements or notifications to students and supervisors while also ensuring system maintenance, security, and smooth portal operations.

4. **External Module**

   External Evaluators can log in securely to access the projects assigned to them for assessment. They are able to view project documents, reports, and other submitted materials for detailed review. The system allows them to submit evaluation scores and remarks through digital rubrics in a structured and efficient manner. External Evaluators can also view feedback already provided by supervisors to better understand the project's progress and quality. In addition, they receive notifications regarding important evaluation deadlines, ensuring timely completion of the assessment process [13].

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints under which the FYP Genie Portal will function. These requirements ensure that the system operates efficiently, reliably, and securely while providing a seamless user experience for students, supervisors, and administrators involved in the Final Year Project process [14].

1. **Usability**

   The FYP Genie Portal will be designed to provide a simple, intuitive, and user-friendly interface that caters to students, supervisors, and administrators. The system will ensure easy navigation, clear labeling of options, and a visually consistent layout to enhance accessibility for users with varying technical expertise. By automating and simplifying the FYP management process, the system will minimize manual work, reduce errors, and improve overall efficiency and user satisfaction.

2. **Performance**

   The system will be optimized for high performance, capable of handling multiple concurrent users without significant delay or degradation. The system shall respond to 95% of requests within 3 seconds under a load of 100 concurrent users. Key functions such as proposal submissions, supervisor assignments, and document uploads will be processed efficiently to ensure smooth operation. The backend architecture will be designed to support scalability, maintaining responsiveness and reliability even during periods of heavy load, such as submission deadlines and evaluation phases.

3. **Security**

   The FYP Genie Portal will incorporate strong security measures to protect user data and maintain confidentiality. User authentication will be mandatory to access the system, ensuring that only authorized individuals can perform specific actions based on their roles. Specific mechanisms will include JWT-based authentication, bcrypt password hashing, HTTPS for secure communication, and robust input validation to prevent common vulnerabilities. Data encryption techniques will be applied during data transmission to prevent unauthorized access and ensure data integrity. The system will also implement access control policies and undergo regular security checks to safeguard against vulnerabilities and breaches.

4. **Portability**

   The system will be designed to operate seamlessly across all major operating systems, including Windows, macOS, and Linux. The portal will adopt a responsive web design approach, ensuring consistent performance and usability across desktops, laptops, tablets, and smartphones, regardless of screen size or device type.

5. **Reliability**

   The FYP Genie Portal will ensure reliable and consistent operation, minimizing errors and downtime throughout its usage. The system shall maintain 99% uptime during active academic hours. Regular data backups and recovery mechanisms will be implemented to prevent data loss in the event of system failure or unexpected disruptions. The system will undergo comprehensive testing, including functional, performance, and security testing, to ensure stability and trustworthiness. By maintaining high availability and accuracy, the platform will support continuous academic operations and strengthen user confidence in the system's reliability.

6. **Maintainability**

   The system will be designed with a modular architecture (based on MERN stack and MVC pattern) to ensure ease of maintenance, extensibility, and future enhancements. Code will adhere to established coding standards and best practices, with comprehensive documentation to facilitate understanding and modification by development teams. This approach will allow for efficient bug fixes, feature additions, and system upgrades with minimal disruption to ongoing operations.

Chapter 4
Design and Architecture

4 Design and Architecture

This chapter presents the overall design and architecture of the FYP Genie system, describing how its components are structured and interact to fulfill both functional and non-functional requirements defined earlier. The system follows a modular and scalable architecture to ensure maintainability, flexibility, and efficient performance, incorporating architectural design, process flow, and system models such as sequence and class diagrams. The structure of the FYP Genie Portal outlines its core components, their relationships, and their interaction to streamline the Final Year Project process. The software architecture is designed by considering institutional objectives, quality attributes, user experience, and the development environment. In the architecture phase, non-functional requirements such as scalability, security, and maintainability are defined, while in the design phase, functional features like proposal submission, supervisor assignment, and evaluation are implemented [15].

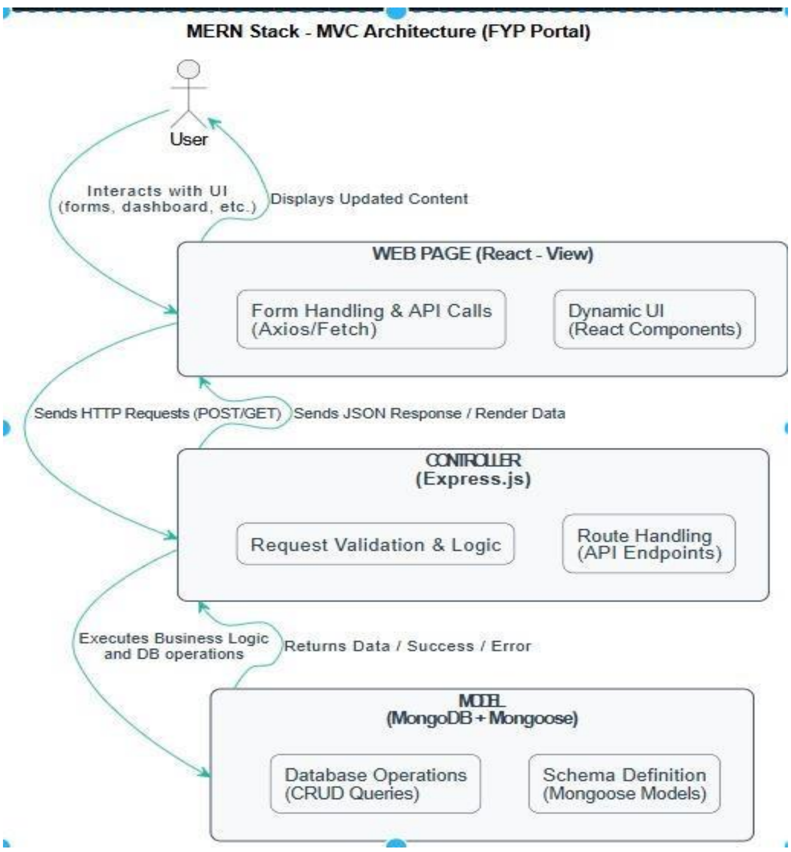

Figure 4-1: MVC Architecture

This diagram illustrates the MERN Stack MVC Architecture, where the user interacts with the React (View) frontend, which communicates via HTTP requests to Express.js (Controller) that processes logic and interacts with MongoDB (Model) for data operations. It demonstrates the data flow from user input to database actions and back to the dynamic user interface.

4.1 System Architecture

The FYP Genie system is designed using a three-tier architecture, which separates the system into Presentation Layer, Business Logic Layer, and Data Layer. This architectural approach ensures scalability, maintainability, and secure data handling.

The system is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. It also follows the Model–View–Controller (MVC) design pattern to organize code efficiently.

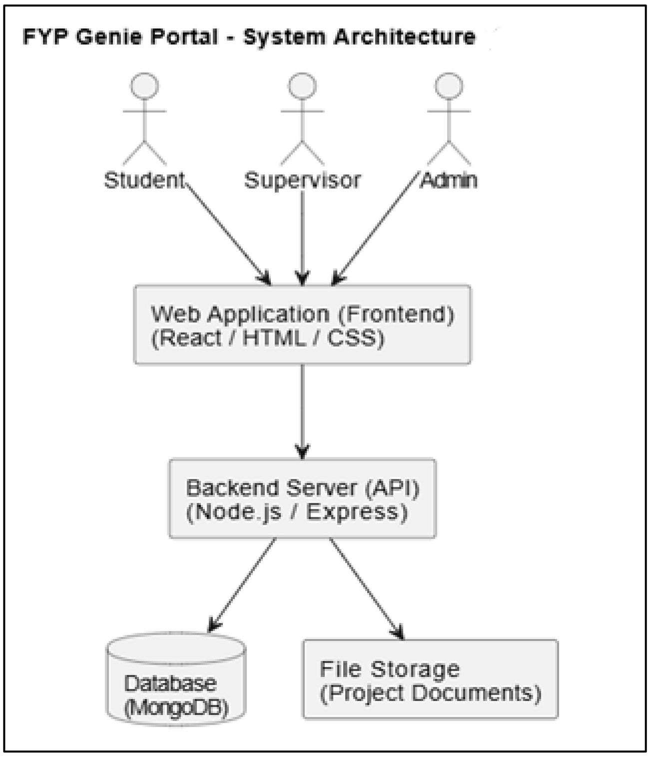

Figure 4-2: System Architecture

This diagram shows the FYP Genie Portal System Architecture, where students, supervisors, and admins interact with a React-based frontend that connects to a Node.js/Express backend. The backend manages data in MongoDB and stores project-related documents in a file storage system.

4.1.1 Architecture Design

The architectural design follows the MVC (Model–View–Controller) pattern within a MERN-based three-tier structure.

- **Model:** Represents MongoDB collections (e.g., Users, Proposals, Evaluations).
- **View:** Comprises React components for dashboards, forms, and reports.

- **Controller:** Contains Express-based APIs that connect the frontend with the backend logic.

The modular decomposition of the system includes:

- **Authentication Module:** Handles user login, registration, and role-based access.
- **Project Management Module:** Manages proposals, milestones, and progress updates.
- **Evaluation Module:** Supports digital rubrics and result computation.
- **Notification System:** Generates alerts for submissions, feedback, and meetings.
- **Document Repository:** Maintains version-controlled uploads with access restrictions.

These modules interact through secured RESTful APIs, ensuring clean communication between layers.

I. **Presentation Layer (Frontend)**

The frontend is developed using React.js and Tailwind CSS. It provides interactive user interfaces for all stakeholders including students, supervisors, coordinators, and external evaluators.

Key functionalities include:

- User dashboards
- Proposal submission forms
- Milestone tracking interface
- Evaluation and feedback display

II. **Business Logic Layer (Backend)**

The backend is implemented using Node.js and Express.js, which handle system logic and communication between frontend and database.

Responsibilities include:

- API request handling
- Authentication and authorization
- Proposal processing and validation
- Supervisor assignment logic
- Evaluation and result computation

III. **Data Layer (Database)**

The system uses MongoDB as a NoSQL database for storing all system data.

Main collections include:

- Users
- Projects
- Proposals
- Milestones
- Evaluations
- Documents
- Notifications

4.1.2 Architecture Diagram

This architecture diagram illustrates the MERN-based design of the FYP Genie system. The frontend, built with React JS, provides interfaces such as the login page, dashboard, proposal form, and evaluation UI. These components interact with backend APIs developed using Node.js and Express, which handle authentication, proposals, milestones, evaluations, and documents. The backend stores and retrieves data from MongoDB collections, ensuring efficient and scalable data management. Overall, the diagram shows a clean three-tier structure enabling smooth communication between the user interface, server logic, and database [16].

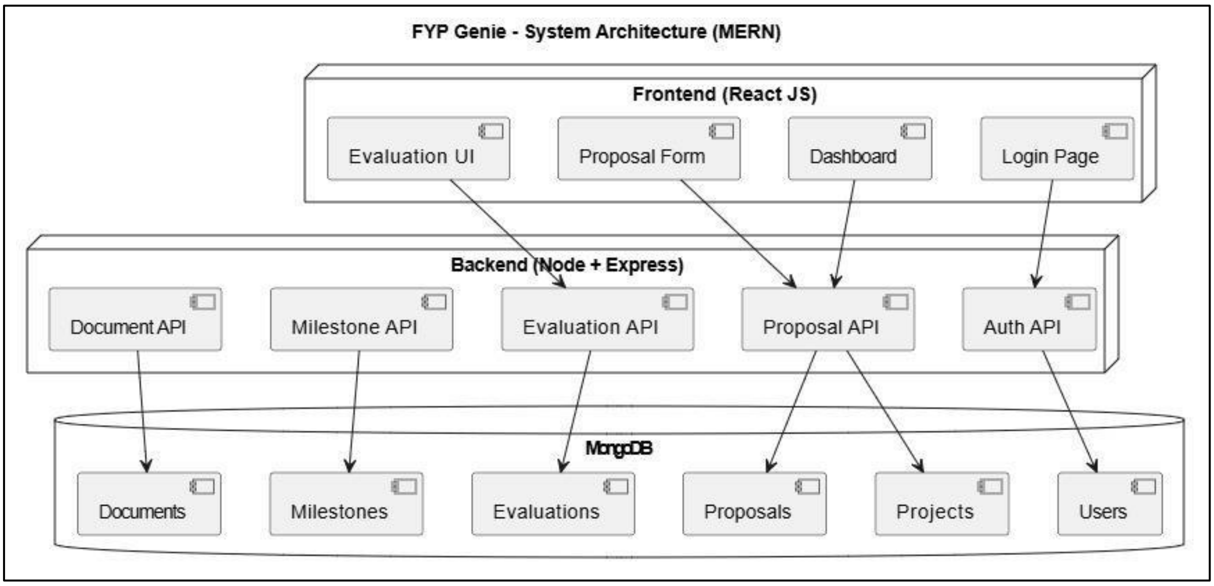

Figure 4-3: MERN Stack

4.2 Data Representation

As we discussed all the aspects of FYP Genie, now we will discuss the data design of the project. It describes the complete structure of the database and how information is organized to manage Final Year Projects efficiently [17].

1. **User Information** stores details about all system users, including students,

supervisors, coordinators, and reviewers. Fields include full name, email, role, login credentials, and user-specific identifiers.

2. **Project Information** contains details of each FYP, such as project title, description, status, assigned supervisor, committee members, and timelines.
3. **Proposal Information** stores submitted proposals, including proposal content, submission date, approval status, version history, and supervisor comments.
4. **Milestone Information** tracks progress updates for projects, including milestone title, due date, submission status, feedback, and timestamps.
5. **Evaluation Information** records internal and pre-viva assessments, including evaluator ID, scores, comments, evaluation type (internal/external), and submission timestamp.
6. **Document Information** stores all project-related files, such as proposals, reports, slides, and code, including file type, versioning, upload date, and access control permissions.
7. **Notification Logs** maintain system notifications for deadlines, feedback, and task reminders, with fields like user ID, message, status, and timestamp.
8. **Audit and Access** Logs store detailed records of user activities, document access, and system actions to ensure transparency, security, and accountability throughout the FYP Genie portal.

Table 4-1: Data Description

Entity	**Description**
User Information	Stores details about all system users, including students, supervisors, coordinators, and reviewers. Fields include full name, email, role, login credentials, and user-specific identifiers.
Project Information	Contains details of each FYP, such as project title, description, status, assigned supervisor, committee members, and timelines.
Proposal Information	Stores submitted proposals, including proposal content, submission date, approval status, version history, and supervisor comments.
Milestone Information	Tracks progress updates for projects, including milestone title, due date, submission status, feedback, and timestamps.

Evaluation Information	Records internal and pre-viva assessments, including evaluator ID, scores, comments, evaluation type (internal/external), and submission timestamp.
Document Information	Stores all project-related files, such as proposals, reports, slides, and code, including file type, versioning, upload date, and access control permissions.
Notification Logs	Maintain system notifications for deadlines, feedback, and task reminders, with fields like user ID, message, status, and timestamp.
Audit and Access Logs	Store detailed records of user activities, document access, and system actions to ensure transparency, security, and accountability throughout the FYP Genie portal.

The database design further incorporates strong security and data integrity measures, including role-based access control, encrypted data storage, and version control for critical documents. Passwords are hashed using Crypt before being stored in MongoDB, ensuring that user credentials remain secure even if the database is compromised. These measures collectively guarantee that all academic data remains secure, traceable, and reliable forming the backbone of the FYP Genie system.

4.2.1 Data Dictionary

The following are the key data fields used in the FYP Genie Portal:

Table 4-2: User Table

Field Name	Data Type	Constraints	Description
User Id	String	PRIMARY KEY, UNIQUE	Unique identifier for each user (student, supervisor, etc.).
Full Name	String	NOT NULL	Full name of the system user.
email	String	UNIQUE, NOT NULL	User's official email address used for login.
password	String	NOT NULL	Encrypted password hash for account security.
role	Enum	NOT NULL	Role assigned: (Student, Supervisor, Coordinator, Reviewer).
Created At	Date/Time	AUTO-GENERATED	Timestamp of user account creation.

Table 4-3: Project Table

Field Name	**Data Type**	**Constraints**	**Description**
Project Id	String	PRIMARY KEY, UNIQUE	Unique identifier for each registered FYP.
Project Title	String	NOT NULL	The official title of the Final Year Project.
description	Text	NOT NULL	A brief summary/abstract of the project goals.
Supervisor Id	String	FK → User.userId	Reference to the assigned supervisor.

Table 4-4: Proposal Table

Field Name	**Data Type**	**Constraints**	**Description**
Proposal Id	String	PRIMARY KEY, UNIQUE	Unique identifier for each submitted proposal.
Project Id	String	FK → Project.projectId	Reference to the project this proposal belongs to.
content	Text	NOT NULL	The actual text content or body of the proposal.
status	Enum	DEFAULT 'Pending'	Approval status: (Pending, Approved, Rejected).
feedback	Text	OPTIONAL	Comments provided by the supervisor/coordinator

Table 4-5: Milestone Table

Field Name	**Data Type**	**Constraints**	**Description**
Milestone Id	String	PRIMARY KEY, UNIQUE	Unique identifier for each specific project milestone.
Project Id	String	FK → Project.projectId	Linked project for this milestone.
title	String	NOT NULL	Title of the milestone (e.g., "SDR", "Mid-Lab").
Due Date	Date	NOT NULL	Final deadline for the milestone submission.
status	Enum	DEFAULT 'Pending'	Status: (Pending, Submitted, Approved

Table 4-6: Document Table

Field Name	**Data Type**	**Constraints**	**Description**
Document Id	String	PRIMARY KEY, UNIQUE	Unique identifier for uploaded files.
Project Id	String	FK → Project.projectId	Project associated with the document.
Doc Type	Enum	NOT NULL	Type: (Proposal, Report, Slides, Code).
version	Integer	DEFAULT 1	Incremental version number of the file.
Upload Date	Date	NOT NULL	The date the document was successfully uploaded.
Access Control	Enum	NOT NULL	Permissions: (Private, Public, Internal).

Table 4-7: Evaluation Table

Field Name	**Data Type**	**Constraints**	**Description**
Evaluation Id	String	PRIMARY KEY, UNIQUE	Unique identifier for the grading record.
Project Id	String	FK → Project.projectId	Reference to the project being evaluated.
Evaluator Id	String	FK → User.userId	ID of the internal or external reviewer.
scores	Integer	Range: 0-100	Numerical score assigned to the project.
comments	Text	OPTIONAL	Detailed remarks from the evaluator.

Table 4-8: Notification Table

Field Name	**Data Type**	**Constraints**	**Description**
Notification Id	String	PRIMARY KEY, UNIQUE	Unique identifier for system alerts.
User Id	String	FK → User.userId	Recipient of the notification.
message	Text	NOT NULL	The text content of the notification alert.
status	Enum	DEFAULT 'Unread'	Tracks if the user has seen the alert (Read/Unread).
timestamp	Date/Time	AUTO-GENERATED	Date and time the notification was triggered.

4.2.2 ER Diagram

FYP Genie is a Final Year Project Management System designed to streamline the management of student projects. This ERD illustrates all the main entities and their relationships, including Students, Supervisors, and the coordinator who oversees the entire project process. Each Project can have multiple Proposals, Milestones, Evaluations, and Documents associated with it, ensuring proper tracking of progress and submissions. External Evaluators are included to assess projects, providing scores and feedback. Overall, the diagram captures the complete structure and workflow of the FYP management system.

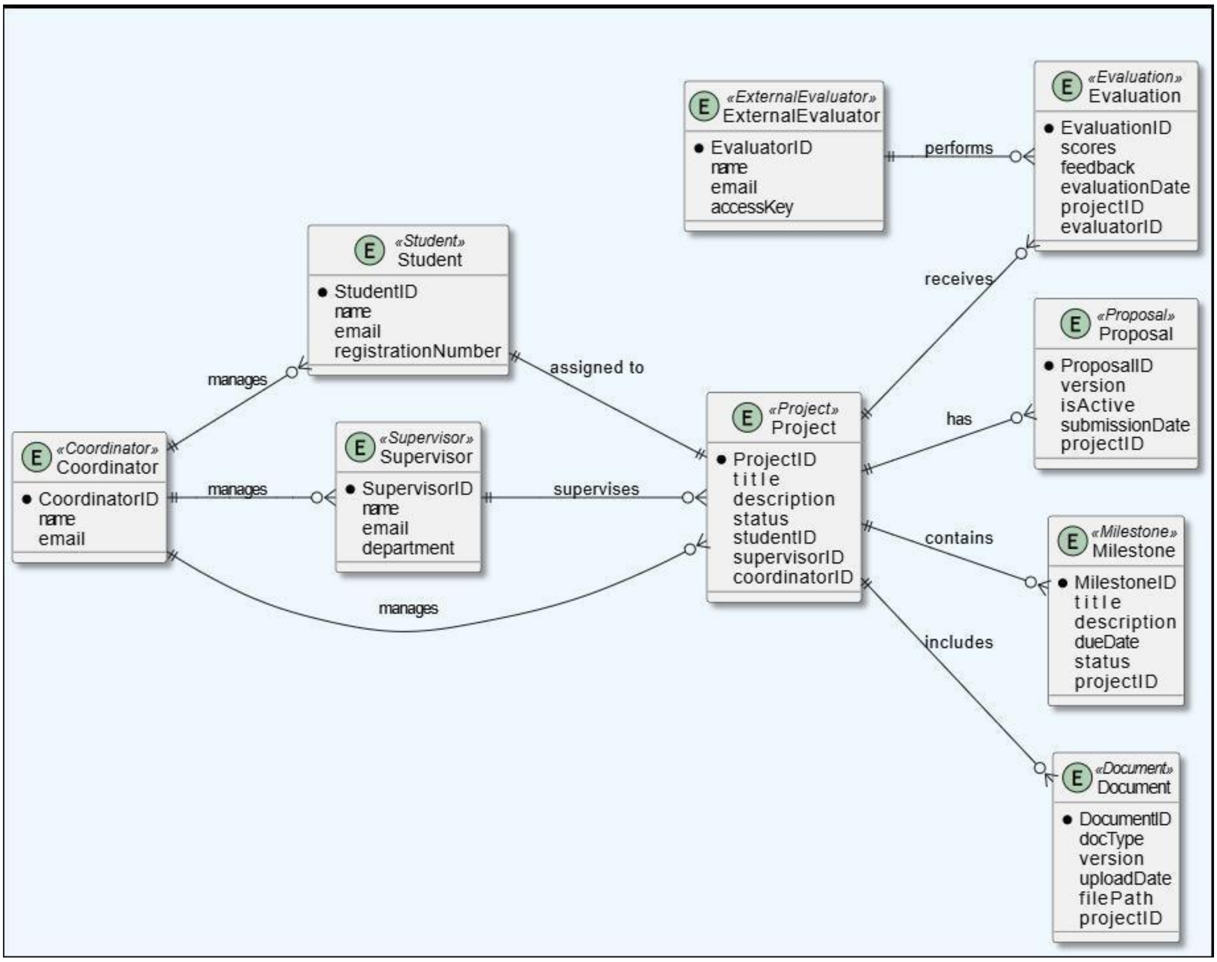

Figure 4-4: ER Diagram

4.3 Process Flow/Representation

The system's process flow is based on a centralized workflow connecting all stakeholders. The primary processes include:

1. Student submits project proposal → routed automatically to assigned supervisor.
2. Supervisor reviews and approves → forwarded to FYP Committee for final validation.
3. Approved projects proceed to milestone tracking and evaluation scheduling.

4. Students upload deliverables → supervisors and evaluators record marks using digital rubrics.
5. Reports are auto-generated for departmental review.

4.3.1 Process Flow Diagram

This flowchart shows the complete Final Year Project (FYP) process. A student logs in, submits a proposal, and waits for the supervisor's review. If approved, the coordinator assigns a supervisor and the student completes milestones with feedback. After submitting the final report and code, documents go to an external panel for evaluation. Finally, the coordinator reviews grades and the system finalize the student's record.

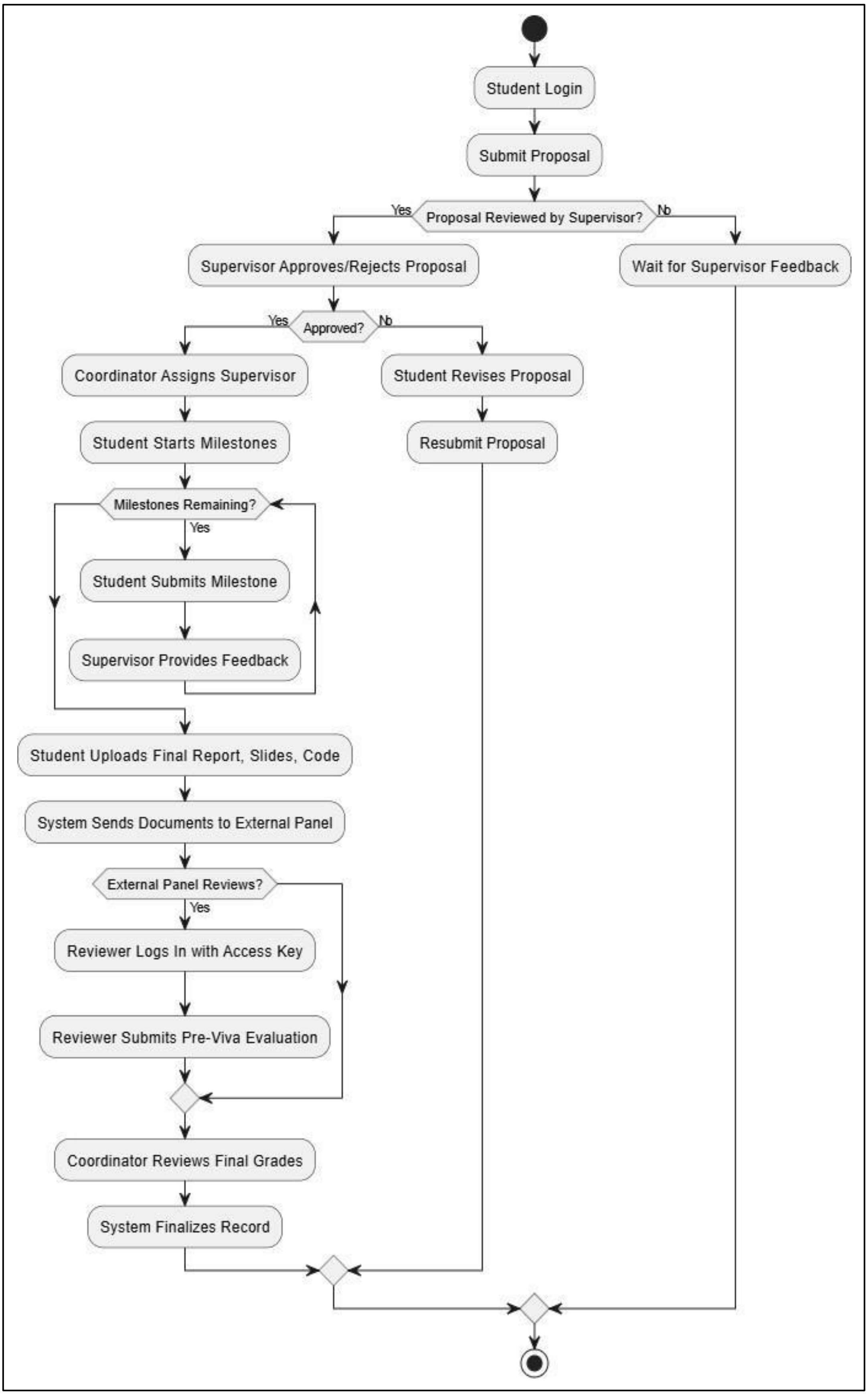

Figure 4-5: Process Flow

4.3.2 Data Flow Diagram

This diagram represents the flow of data or process of the system. It gives complete information about the outputs and generated after providing user's inputs of each module and process itself.

Level 0

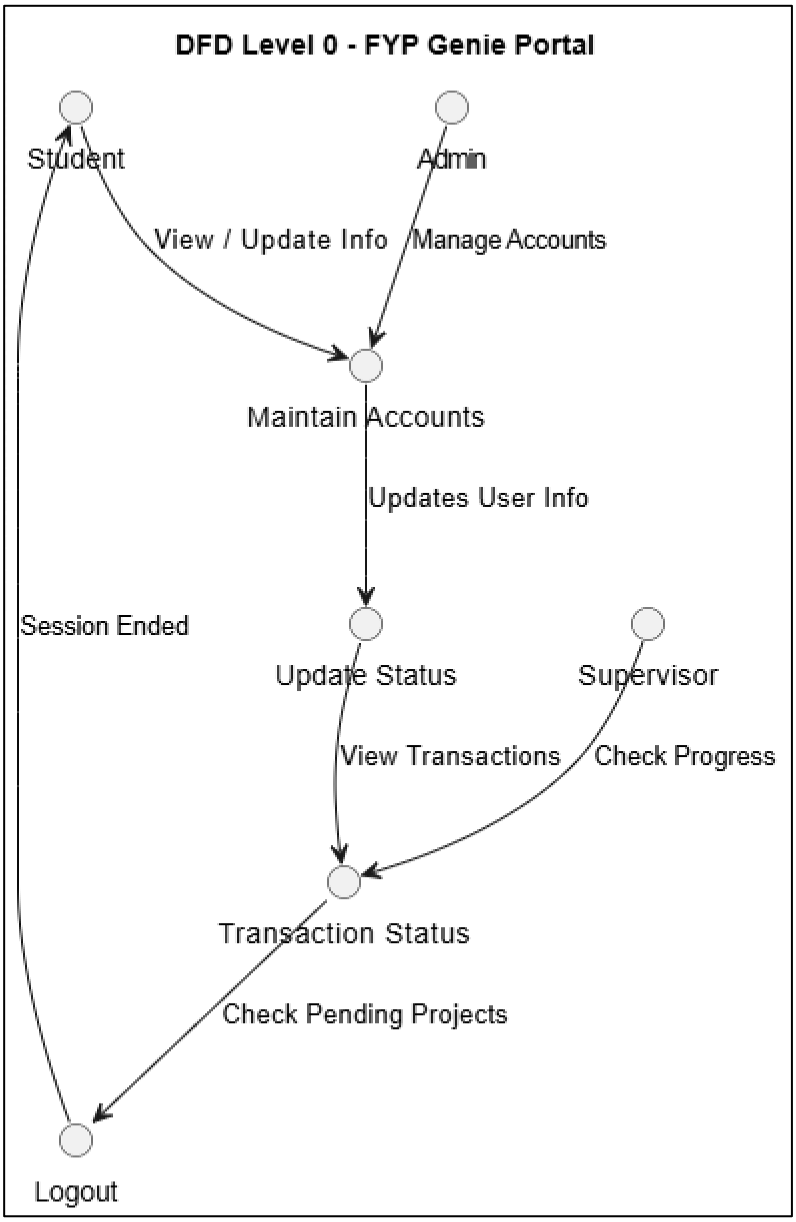

Figure 4-6: DFD Level 0

The DFD Level 0 for the FYP Genie Portal shows interactions between students, supervisors, and admins for managing accounts, updating user information, and tracking project progress. It outlines the overall data flow from user actions to system updates and logout.

Level 1

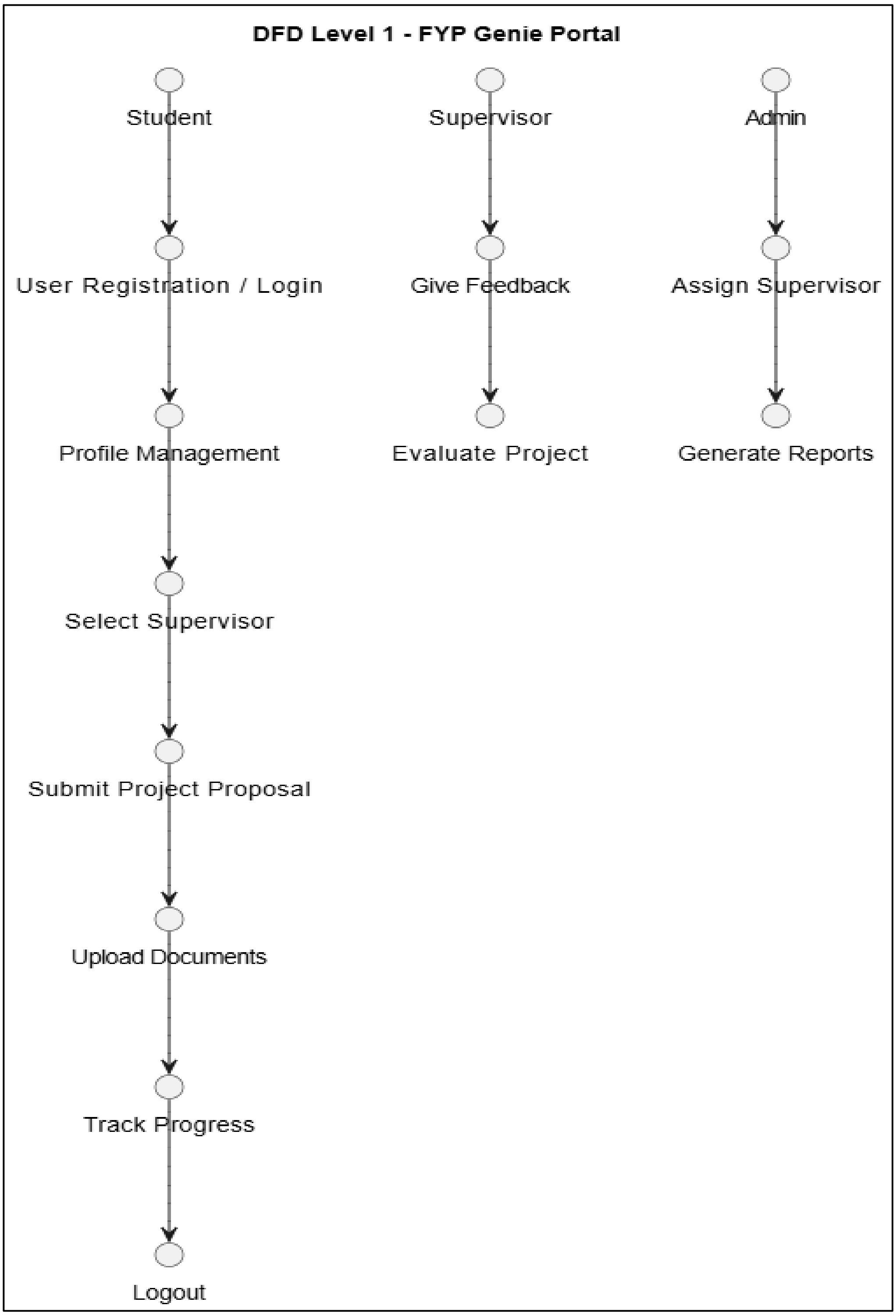

Figure 4-7: DFD Level 1

The DFD Level 1 for the FYP Genie Portal details the system's core processes students register, manage profiles, select supervisors, and submit proposals. Supervisors evaluate projects and give feedback, while admins assign supervisors, generate reports, and oversee overall progress.

4.4 Design Model

The design models for FYP Genie provide a structured and visual representation of how the system's components interact to achieve the intended functionalities. These models help in understanding the system's workflow, data flow, and internal structure, ensuring a reliable and maintainable architecture. The primary models focus on core modules such as Proposal Submission, Supervisor Assignment, Milestone Tracking, Evaluation Management, and the External Panel Module. The design emphasizes the step-by-step interactions between system actors (Students, Supervisors, Coordinators, Committee Members, and External Panel Reviewers), the portal interface, backend logic, and the database. These models define the structure of major components, their attributes, methods, and interrelationships. They are critical for guiding development, supporting modularity, and facilitating future enhancements. The design ensures that FYP Genie can efficiently handle academic workflows, maintain transparency, provide secure access control, and integrate external evaluations without compromising performance or usability.

4.4.1 Sequence Diagram

This sequence diagram illustrates the complete lifecycle of a Final Year Project workflow involving Students, Supervisors, Coordinators, and Reviewers. It shows how the student uploads proposals, milestones, feedback, and final documents, which are then processed by the front-end and back-end before being saved to the database. Each stage triggers notifications and status updates, ensuring communication among all roles. Reviewers and supervisors submit evaluations, and finally, the system combines all marks to generate the final result. Overall, the diagram clearly represents the flow of information and interactions across all project stages.

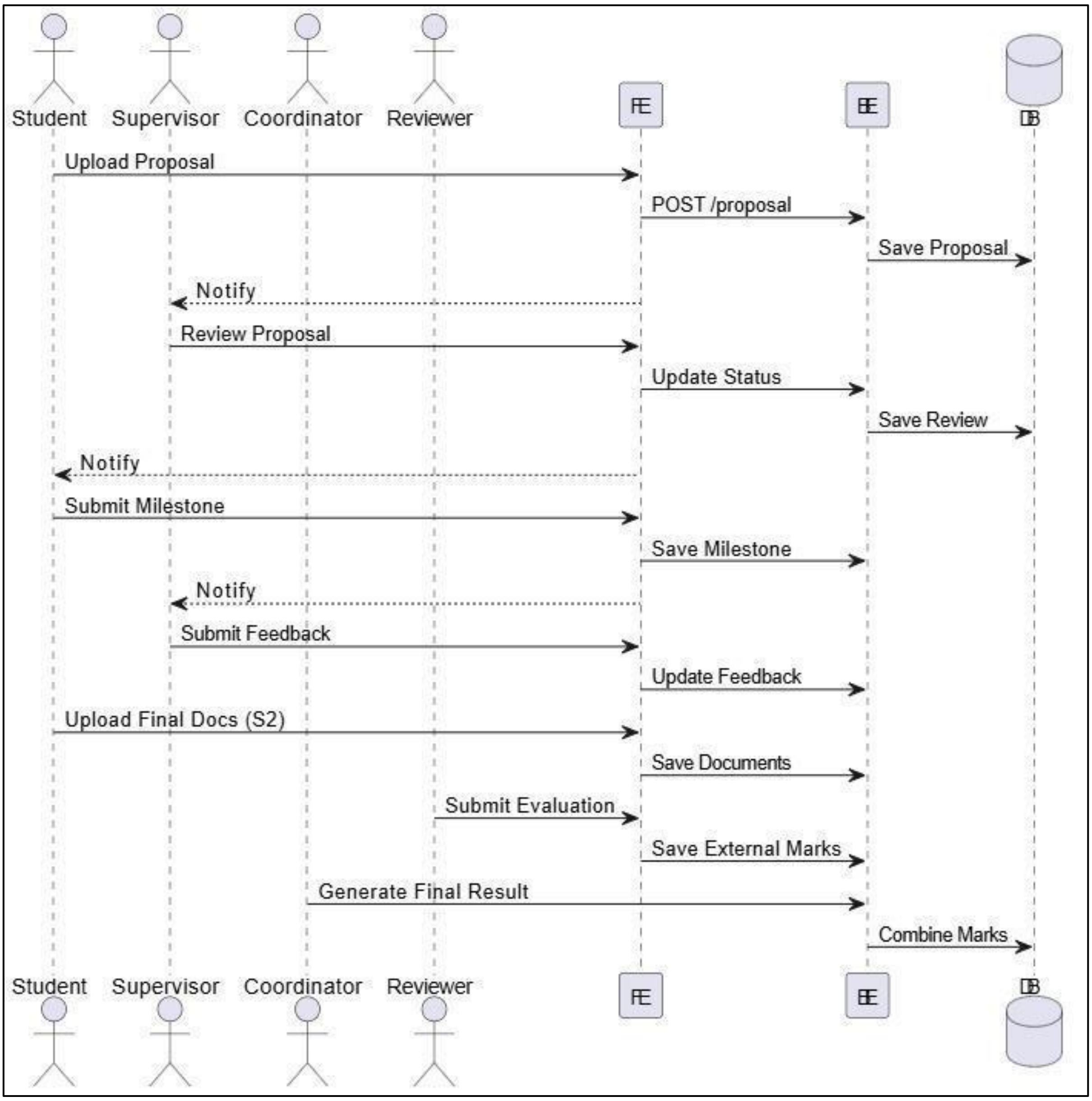

Figure 4-8: Sequence Diagram

4.4.2 Class Diagram

This class diagram represents the overall structure of the FYP Management System (FYP Genie). The system is built around the project entity, which maintains relationships with proposals, milestones, evaluations, and documents. Different user roles Student, Supervisor, Coordinator, and External Evaluator inherit from the main User class, ensuring clean role-based access. A project can have multiple proposal versions, milestones, evaluations, and uploaded documents. Supervisors oversee multiple projects, while each student is linked to a single project. External evaluators provide independent assessments, ensuring transparency and standardized evaluation.

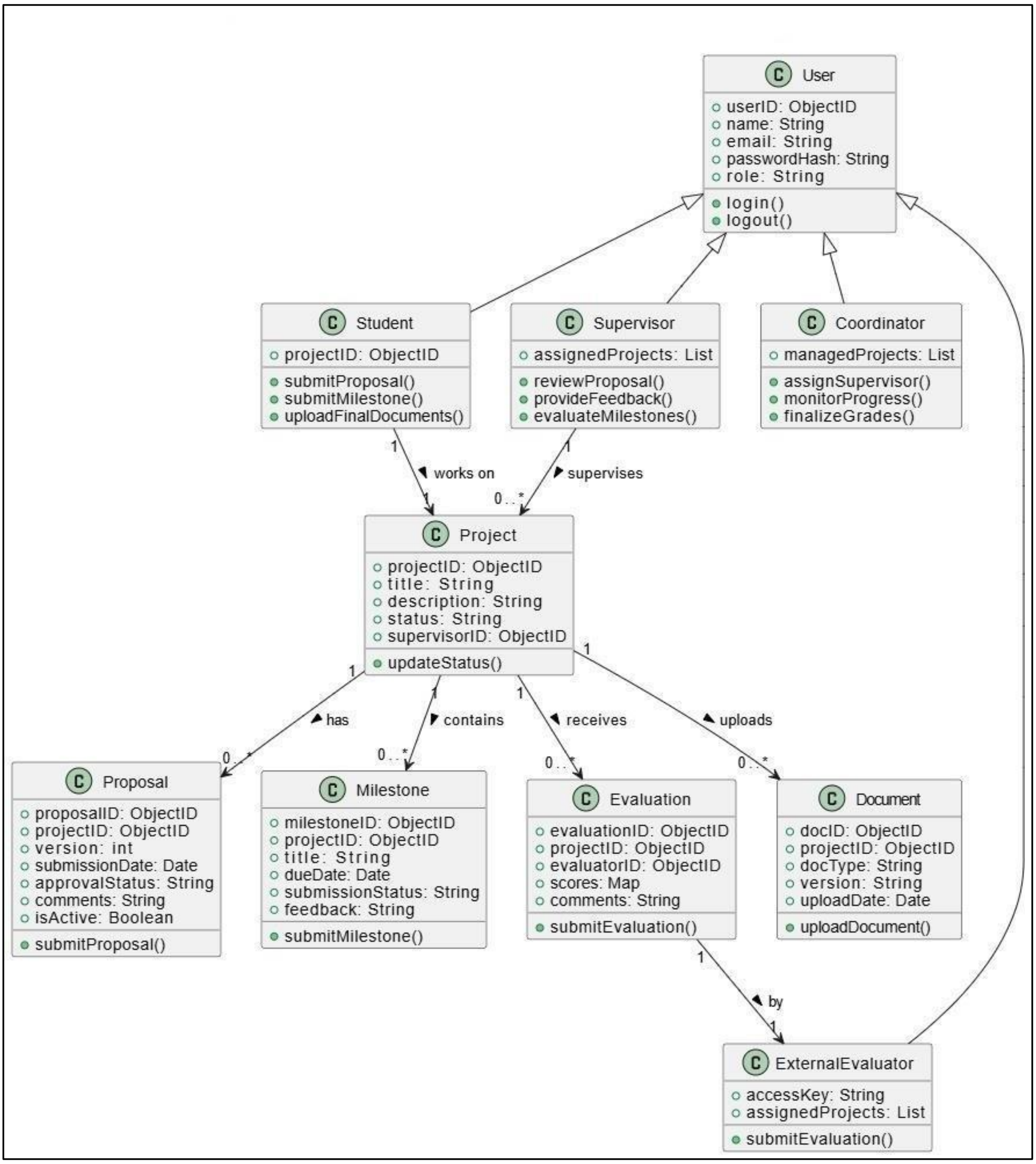

Figure 4-9: Class Diagram

Chapter 5
Implementation

5 Implementation

This chapter presents the complete implementation details of the FYP Genie system. It explains how the system architecture and design models described in Chapter 4 are translated into a working software system. The implementation focuses on modular development, efficient data handling, secure communication, and scalable system integration.

The system is developed using a full-stack JavaScript environment, ensuring consistency between frontend and backend operations. Each module is implemented independently and integrated using RESTful APIs, following industry best practices for web application development.

The FYP Genie system is implemented using the MERN stack, which provides a full-stack JavaScript-based development environment.

1 Technologies Used

- **Frontend:** React.js, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **API Communication:** RESTful APIs
- **Version Control:** GitHub

2 Development Tools

- Visual Studio Code
- MongoDB Compass
- Postman (API testing)

3 System Module Implementation

It presents a comprehensive overview of the major modules implemented in the FYP Genie system along with their respective functionalities, technologies used, and underlying algorithms. Each module is designed to handle a specific part of the Final Year Project workflow, ensuring clear separation of concerns and efficient system organization. The Authentication and User Management module ensures secure access and role-based control, while the Proposal Management module facilitates structured submission and review of project proposals. The Supervisor Assignment module automates the allocation process using a scoring-based algorithm to maintain fairness and balance. Similarly, the Milestone Tracking and Evaluation modules ensure continuous monitoring and accurate assessment of student progress. The Document Management module provides version control and secure storage of project files, whereas the Notification module enhances communication through timely alerts.

Table 5-1: System Module Implementation

Module Name	Description / Functionality	Technologies Used	Associated Algorithm / Logic
Authentication & User Management Module	Manages user registration, login, logout, role-based access control, and session handling for students, supervisors, coordinators, and external evaluators.	React.js, Node.js, Express.js, MongoDB, JWT, bcrypt	User Registration Algorithm, User Login Algorithm
Proposal Management Module	Allows students to submit project proposals, enables supervisors to review and approve/reject proposals, and maintains proposal history with status updates.	React.js Forms, Express APIs, MongoDB	Proposal Submission Algorithm, Proposal Review Algorithm
Supervisor Assignment Module	Automatically assigns supervisors based on expertise, workload, and project type to ensure fair distribution.	Node.js, Express.js, MongoDB	Supervisor Assignment Algorithm
Milestone Tracking Module	Tracks student progress through milestones, compares deadlines with submission dates, and updates status automatically.	React Dashboard, Node APIs, MongoDB	Milestone Progress Tracking Algorithm
Evaluation Management Module	Handles internal and external evaluations, stores marks, comments, and generates final results.	React.js, Express.js, MongoDB	Evaluation Submission Algorithm, Final Result Calculation Algorithm
Document Management Module	Manages upload, download, and version control of	Node.js File Handling, MongoDB	Document Upload & Version Control Algorithm

	proposals, reports, slides, and source code files.		
Notification & Reminder Module	Sends real-time notifications for deadlines, proposal approvals, milestone updates, and evaluation alerts.	React Notifications, Express.js, MongoDB	Notification Generator Algorithm
Coordinator Dashboard Module	Provides administrative control for managing students, supervisors, groups, viva schedules, and reports.	React.js Dashboard, APIs	Dashboard Management Logic
Supervisor Dashboard Module	Allows supervisors to review submissions, provide feedback, check plagiarism reports, and schedule viva.	React.js Dashboard, APIs	Submission Review Logic
External Evaluator Module	Provides secure read-only access to final documents and allows evaluators to submit assessments.	React.js, Node.js, MongoDB	External Evaluation Algorithm

5.1 Algorithms

The core functionality of the FYP Genie system is driven by a set of well-defined algorithms that automate critical operations involved in Final Year Project management. These algorithms are designed to reduce manual effort, eliminate human error, and ensure consistency, fairness, and efficiency across the system.

Unlike traditional systems where tasks such as supervisor assignment, progress monitoring, and evaluation are handled manually, FYP Genie leverages algorithmic logic to make data-driven decisions. Each algorithm operates on structured data stored in the database and interacts with different system modules through backend services.

The algorithms are designed with the following objectives:

- **Automation:** Minimize manual intervention in repetitive tasks
- **Accuracy:** Ensure correct and consistent outcomes
- **Fairness:** Provide unbiased decision-making (e.g., supervisor allocation)

- **Efficiency:** Reduce processing time and improve system responsiveness
- **Scalability:** Handle increasing number of users and projects

These algorithms collectively form the backbone of the system, enabling seamless coordination between students, supervisors, coordinators, and evaluators.

5.1.1 Supervisor Assignment Algorithm

The supervisor assignment process is one of the most critical components of the system, as it directly impacts project quality and academic supervision. In traditional systems, this task is often performed manually by coordinators, which can lead to imbalance, bias, or inefficient allocation.

The proposed algorithm addresses these issues by implementing a multi-criteria decision-making approach. It evaluates each available supervisor based on multiple parameters such as domain expertise, current workload, and project compatibility.

The algorithm works iteratively by scanning the list of all supervisors and calculating a numerical score for each one. The score reflects how suitable a supervisor is for a particular project. Supervisors with higher expertise in the project domain receive a higher score, while those with heavier workloads are penalized to avoid overburdening.

This approach ensures:

- Balanced workload distribution among supervisors
- Better alignment between project requirements and supervisor expertise
- Reduced manual effort for coordinators

Overall, the algorithm introduces fairness, transparency, and efficiency into the supervisor allocation process.

```
FUNCTION assignSupervisor(project):
    bestSupervisor ← NULL
    bestScore ← -∞
    FOR each supervisor IN supervisorList DO
        expertiseScore ← calculateExpertiseMatch(supervisor, project)
        workload ← getCurrentWorkload(supervisor)
        projectTypeMatch ← checkProjectType(supervisor, project)
        score ← expertiseScore - workload + projectTypeMatch
        IF score > bestScore THEN
            bestScore ← score
            bestSupervisor ← supervisor
        ENDIF
    ENDFOR
    ASSIGN project TO bestSupervisor
```

```
  UPDATE workload(bestSupervisor)
END FUNCTION
```

5.1.2 Milestone Progress Tracking Algorithm

The milestone tracking algorithm is responsible for continuously monitoring the progress of student projects. In manual systems, tracking deadlines and submissions is often inconsistent and prone to delays.

This algorithm ensures real-time tracking by comparing each milestone's submission status against its predefined deadline. It operates dynamically, meaning it updates the status automatically without requiring manual input.

The system categorizes milestones into different states:

- **Pending:** Deadline not reached and no submission
- **Submitted:** Work submitted before deadline
- **Late:** Submission after deadline
- **Overdue:** No submission even after deadline

Additionally, the algorithm triggers notifications when deadlines are approaching or missed. This proactive monitoring helps students stay on track and allows supervisors to intervene when necessary.

This mechanism improves:

- Time management
- Accountability
- Continuous project monitoring

```
FUNCTION trackMilestoneProgress(milestone):
   currentDate ← GET current system date
   IF milestone.submissionDate IS NULL THEN
      IF currentDate > milestone.dueDate THEN
         milestone.status ← "Overdue"
         SEND notification to student
      ELSE
         milestone.status ← "Pending"
      ENDIF
   ELSE
      IF milestone.submissionDate ≤ milestone.dueDate THEN
         milestone.status ← "Submitted"
      ELSE
         milestone.status ← "Late Submission"
         SEND warning notification
```

```
    ENDIF
  ENDIF
  UPDATE milestone in database
END FUNCTION
```

5.1.3 Notification and Reminder Algorithm

Communication is a key challenge in academic project management. The notification algorithm addresses this by implementing an event-driven communication system.

Whenever a significant event occurs in the system (e.g., proposal submission, approval, deadline nearing, evaluation completion), the algorithm generates a corresponding notification. These notifications are stored in the database and displayed to users in real time.

The algorithm ensures that:

- Relevant users are identified correctly
- Messages are generated based on event type
- Notifications are timestamped and tracked (read/unread)

This system reduces dependency on manual communication (emails, verbal updates) and ensures that all stakeholders are informed about important activities.

```
FUNCTION generateNotification(eventType, user):
  notification ← CREATE new notification object
  notification.userID ← user.id
  notification.message ← ""
  IF eventType = "Proposal Submitted" THEN
    notification.message ← "New proposal submitted"
  ELSE IF eventType = "Deadline" THEN
    notification.message ← "Milestone deadline approaching"
  ELSE IF eventType = "Approved" THEN
    notification.message ← "Your submission has been approved"
  ELSE IF eventType = "Rejected" THEN
    notification.message ← "Your submission has been rejected"
  ENDIF
  notification.status ← "Unread"
  notification.timestamp ← current time
  SAVE notification to database
END FUNCTION
```

5.1.4 Document Management and Version Control Algorithm

Managing project documents is a complex task, especially when multiple versions are involved. The document management algorithm ensures that all files are stored systematically with proper version tracking.

Whenever a document is uploaded, the algorithm checks for existing versions and assigns a new version number accordingly. Instead of overwriting files, it preserves previous versions, allowing users to track changes over time.

This approach provides:

- Data integrity (no loss of previous data)
- Traceability (history of document updates)
- Controlled access (based on user roles)

It is particularly useful for tracking revisions in proposals, reports, and final submissions.

```
FUNCTION uploadDocument(document, user):
   existingVersions ← GET all versions of document
   IF existingVersions IS EMPTY THEN
       versionNumber ← 1
   ELSE
       versionNumber ← MAX(existingVersions) + 1
   ENDIF
   newDocument.version ← versionNumber
   newDocument.uploadedBy ← user.id
   newDocument.timestamp ← current time
   STORE newDocument in database
  RETURN success message
END FUNCTION
```

5.1.5 External Evaluation Algorithm

The evaluation algorithm manages the process of assessing student performance. It ensures that evaluations are conducted in a structured and controlled manner.

Before allowing evaluation, the algorithm verifies whether the project is eligible (e.g., submitted and approved). Evaluators then input marks and feedback, which are stored securely in the database.

To maintain integrity:

- Evaluations are time-stamped
- Submissions may be locked after evaluation
- Only authorized users can evaluate

This structured approach ensures fairness, transparency, and proper documentation of academic assessments.

```
FUNCTION evaluateProject(project, evaluator):
   IF project.status ≠ "Submitted for Evaluation" THEN
      RETURN "Evaluation not allowed"
   ENDIF
   marks ← INPUT evaluation marks
   comments ← INPUT evaluator feedback
   evaluationRecord.projectID ← project.id
   evaluationRecord.evaluatorID ← evaluator.id
   evaluationRecord.marks ← marks
   evaluationRecord.comments ← comments
   evaluationRecord.timestamp ← current time
   SAVE evaluationRecord in database
   LOCK project submissions
  SEND notification to coordinator
END FUNCTION
```

5.1.6 Authentication Algorithm

Security is a critical aspect of any web-based system. The authentication algorithm ensures that only authorized users can access the system.

It works by validating user credentials against stored records. Passwords are stored in encrypted (hashed) form, and secure comparison methods are used during login. Upon successful authentication, a JWT token is generated, which is used to manage user sessions.

This approach provides:

- Secure login mechanism
- Protection against unauthorized access
- Session management through tokens

```
FUNCTION loginUser(email, password):
   user ← FIND user BY email
   IF user DOES NOT EXIST THEN
      RETURN "Invalid Email"
   ENDIF
   IF compareHash(password, user.password) = TRUE THEN
      token ← GENERATE JWT token
      RETURN token
   ELSE
      RETURN "Invalid Password"
   ENDIF
END FUNCTION
```

5.1.7 Proposal Submission Algorithm

The proposal submission algorithm handles the initial stage of the FYP lifecycle. It ensures that all proposals are properly recorded, tracked, and forwarded for review.

When a student submits a proposal:

- A new record is created with status "Pending"
- The proposal is stored with timestamp and student ID
- A notification is sent to the supervisor

This structured workflow ensures that no proposal is lost and all submissions are properly reviewed.

```
FUNCTION submitProposal(student, proposalData):
   proposal.studentID ← student.id
   proposal.content ← proposalData
   proposal.status ← "Pending"
   proposal.timestamp ← current time
   SAVE proposal in database
   SEND notification to supervisor
  RETURN "Proposal Submitted Successfully"
END FUNCTION
```

5.2 External APIs

The FYP Genie system uses RESTful APIs (Application Programming Interfaces) to enable communication between the frontend (React.js) and backend (Node.js/Express.js). APIs act as an intermediate layer that allows different components of the system to interact in a structured and secure manner.

Instead of directly accessing the database, the frontend sends HTTP requests to the backend through defined API endpoints. The backend processes these requests, applies business logic, interacts with the database, and returns appropriate responses in JSON format. This architecture ensures separation of concerns, making the system more scalable, maintainable, and secure.

5.2.1 API Architecture and Workflow

The API implementation follows a client-server architecture, where:

1. The client (frontend) sends a request (e.g., login, submit proposal)
2. The server (backend) receives the request through an API endpoint
3. The request is validated and processed using business logic
4. The backend interacts with the database (MongoDB)
5. A response (success/error/data) is sent back to the frontend

This structured workflow ensures smooth communication and efficient data handling across the system.

5.2.2 Types of API Requests

The system uses standard HTTP methods:

- **GET:** Retrieve data (e.g., fetch proposals, user data)
- **POST:** Submit new data (e.g., proposal submission, registration)
- **PUT:** Update existing data (e.g., update milestone status)
- **DELETE:** Remove data (if required)

Each request is designed to perform a specific function within the system.

5.2.3 Key API Modules

The API layer is divided into different modules based on system functionality:

1. **Authentication APIs**

   These APIs handle user login, registration, and session management.

   Examples:

   - /api/auth/register → Register new user
   - /api/auth/login → Authenticate user

   These APIs validate user credentials, hash passwords, and generate JWT tokens for secure access.

2. **Proposal Management APIs**

   These APIs manage submission and review of project proposals.

   Examples:

   - /api/proposals/submit → Submit proposal
   - /api/proposals/review → Approve/Reject proposal

   They ensure proper storage, status updates, and routing to supervisors.

3. **Supervisor Assignment APIs**

   These APIs automate supervisor allocation.

   Example:

   - /api/assign-supervisor

   This endpoint triggers the assignment algorithm and updates project records accordingly.

4. **Milestone Tracking APIs**

   These APIs monitor project progress.

Examples:

- /api/milestones/create
- /api/milestones/update

They track deadlines, submissions, and update milestone status dynamically.

5 **Evaluation APIs**

These APIs handle project assessment.

Examples:

- /api/evaluation/submit
- /api/evaluation/result

They store marks, feedback, and calculate final results.

6 **Document Management APIs**

These APIs manage file uploads and storage.

Examples:

- /api/documents/upload
- /api/documents/download

They handle version control and secure file access.

7 **Notification APIs**

These APIs generate and retrieve notifications.

Examples:

- /api/notifications/send
- /api/notifications/get

They ensure users receive updates about important events.

5.2.4 Data Validation and Error Handling

Each API includes validation mechanisms to ensure data integrity. Before processing any request:

- Input fields are checked for completeness
- Invalid data is rejected with error messages
- Duplicate entries are prevented

Error handling ensures that the system responds gracefully in case of failures, such as invalid login credentials or missing data.

5.2.5 Security in API Implementation

Security is a key aspect of API design in FYP Genie. The system implements:

- **JWT Authentication:** Ensures only authenticated users access protected routes

- **Middleware Authorization:** Restricts access based on user roles
- **Password Encryption:** Uses hashing (bcrypt) for secure storage
- **Protected Routes:** Sensitive APIs require valid tokens

These measures prevent unauthorized access and ensure data confidentiality.

5.2.6 API Flow Architecture Diagram

This diagram illustrates the API flow architecture of the FYP Genie system. The frontend sends HTTP requests to the backend through defined API endpoints. These requests pass through middleware layers where authentication, authorization, and validation are performed. After validation, the business logic layer processes the request using system algorithms such as supervisor assignment or evaluation handling. The processed data is then stored or retrieved from the MongoDB database. Finally, the server sends a structured JSON response back to the frontend, where the user interface is updated accordingly. This layered architecture ensures modularity, security, and efficient communication between system components.

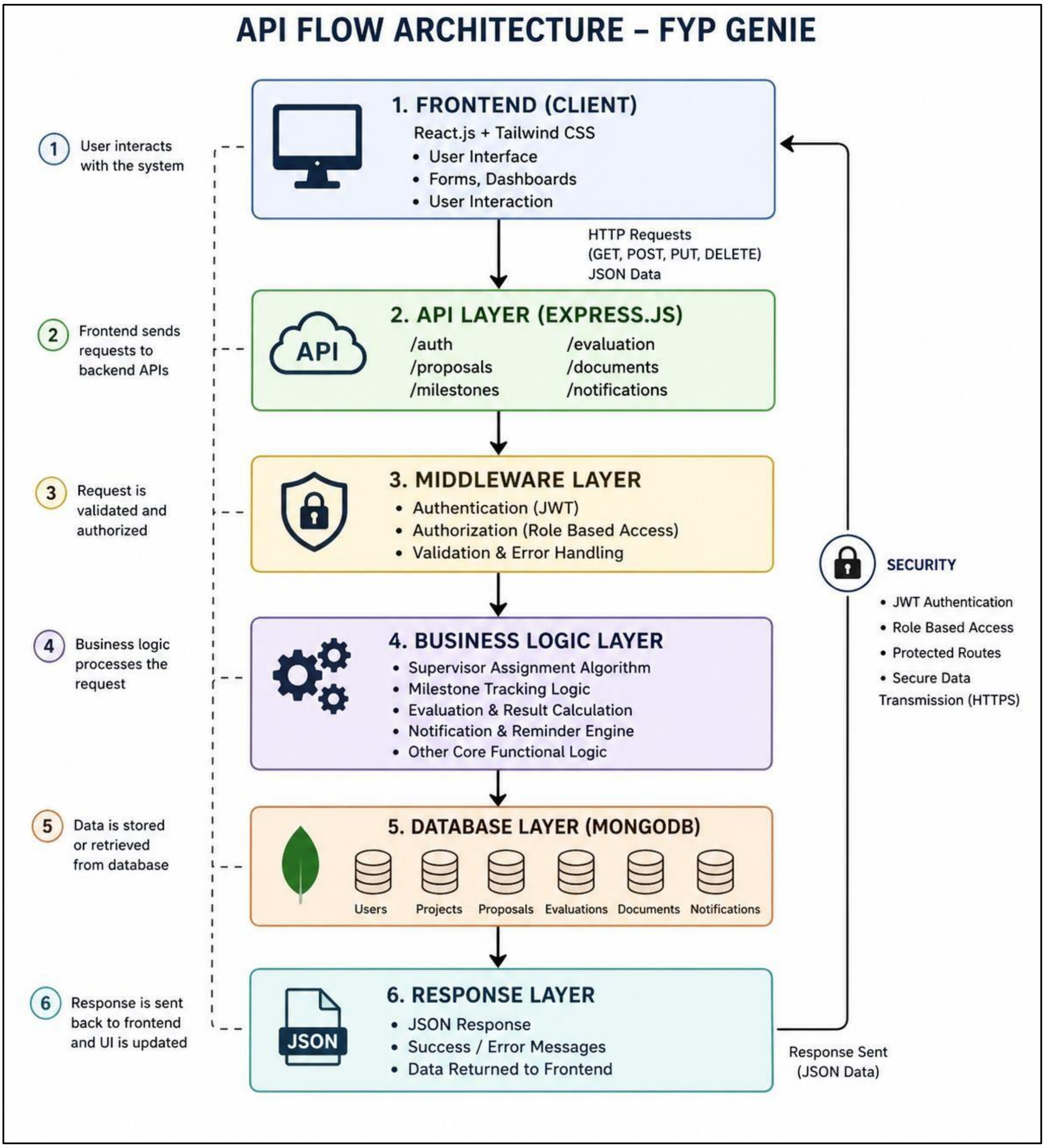

Figure 5-1: API Flow Architecture

5.3 User Interface

The User Interface Design defines the user-facing components of the FYP Genie portal. It focuses on providing a smooth, intuitive, and user-friendly experience for all stakeholders, including students, supervisors, coordinators, committee members, and reviewers.

Key design principles include:

1. **Role-Based Dashboards:** Each user sees a customized dashboard showing relevant information and actions, such as pending proposals, milestone status, evaluation requests, or notifications.
2. **Clean Navigation:** A simple menu structure allows easy access to proposals, milestones, evaluations, documents, and notifications.
3. **Forms and Submissions:** Structured forms for proposal submission, milestone updates, and evaluations guide users through required fields with validation to minimize errors.
4. **Real-Time Feedback:** Users receive immediate confirmation and alerts for submissions, approvals, or reminders, enhancing transparency and efficiency.
5. **Document Access and Management:** Uploading, downloading, and version control are intuitive, ensuring users can manage project files without confusion.
6. **External Reviewer Interface:** A secure, read-only interface allows reviewers to access final submissions and submit evaluations effortlessly.
7. **Consistency and Responsiveness:** UI components are consistent across modules, mobile-friendly, and responsive to different screen sizes.

This design ensures that all users can interact with the system efficiently, reduces learning curves, and supports the automated FYP workflow from proposal submission to final evaluation.

5.3.1 System Landing Page

The main landing page of the COMSATS FYP Portal, featuring a scenic background of the university campus. At the top center, the official COMSATS University Islamabad logo is prominently displayed above the welcoming text. A call to action instructs users to "Select your role to continue," guiding them toward the login process. Below this, four distinct blue buttons are available for different user roles: Student, Supervisor, Coordinator, and External Panel. The layout is simple and visually appealing, using a clear hierarchy to help users navigate to their respective sections. The overall design emphasizes accessibility and professional branding for the university's project management system. This page serves as the primary entry point for all stakeholders involved in Final Year Projects.

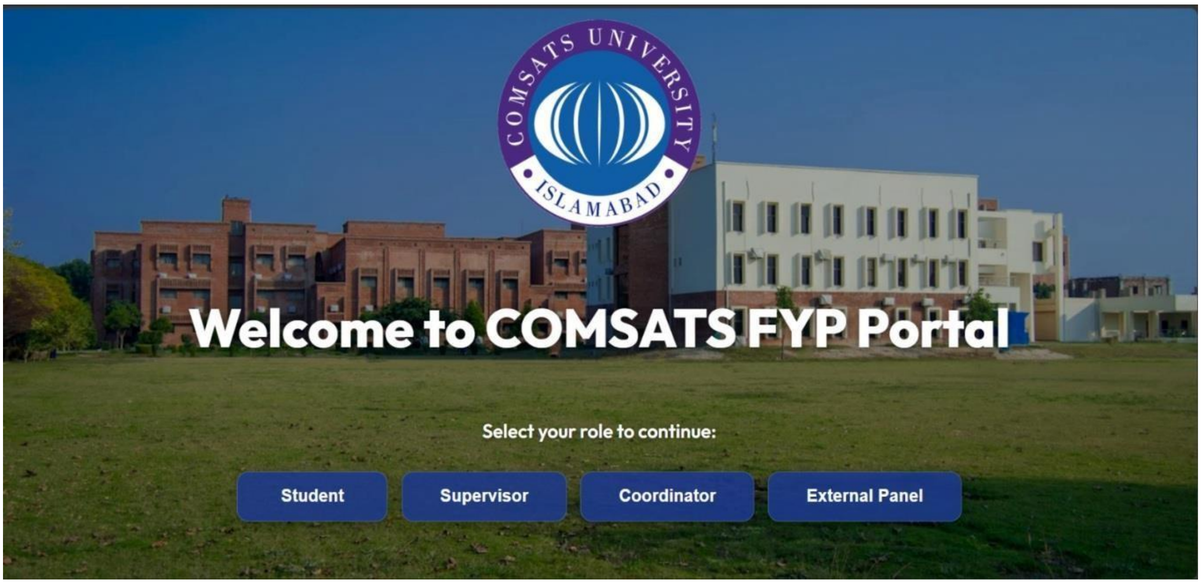

Figure 5-2: Landing Page

5.3.2 Student Modules: Login, Profile and Dashboard Screens

The Student Module in FYP Genie provides essential functionalities including login, profile management, and dashboard access. The login feature allows students to securely access the system using their credentials. The profile screen enables students to view and update personal and academic information. The dashboard acts as the main interface where students can submit proposals, upload milestones, and track project progress. It also displays notifications, feedback, and important updates related to their Final Year Project.

Login

This image displays the login interface for the FYP Management Portal at COMSATS University Islamabad, Sahiwal campus. The layout is clean and professional, featuring the university's official logo at the top center. Four navigation tabs Student, Supervisor, Coordinator, and External Panel allow users to select their specific role, with the "Student" tab currently active. The login form includes a field for the university email address, showing a specific student ID, and a masked password field below it. A verified reCAPTCHA box is present to ensure security by confirming the user is not a robot. At the bottom, a prominent blue button labeled "Login as Student" serves as the primary action for system access. Finally, a small footer note mentions that credentials are provided by the University Administration.

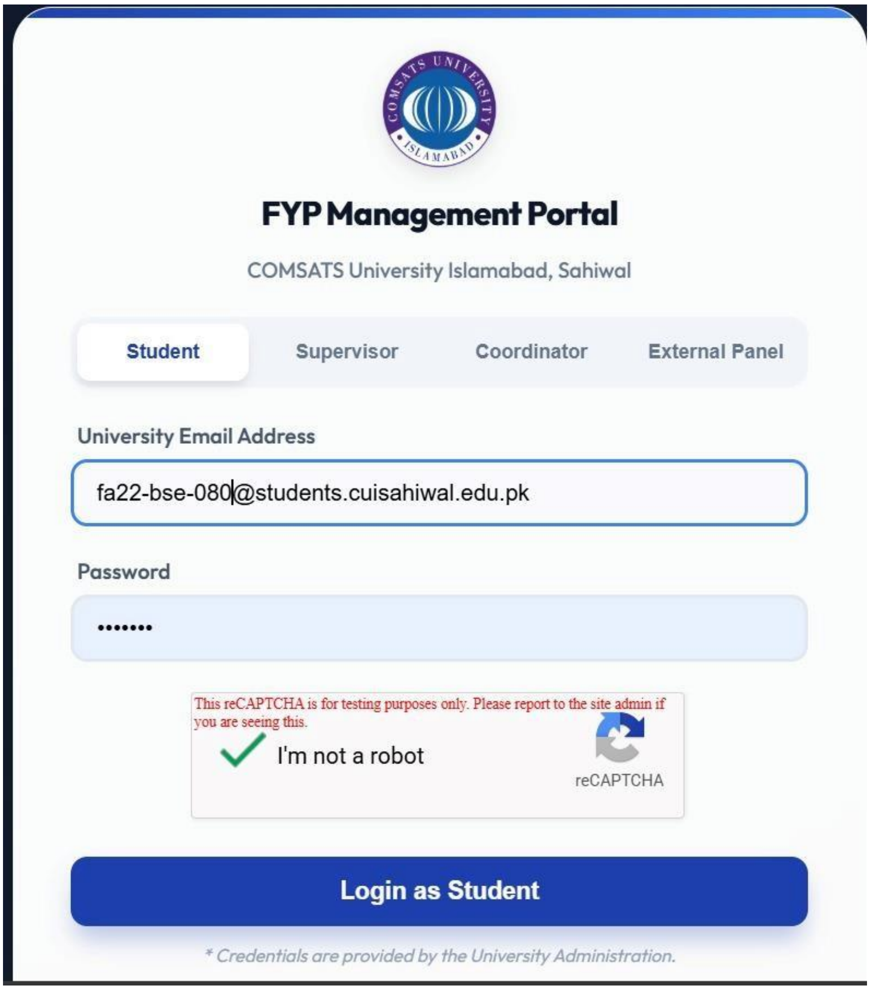

Figure 5-3: Student Login

Profile

The Student Profile page displays the personal and academic information of a registered student in a clean and organized layout. It includes a profile picture along with key details such as name, email address, enrollment ID, department, and current semester. The interface is designed to provide quick access to essential student information in a single view. An "Edit Profile" button is available at the bottom, allowing students to update their personal and academic details when needed. The design is user-friendly and ensures easy readability of all profile data. Overall, this page helps in maintaining accurate and up-to-date student records within the FYP Genie system.

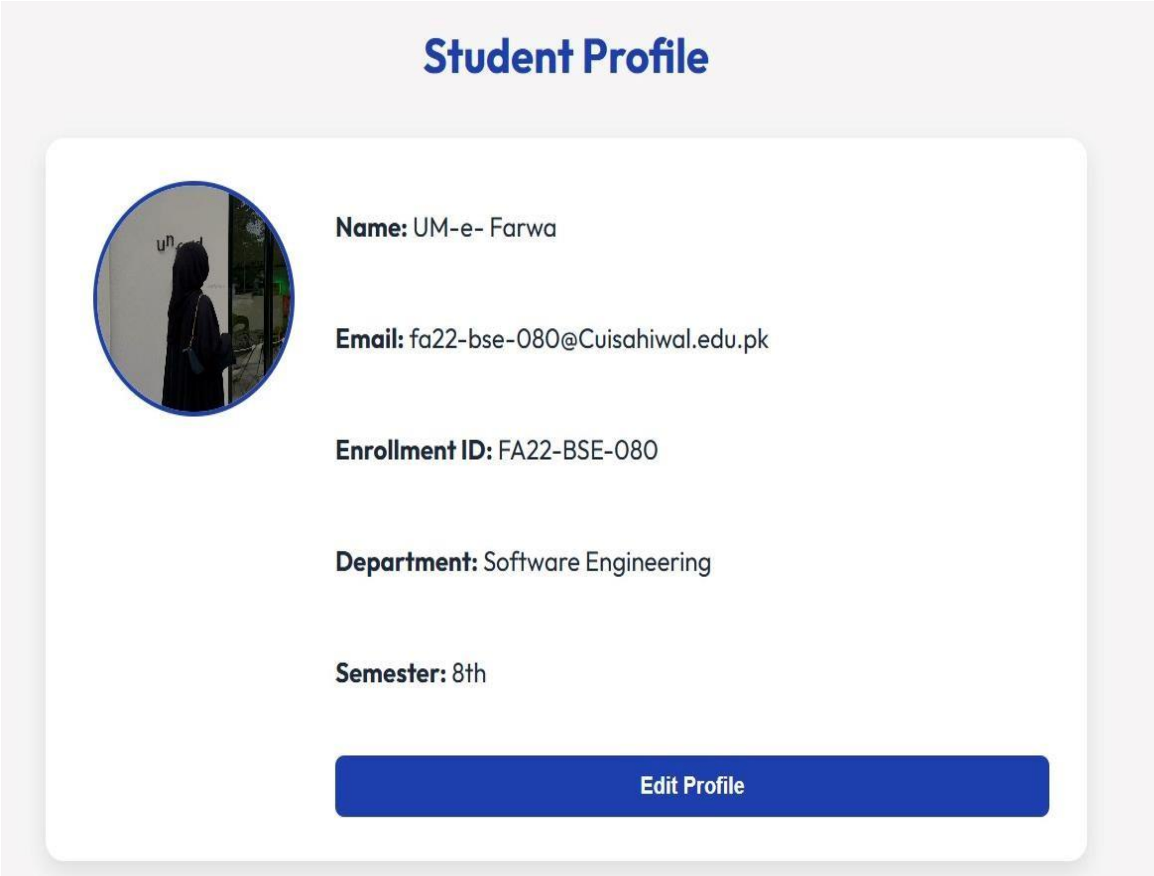

Figure 5-4: Student Profile

Home Page

The Student Portal dashboard provides a centralized interface for students to manage their Final Year Project activities efficiently. It allows students to track project milestones, communicate with supervisors, and access different system modules from a single platform. The sidebar navigation includes options such as Home, Groups, Faculty, Appointments, Tasks, and Deadlines for easy access to key features. The main dashboard displays portal-wide phase deadlines and indicates whether milestones are assigned or pending. Quick-access tiles such as Dashboard, Manage Group, semester sections, Appointments, Tasks, and AI Assistant improve usability and navigation. Overall, the interface is designed to enhance user experience by providing organized, fast, and structured access to all FYP-related activities.

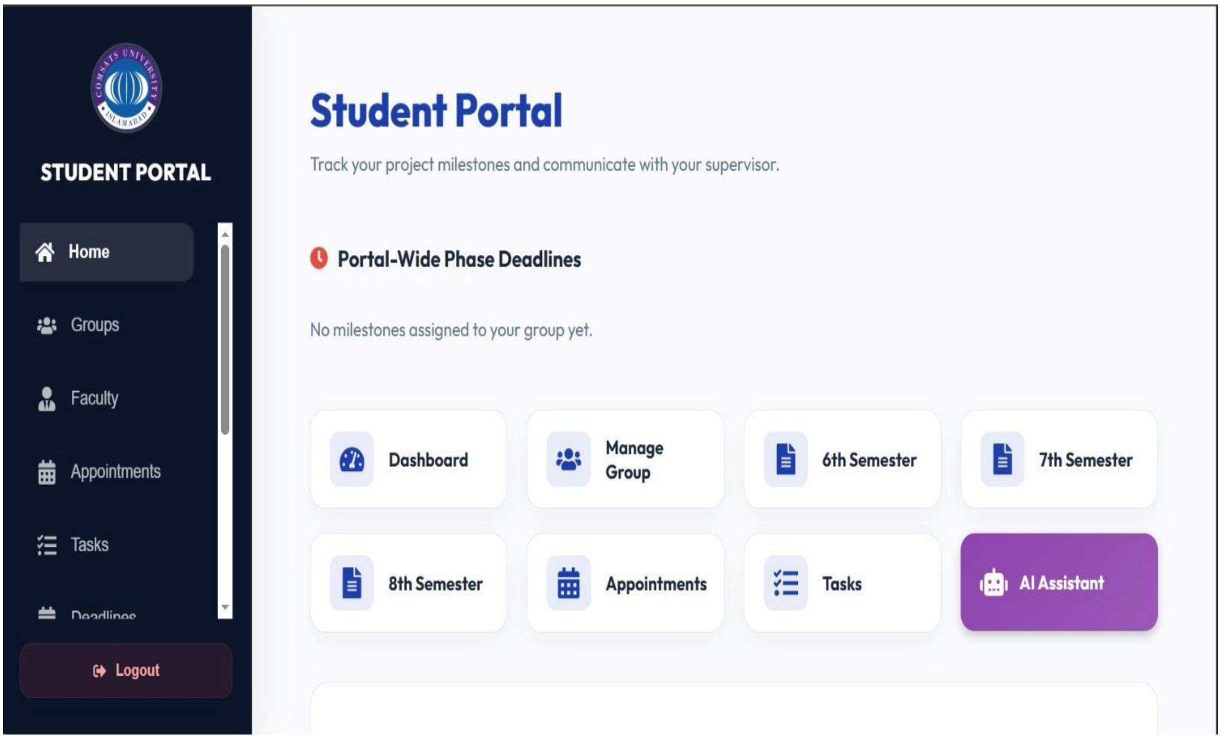

Figure 5-5: Home Page

Group Hub

The FYP Group Hub section of the Student Portal, where students can manage their project teams and supervisor requests. The main dashboard displays an "Accepted" project card for an AI Based project with a specific Group ID. It lists two group members, Zainab and Ali, along with their respective student registration numbers. A dedicated sidebar on the right indicates that the group is supervised by Usman Nasir and provides quick access to Personal Chat and Group Chat features. Additionally, the project idea is noted as "MERN," and there is a "New Group" button at the top right for creating additional teams. The left-hand sidebar maintains consistent navigation for the user across the entire portal.

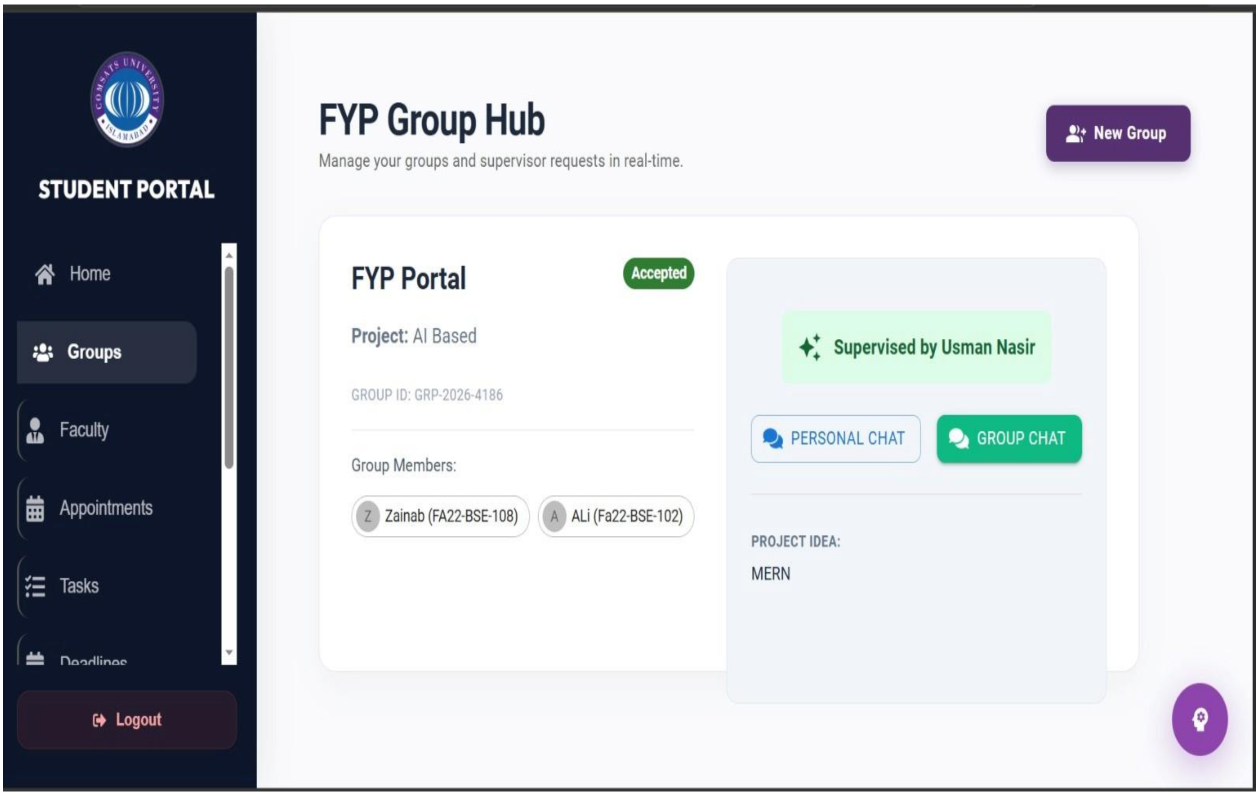

Figure 5-6: Group Hub

Faculty Directory

The Faculty Directory within a Student Portal, specifically designed for browsing potential FYP supervisors. The main dashboard features several profile cards for faculty members like Dr. Hassan and Dr. Fatima, listing their departments and specific expertise such as AI or Cybersecurity. Each card provides essential contact details, including university email addresses and buttons to either Chat or send a formal Request. A unique "Load" indicator shows how many project groups each professor is currently supervising against their maximum capacity. On the left, a dark sidebar offers easy navigation to other portal sections like Groups, Appointments, and Tasks. The overall design is organized and professional, aimed at streamlining the supervisor selection process for students.

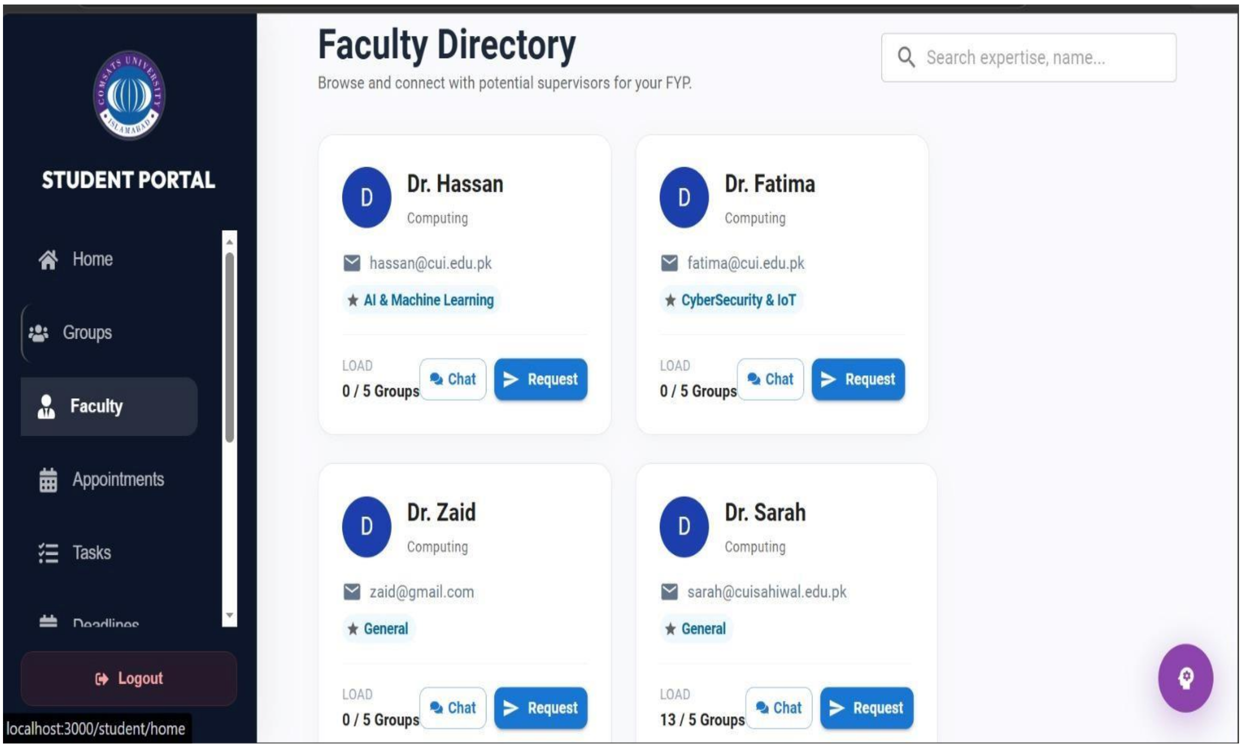

Figure 5-7: Faculty Directory

Appointments

The Book an Appointment page within the Student Portal, used for scheduling meetings with supervisors. On the left side, there is a list of faculty members to choose from, including Dr. Hassan, Dr. Fatima, Dr. Zaid, and Dr. Sarah, who is currently selected. The main section displays Dr. Sarah's live availability, showing different days of the week like Friday, Monday, and Thursday. Some time slots are marked as "OCCUPIED", while an available slot on Monday is shown for 14:00 - 14:39. The sidebar on the left highlights the Appointments tab, maintaining the portal's consistent navigation structure. This interface simplifies the process of coordinating project meetings by providing a clear view of faculty schedules.

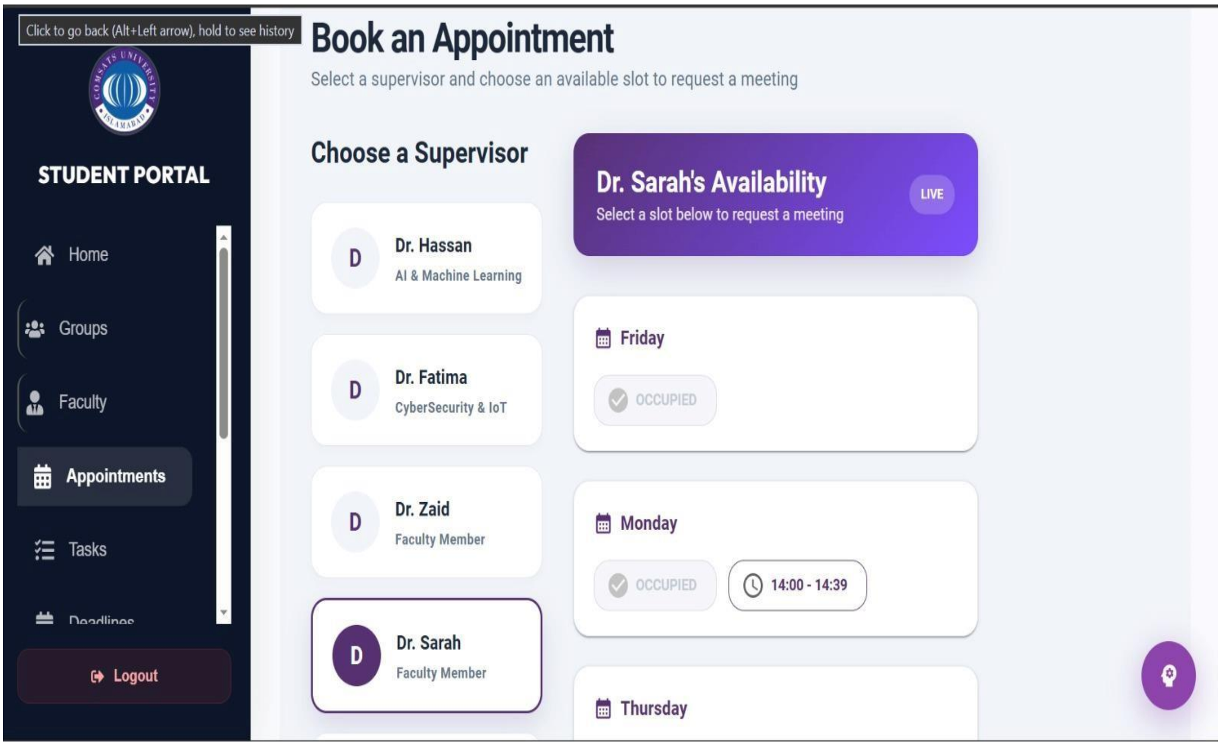

Figure 5-8: Appointment

Tasks

The Task Management page of the Student Portal, where students can view and submit tasks assigned by their project supervisors. The interface features two task cards, both marked with a "Medium" priority and a green "Completed" status badge. One task is titled "FYP portal" with a completion date of 4/29/2026, while the other is "AI based" with a date of 4/28/2026. Notably, the "AI based" task includes a Feedback section from the supervisor stating "need improvement." The navigation sidebar on the left highlights the "Tasks" tab, keeping the layout consistent with other portal sections. Overall, the design allows students to track their progress and supervisor commentary in a structured, real-time environment.

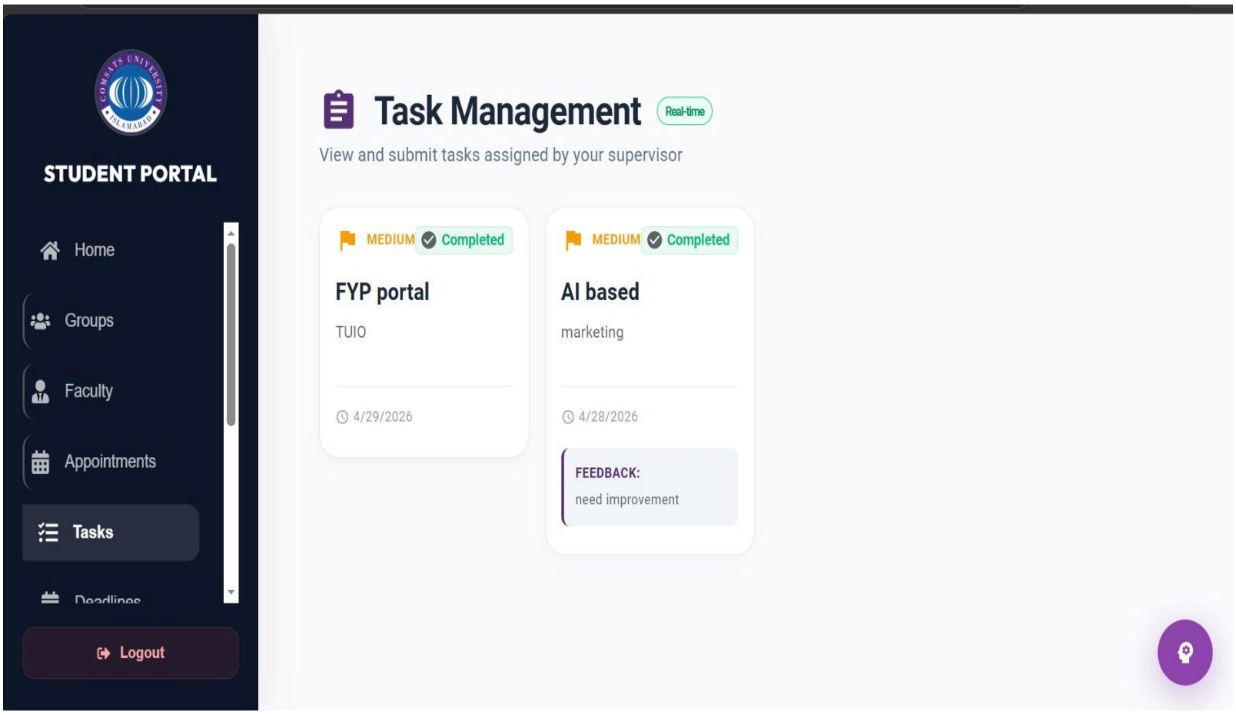

Figure 5-9: Task

Deadlines

This image displays the Deadlines & Submissions section of the Student Portal, designed for tracking academic deliverables. A prominent purple header shows summary statistics, indicating 1 total task with 0 completed. Below, a specific card for the "6th Semester" shows a proposal submission currently marked as "Pending." The task includes a due date of 5/9/2026 and a brief description simply stating "must." A dark "Upload Submission" button is provided at the bottom of the card for students to turn in their work. The consistent dark sidebar on the left allows for easy navigation to other portal features like Tasks and Faculty.

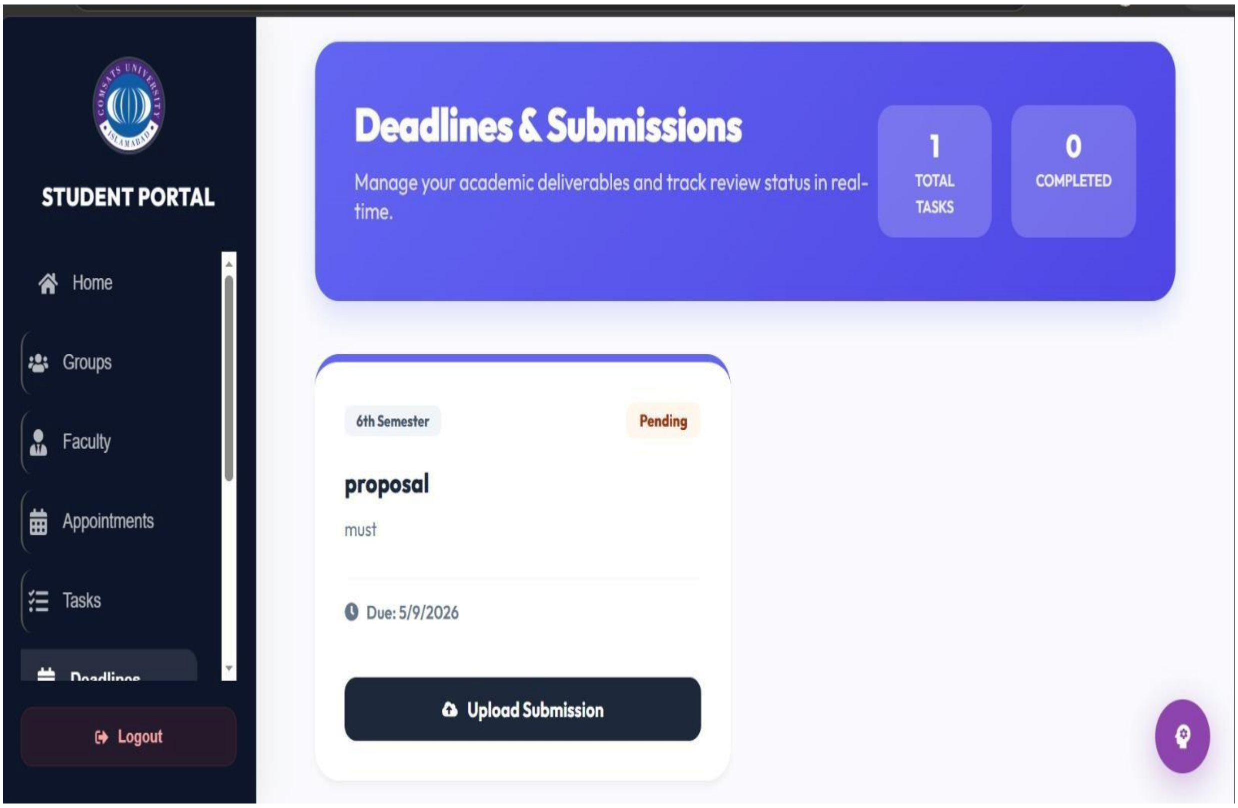

Figure 5-10: Deadlines

Notifications

The Notifications page within the Student Portal, providing real-time updates on academic timelines. The main section lists several alerts regarding updated deadlines for the 6th, 7th, and 8th semesters, each marked with a blue bell icon and a specific timestamp. A badge at the top right indicates there are 13 new notifications for the user to review. On the left, the dark sidebar highlights the "Notifications" tab with a red counter, keeping it consistent with the portal's navigation. Each entry clearly states the new date, such as 5/8/2026 or 5/1/2026, to ensure students stay informed. The overall interface is minimalist and organized, focusing on delivering critical information at a glance.

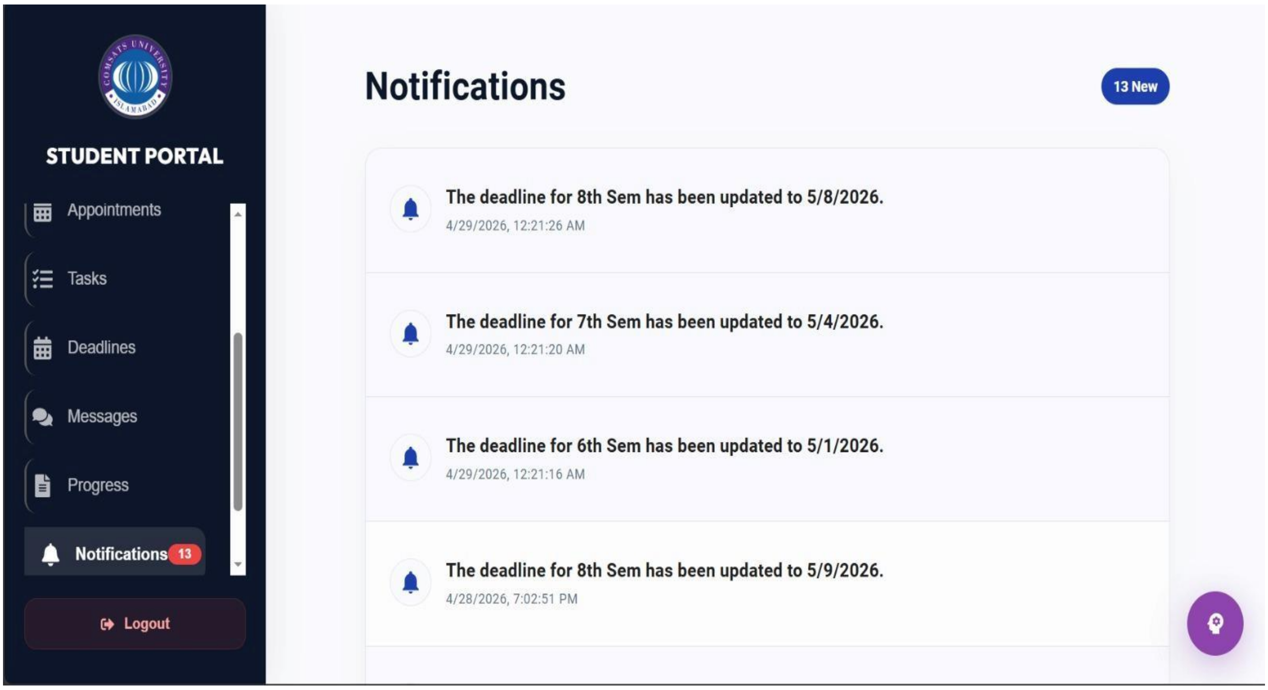

Figure 5-11: Notifications

5.3.3 Supervisor Modules: Login, Profile and Dashboard Screens

The Supervisor Module in FYP Genie provides features for login, profile management, and dashboard access. The login functionality ensures secure access for supervisors using authenticated credentials. The profile section allows supervisors to manage their personal details and areas of expertise. The dashboard provides an overview of assigned projects, student submissions, and pending tasks. It also enables supervisors to review proposals, give feedback, evaluate milestones, and monitor student progress.

Login

The login interface for the FYP Management Portal at COMSATS University Islamabad, Sahiwal campus, specifically for the Supervisor role. The page features the university logo at the top, followed by four tabs: Student, Supervisor, Coordinator, and External Panel, with "Supervisor" currently selected. The login form is filled with a university email address (usmannasir@cuisahiwal.edu.pk) and a hidden password. A verified reCAPTCHA "I'm not a robot" box is visible to ensure security. At the bottom, a prominent blue button is labeled "Login as Supervisor" for system access. A small footer note clarifies that login credentials are provided by the University Administration. The overall design is clean, professional, and consistent with the other portal login screens.

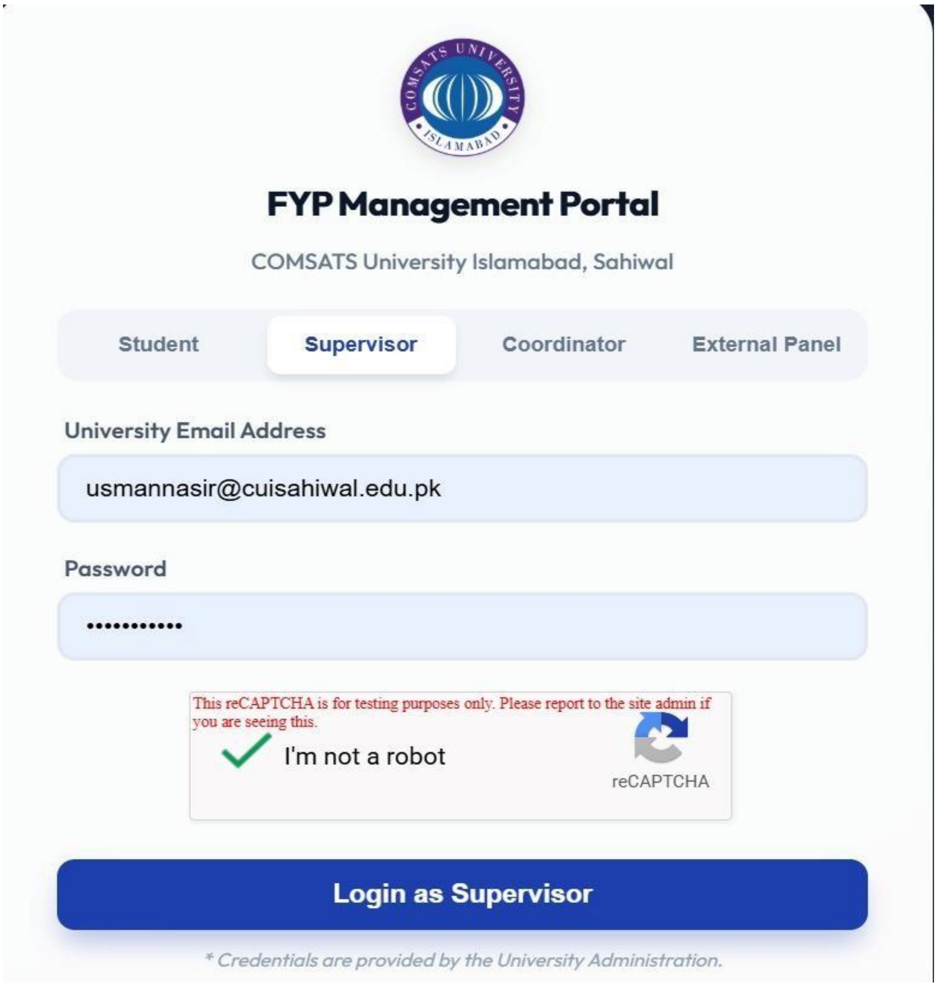

Figure 5-12: Supervisor Login

Profile

This interface displays the Supervisor Profile section of the FYP Genie system. It shows key details such as the supervisor's name, email, department, phone number, and office location. A profile picture is included on the left side for visual identification. The layout is clean and organized, making information easy to read. An "Edit Profile" button is provided to allow supervisors to update their personal information.

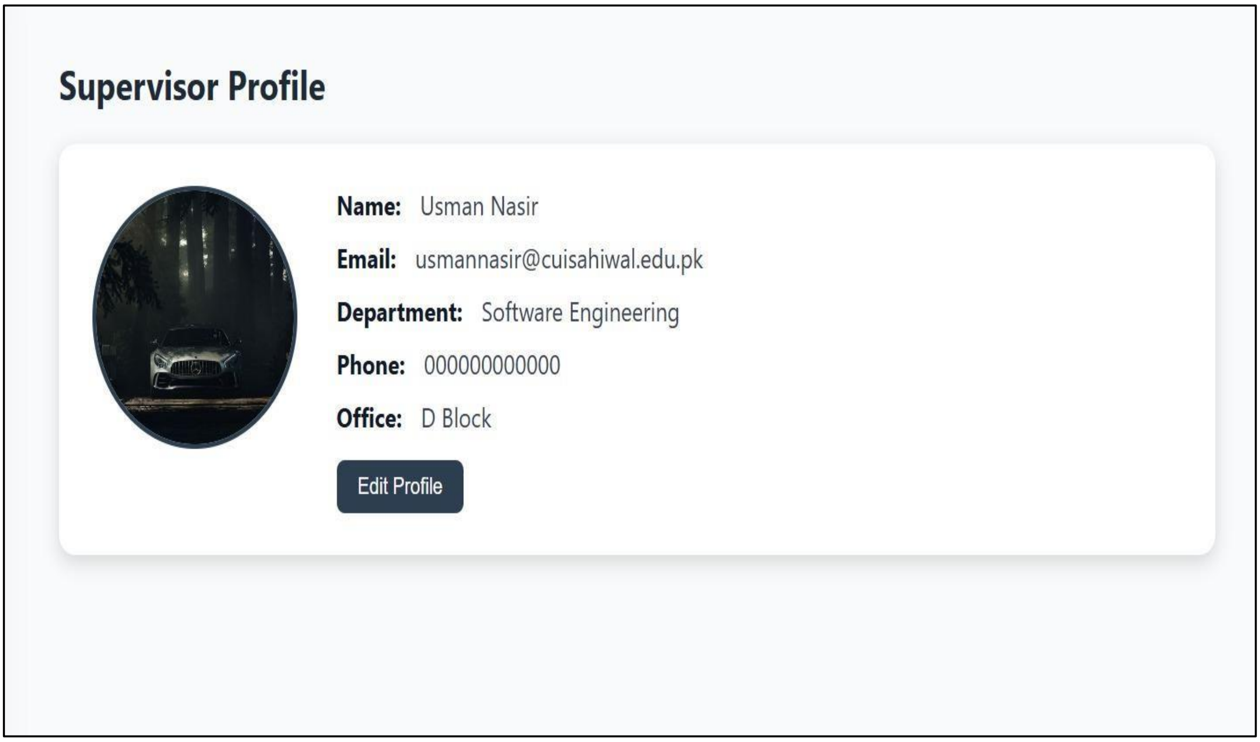

Figure 5-13: Supervisor Profile

Dashboard

This image showcases the Supervisor Dashboard of the FYP Management Portal, serving as a central hub for faculty to manage their assigned project groups. At the top, three summary cards display key statistics: 1 Active Group, 0 Pending Requests, and 0 Submissions. Below these, a grid of functional buttons provides quick access to tools like Manage Requests, Viva Schedule, Plagiarism checks, and a Progress Roadmap. A specialized purple button for an AI Assistant is also prominent, suggesting integrated smart support for supervisors. The left-hand sidebar, labeled "Supervisor Console," offers consistent navigation to sections such as Appointments, Tasks, and Messages. The overall interface is clean, data-driven, and designed to streamline the academic supervision process efficiently.

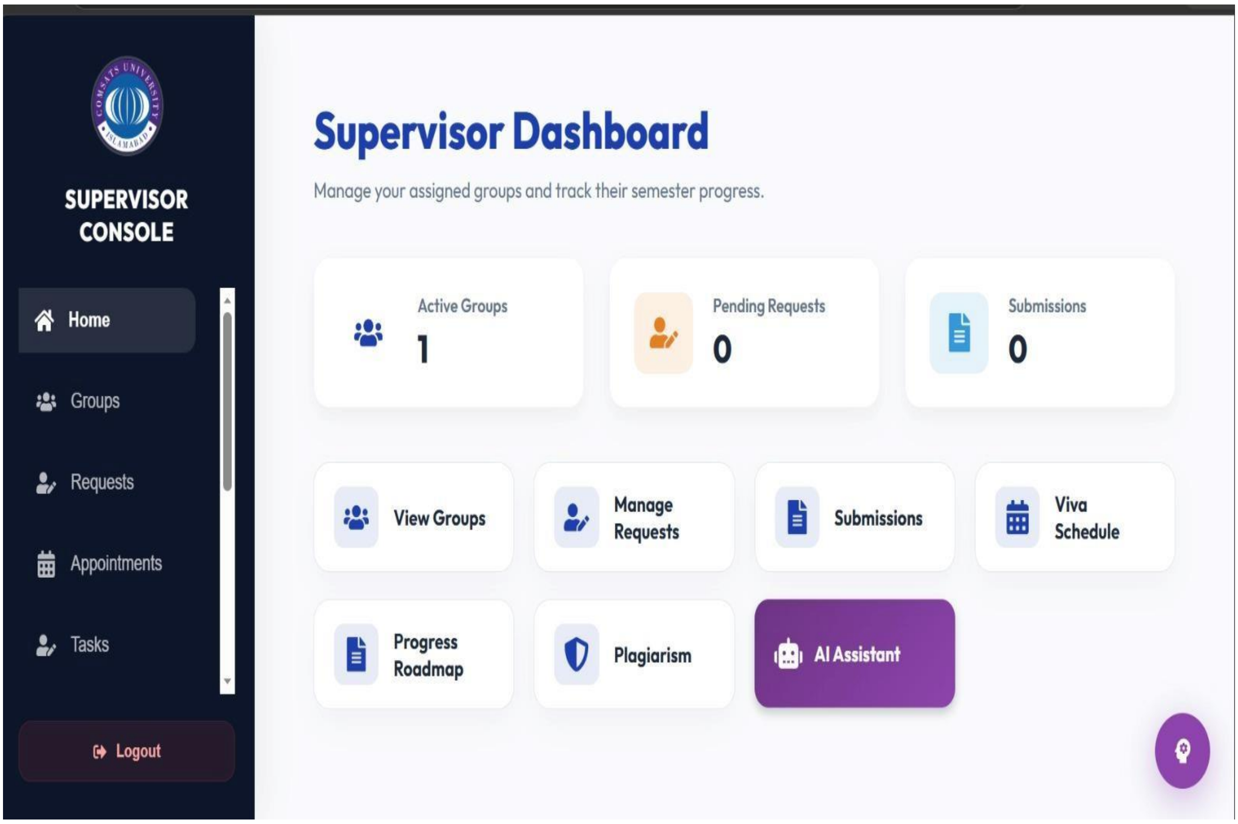

Figure 5-14: Supervisor Dashboard

My Supervised Groups

This image displays the My Supervised Groups page within the Supervisor Console of the FYP Management Portal. The main dashboard features a project card for a group working on an "AI Based" project titled "FYP Portal," which includes a badge indicating it has 2 members. Two primary action buttons are available on the card: "View Details" for a deeper look into the project and "Group Chat" for direct communication with the team. At the top right, there is an option to access "Archived Projects" for reviewing past work. The left-hand sidebar highlights the "Groups" tab, ensuring a consistent navigation experience across the portal. The overall design is minimalist and efficient, focusing on helping supervisors manage and communicate with their assigned teams in real-time.

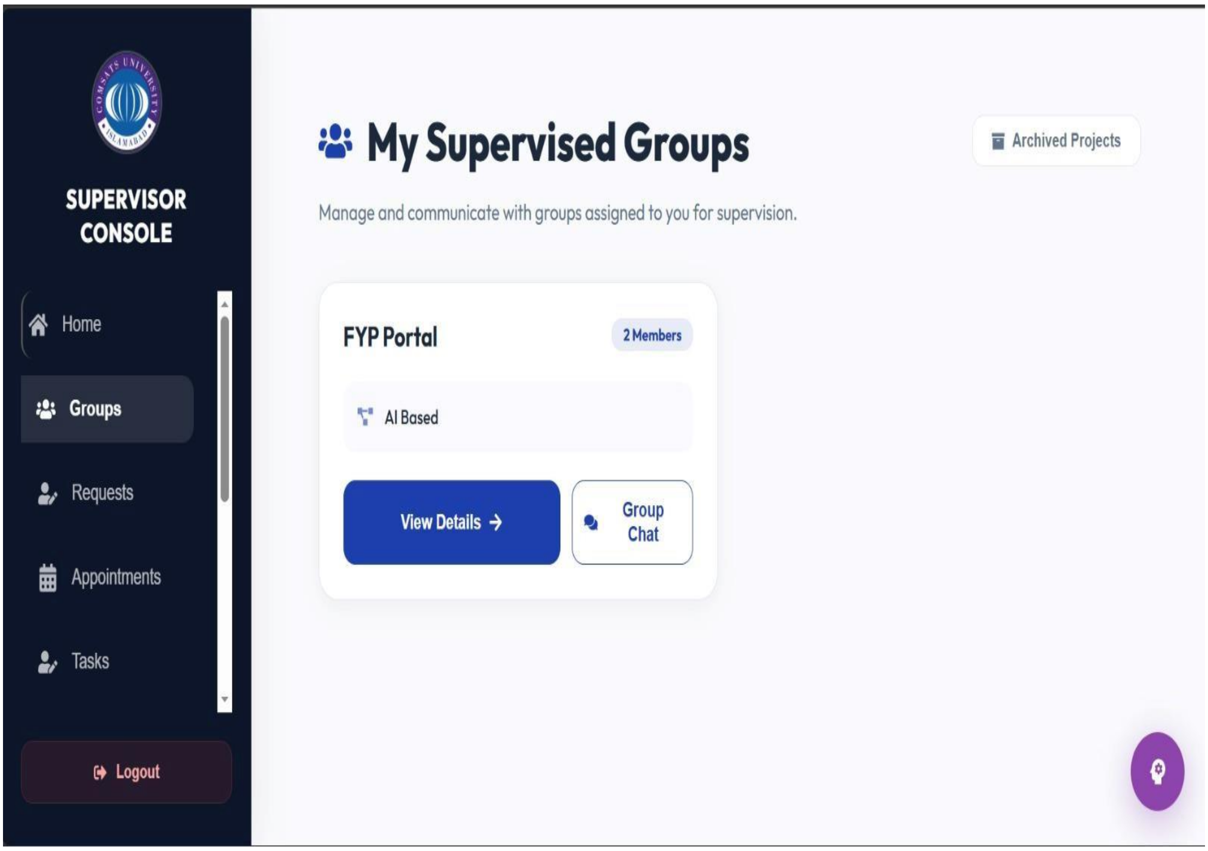

Figure 5-15: My Supervised Groups

Incoming Requests

The first image shows the login screen for the FYP Management Portal at COMSATS University Islamabad, Sahiwal, specifically set to the coordinator tab. It features fields for a university email (coordinator@cuisahiwal.edu.pk) and a masked password, along with a verified reCAPTCHA checkbox. A prominent blue button at the bottom allows the user to "Login as Coordinator". The second image illustrates the Incoming FYP Requests page within the Supervisor Console, where faculty members review student proposals. It displays a request for an "AI Based App" pitched by a user, listing group members Amna and Ahmed with their registration numbers. The supervisor has an input box to provide feedback or conditions, followed by clear Accept and Reject buttons to finalize their decision. This interface is part of the "Requests" section, as indicated by the active tab and notification badge on the dark sidebar.

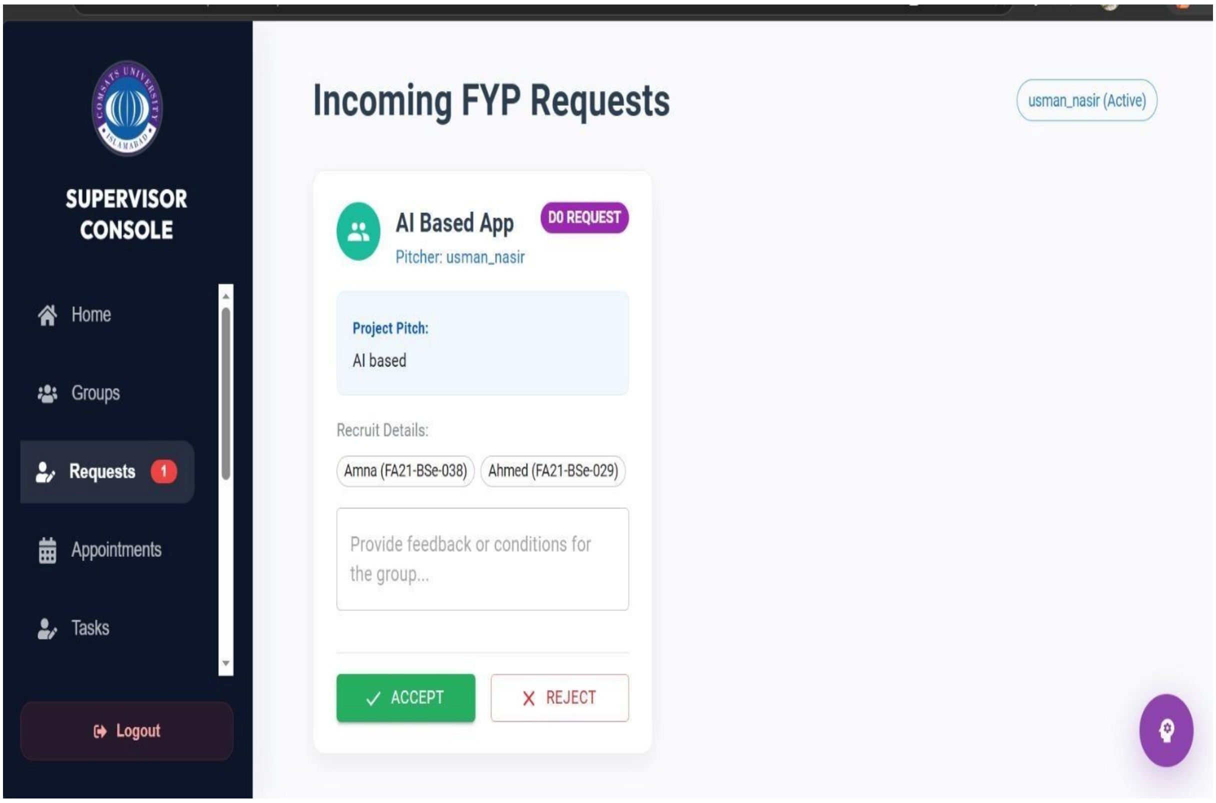

Figure 5-16: Incoming Request

Appointment Manager

The provided images illustrate the comprehensive FYP Management Portal for COMSATS University Islamabad, Sahiwal, detailing the user journey from authentication to project coordination. The interface begins with a professional landing page and role-specific login screens for students, supervisors, and coordinators, featuring secure reCAPTCHA verification. Once logged in, the Student Portal offers a structured Faculty Directory for browsing supervisors by expertise, alongside dedicated sections for tracking group status, deadlines, and task submissions. On the other hand, the Supervisor Console provides a high-level dashboard to monitor active groups, manage incoming project requests, and organize weekly meeting slots through an Appointment Manager. Throughout the portal, a consistent dark-sidebar navigation and clean card-based layout ensure that academic milestones, such as semester deadlines and supervisor feedback, are easily accessible and manageable in real-time.

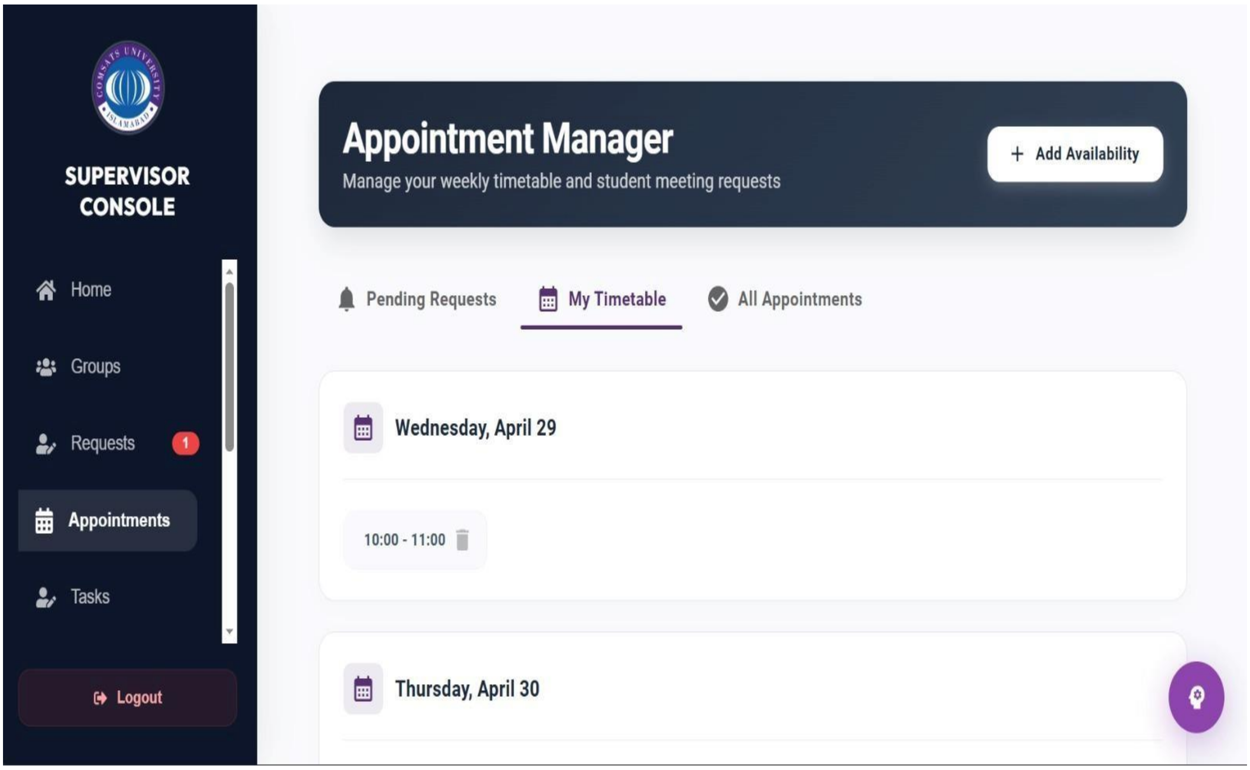

Figure 5-17: Supervisor Appointment Manager

Task Management

The provided images illustrate the comprehensive FYP Management Portal for COMSATS University Islamabad, Sahiwal, detailing the user journey from authentication to project coordination. The interface begins with professional login screens for students, supervisors, and coordinators, featuring secure reCAPTCHA verification and role-specific tabs. Once logged in, the Student Portal offers a structured Faculty Directory for browsing supervisors by expertise, alongside dedicated sections for tracking group status, deadlines, and task submissions. On the other hand, the Supervisor Console provides a high-level dashboard to monitor active groups, assign and track tasks, and organize weekly meeting slots through an Appointment Manager. Throughout the portal, a consistent dark-sidebar navigation and clean card-based layout ensure that academic milestones, such as semester deadlines and real-time supervisor feedback, are easily accessible and manageable.

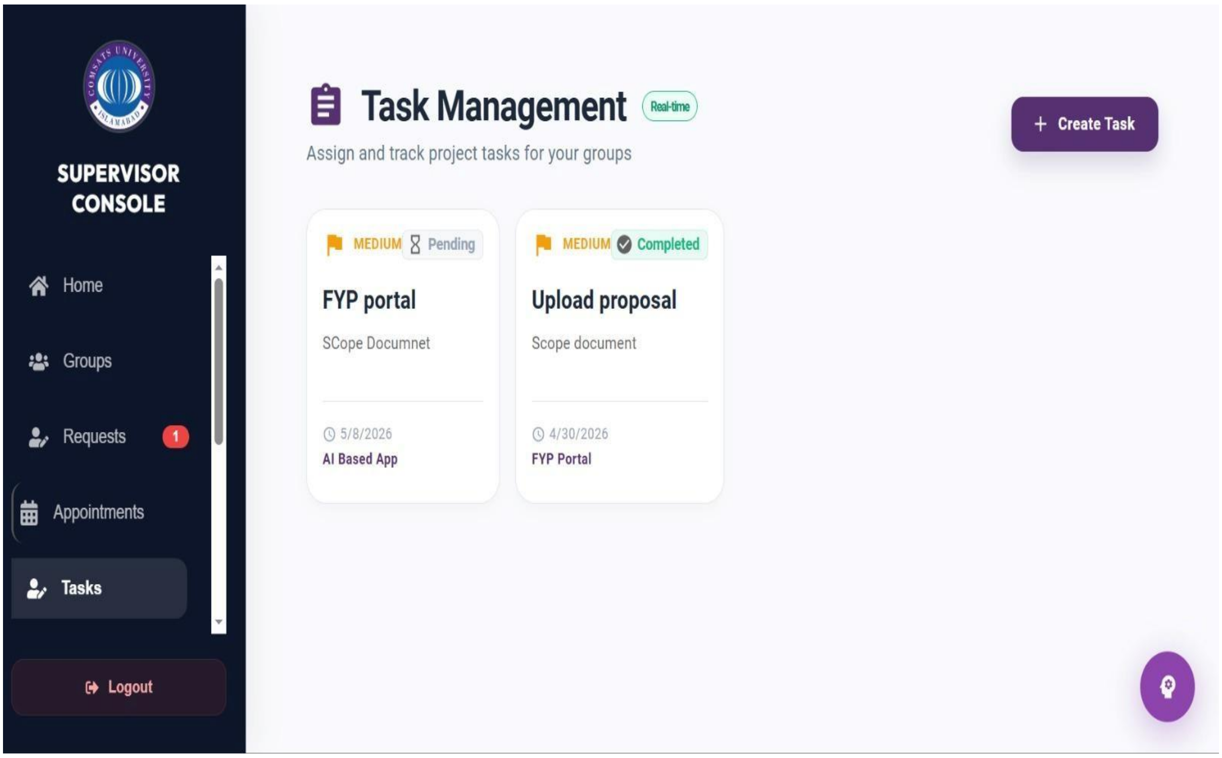

Figure 5-18: Supervisor Task Management

Student Submissions

The uploaded images detail a comprehensive FYP Management Portal for COMSATS University Islamabad, Sahiwal, highlighting the collaborative workflow between students and faculty. The Student Portal serves as a central hub where users can browse the Faculty Directory to select supervisors based on expertise, manage their team dynamics in the FYP Group Hub, and stay updated via a real-time Notifications system. It also includes a dedicated Deadlines & Submissions interface, allowing students to track pending tasks and upload academic deliverables like project proposals. Complementing this, the Supervisor Console provides faculty with a streamlined Student Submissions dashboard to review files, provide feedback, and update project statuses. Across both platforms, a consistent dark-sidebar navigation and clean, data-driven layout ensure that critical milestones and communication channels remain easily accessible for all stakeholders.

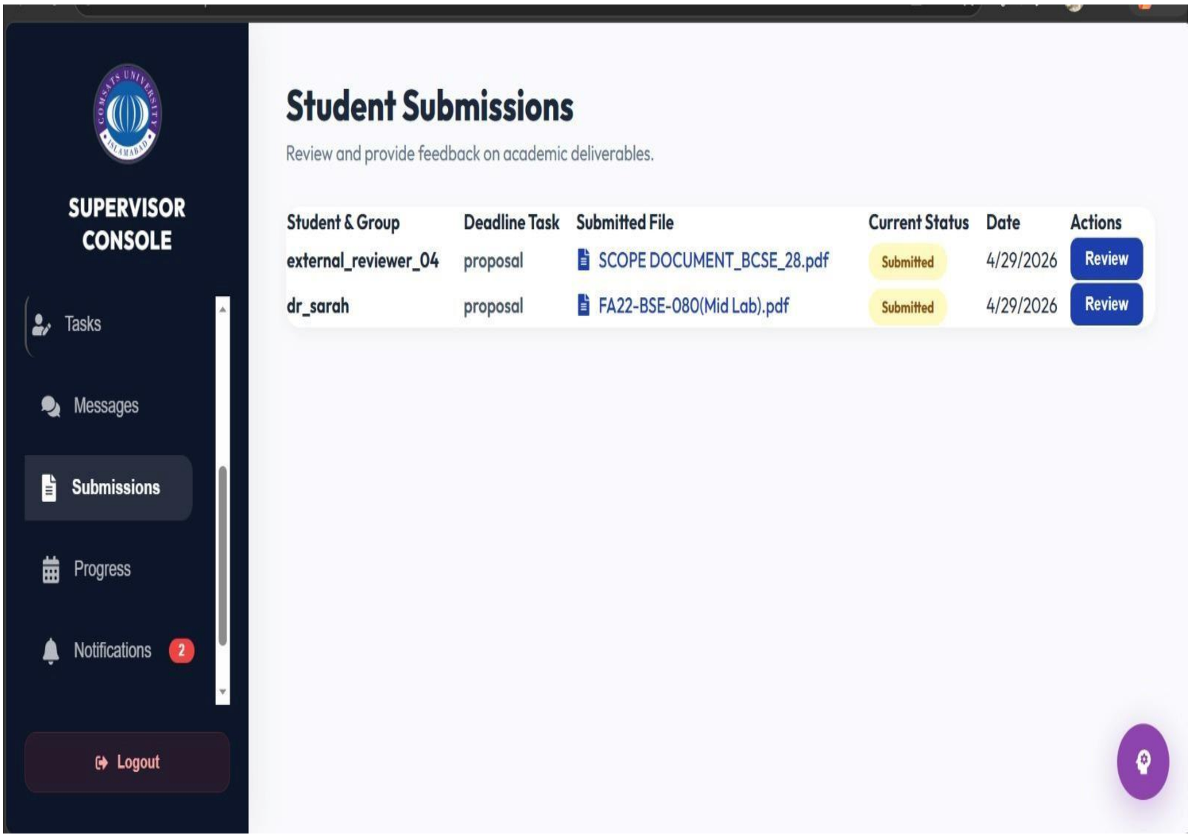

Figure 5-19: Student Submissions

Notification

The COMSATS FYP Management Portal is a specialized digital ecosystem designed to streamline the Final Year Project lifecycle for students and faculty through role-specific dashboards. The Student Portal enables users to browse a searchable Faculty Directory to find supervisors, manage team dynamics in the Group Hub, and track deliverables via dedicated Deadlines & Submissions sections. Meanwhile, the Supervisor Console provides faculty with data-driven tools to oversee active groups, assign tasks with real-time feedback, and review academic submissions like project proposals. Key interactive features include a live Appointment Booking system for scheduling meetings and a robust Notification module that keeps all stakeholders updated on semester milestones. With its modern UI and consistent sidebar navigation, the portal ensures a highly organized and collaborative environment for academic project management.

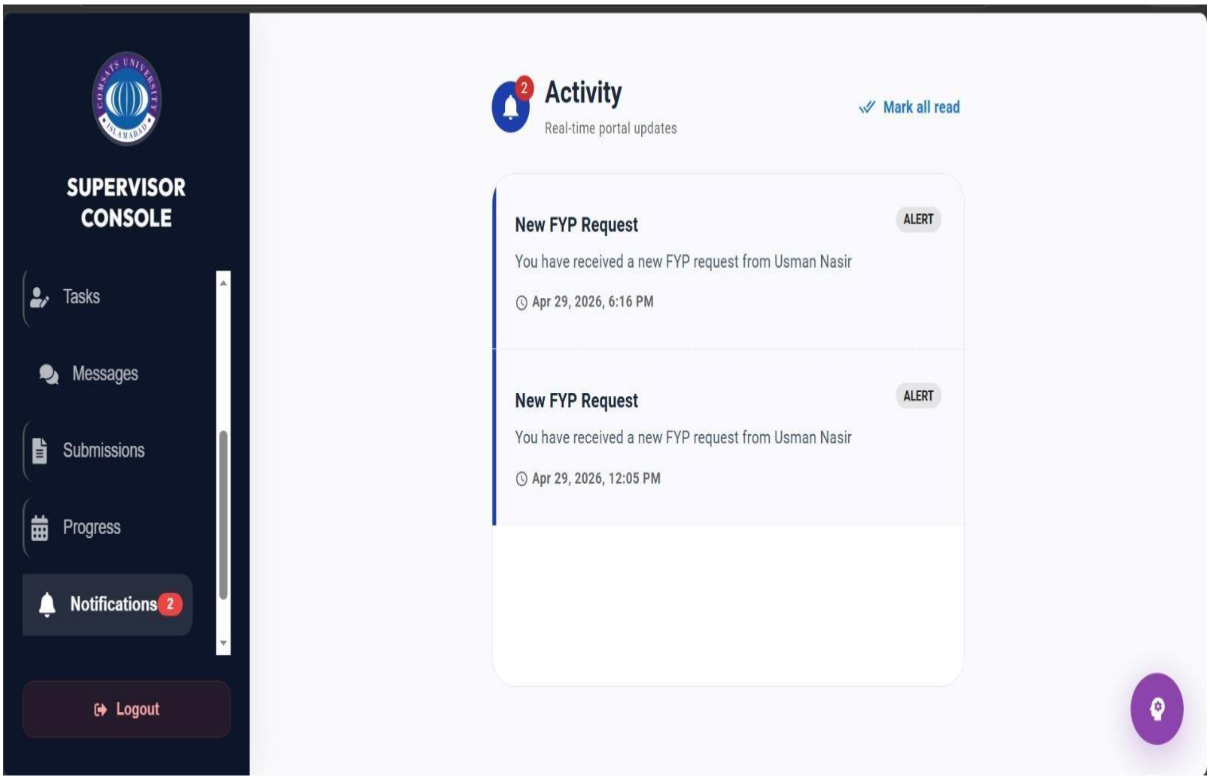

Figure 5-20: Supervisor Notifications

5.3.4 Coordinator Modules: Login, Profile and Dashboard Screens

The Coordinator Module in FYP Genie allows coordinators to securely log in and manage the overall FYP process. The profile screen enables them to view and update personal and administrative details. The dashboard provides access to key functionalities such as managing students, supervisors, and project groups. Coordinators can assign supervisors, schedule evaluations, and monitor project progress. It also displays reports, notifications, and system updates for effective decision-making.

Login

The image displays a login interface for the FYP Management Portal at COMSATS University Islamabad, Sahiwal campus. The design features a clean, professional aesthetic with the university logo centered at the top. Below the title, there are four navigational tabs allowing users to switch between roles: Student, Supervisor, Coordinator, and External Panel. The current selection is set to "Coordinator," with input fields populated for a university email address and a masked password. A Google reCAPTCHA verification box is visible, showing a green checkmark to confirm the user is not a robot. Finally, a prominent blue button labeled "Login as Coordinator" serves as the primary call to action for accessing the system.

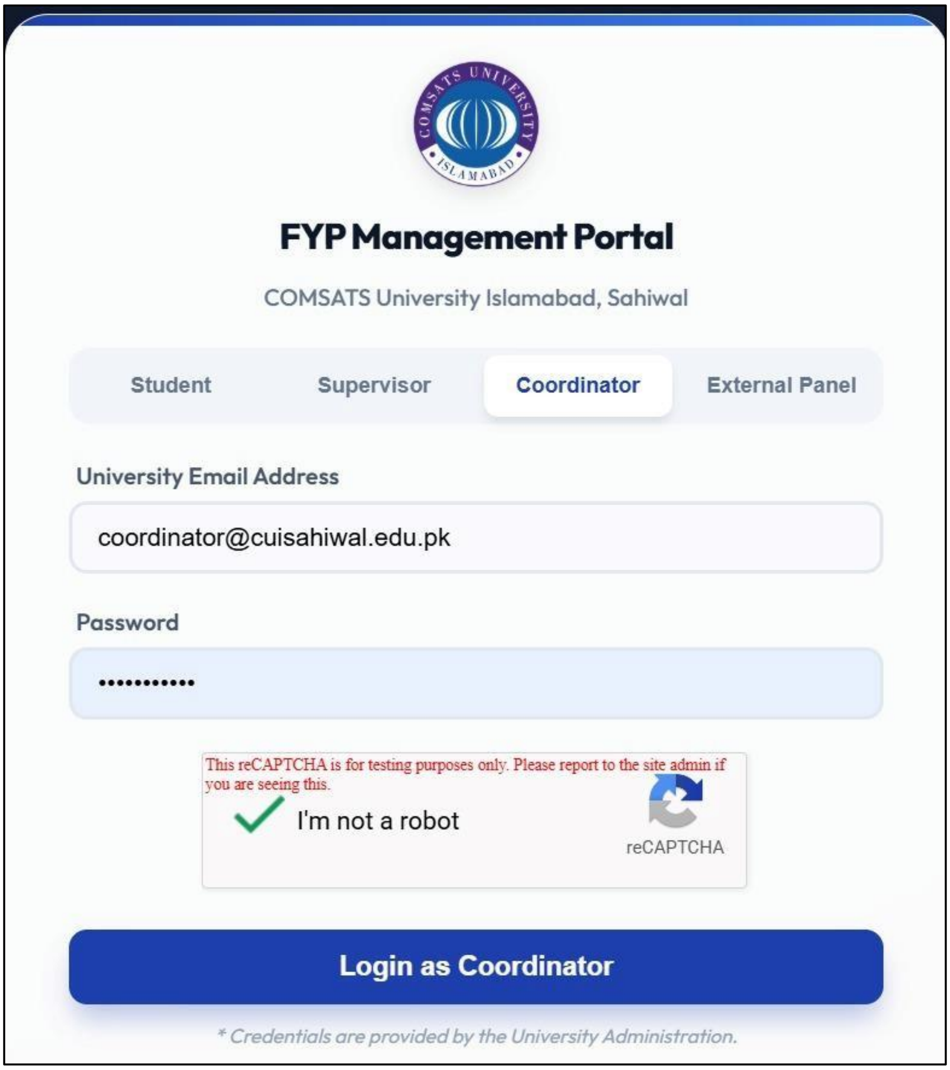

Figure 5-21: Coordinator Login

Student List

The image shows a **"**Students List**"** page within a coordinator dashboard system. On the left side, a sidebar labeled "Coordinator Console" provides navigation options such as Home, Students, Supervisors, Groups, and Requests. The main section displays a table containing student records with columns for ID, Name, Email, and Group. Multiple student entries are listed, including names like Ali Ahmed, Sara Khan, and Usman Ali. Most students currently have their group status marked as "Not assigned." The layout is clean and structured, making it easy to view and manage student information. A logout button is also visible at the bottom of the sidebar for exiting the system.

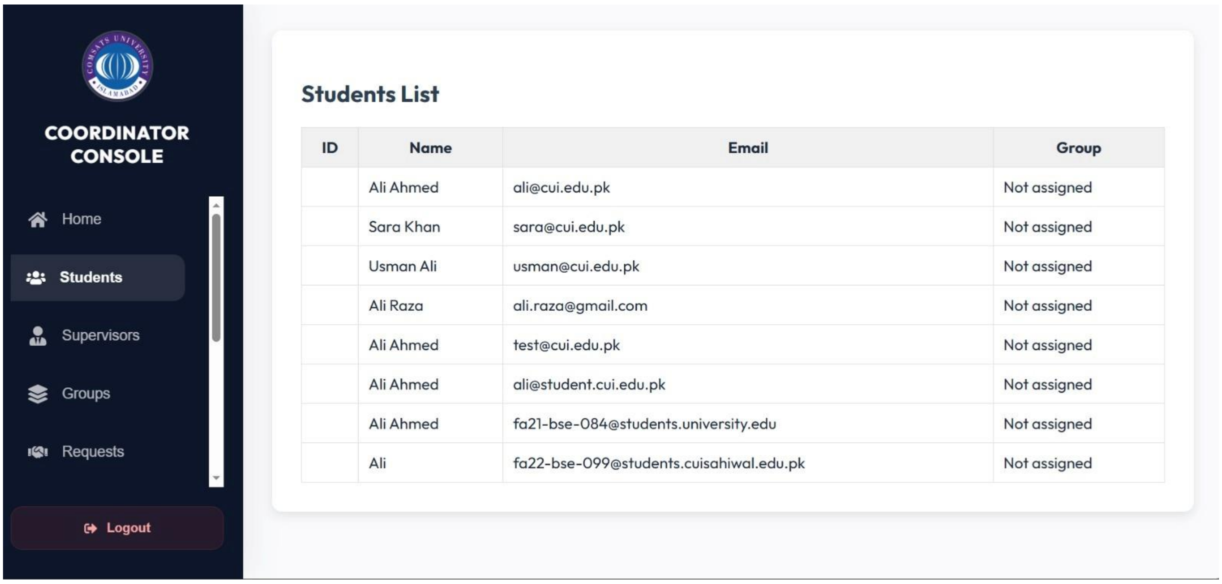

Figure 5-22: Student List

Faculty Network

This screen shows the Faculty Network section of the Coordinator Console, designed to manage and monitor all FYP supervisors in the system. It presents a structured table listing each supervisor along with their department, contact information, and area of expertise such as AI & Machine Learning or Cybersecurity & IoT. The interface also displays each supervisor’s workload, indicating how many groups they are currently supervising out of their capacity. A status label like “Open” or “Full” helps quickly identify whether a supervisor is available for new groups or not. Additionally, a search bar at the top allows coordinators to easily find supervisors by name, department, or expertise, making the overall management process more efficient and organized.

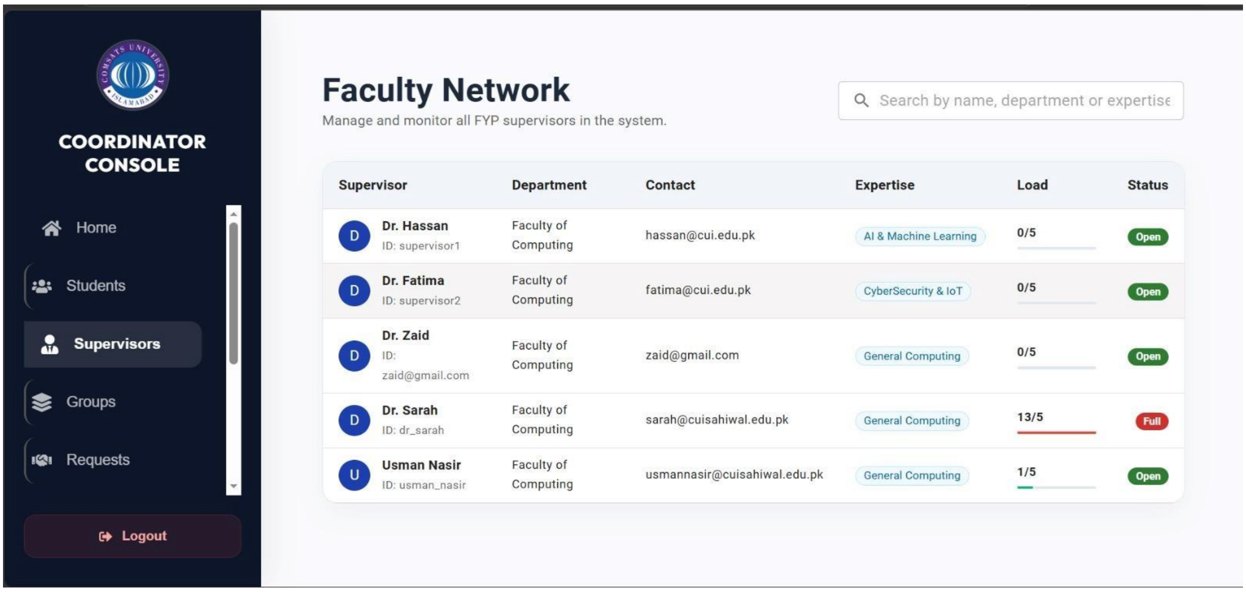

Figure 5-23: Faculty Network

Group Management

This screen represents the Coordinator Console-Groups Management module of the FYP Genie Portal. It provides an overview of all student groups along with their project details in a structured table format. Each row displays key information such as group ID, project title, synopsis, assigned students, and supervisor details. The coordinator can easily track the status of proposals, including whether they are accepted, rejected, or pending. Status indicators are visually highlighted to improve readability and quick decision-making. The sidebar navigation allows access to other modules like Students, Supervisors, Requests, and Home. Overall, this interface enables efficient monitoring and management of FYP groups by the coordinator.

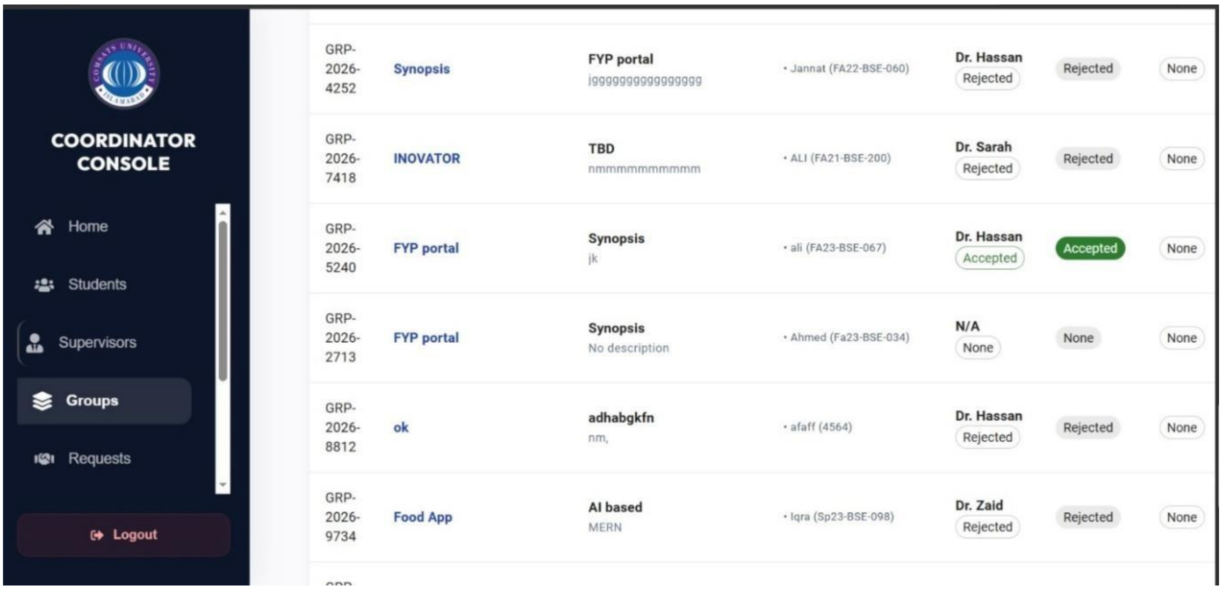

Figure 5-24: Coordinator Console Group Management

Supervision Requests

It provides a real-time view of different project groups and their assigned supervisors. Each row displays key details like group name, supervisor, project idea, and request status. The status labels (e.g., Pending, Accepted) help quickly identify approval progress. There is also a "Requested On" column to track when each request was submitted. Action icons (like the eye symbol) allow the coordinator to view more details of each request. Overall, the system helps streamline and monitor the FYP supervision approval process efficiently.

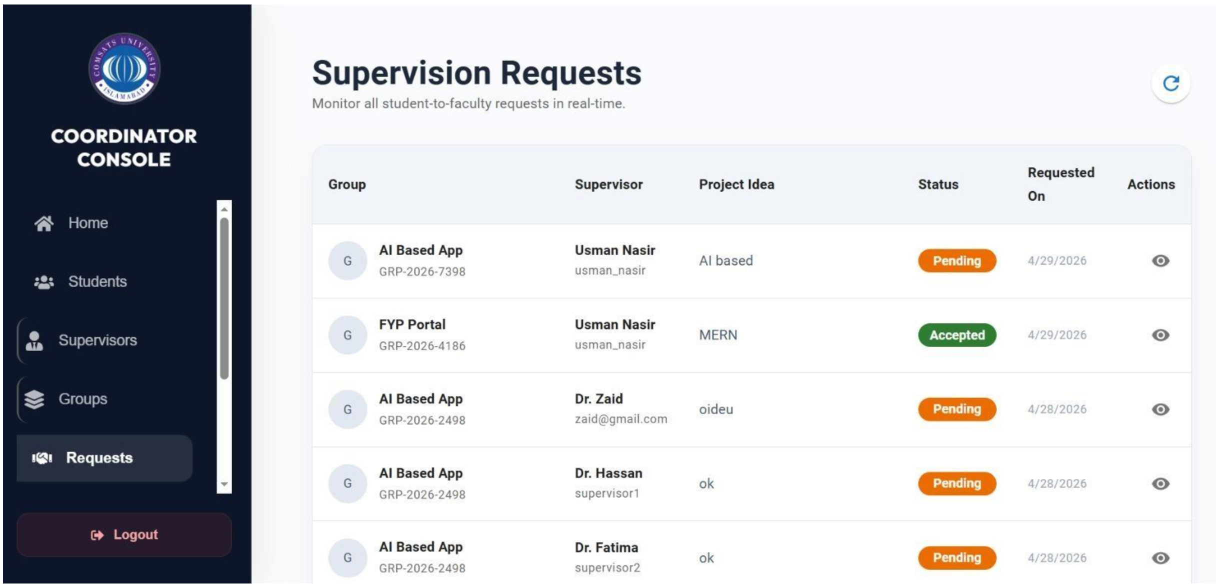

Figure 5-25: Supervision Requests

Academic Timeline

This screen represents the Academic Timeline section of the Coordinator Console, which is used to manage semester milestones and deadlines for FYP activities. It allows coordinators to create new deadlines and monitor student submissions in real time. The “Active Milestones” area highlights current tasks, such as proposal submissions for specific semesters, along with their due dates and submission status. Filters like All, 6th, 7th, and 8th semester help organize milestones efficiently. On the right side, the “Global Activity” panel displays recent actions, such as file uploads by reviewers or supervisors, providing quick access to submitted documents. Overall, this interface helps ensure smooth tracking of academic progress and timely completion of project requirements.

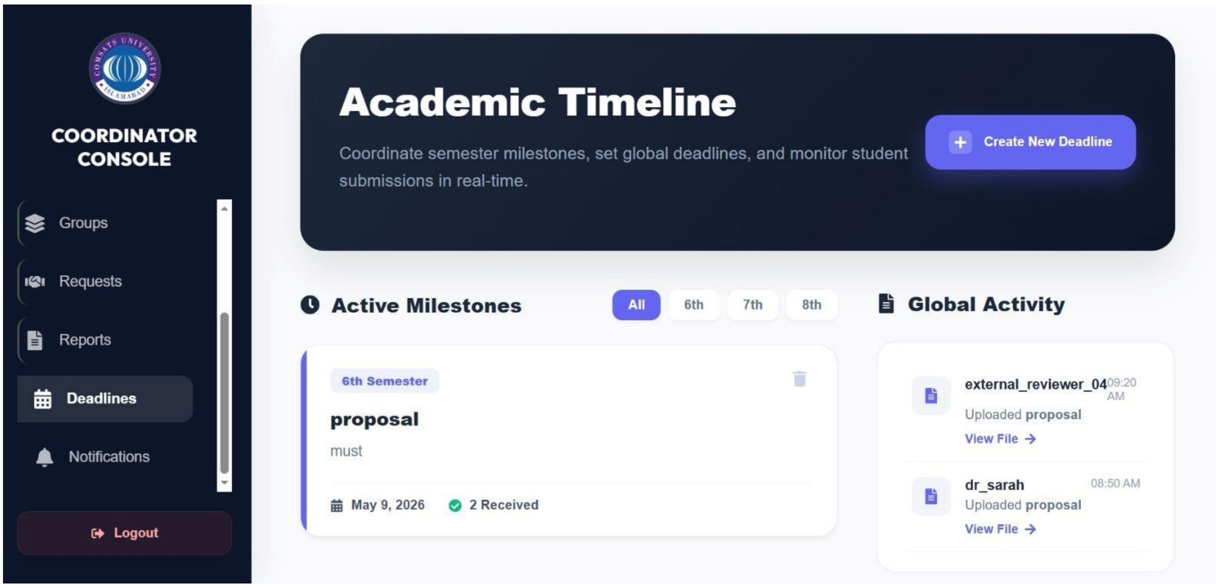

Figure 5-26: Academic Timeline

Global Submission Archive

The image shows a web-based dashboard titled **"**Global Submission Archive"**,** designed for academic coordinators. On the left, there is a dark sidebar labeled "Coordinator Console" with navigation options like Groups, Requests, Reports, Deadlines, and Notifications. The main panel displays submissions categorized by semesters, with the 6th semester currently selected. A search bar is available at the top for finding students or milestones quickly. Below, a table lists student IDs, milestone file names, submission dates, and review statuses. Each entry includes a "Submitted" status badge and a download button for accessing the files. The interface has a clean, modern layout with soft colors and rounded elements for better usability.

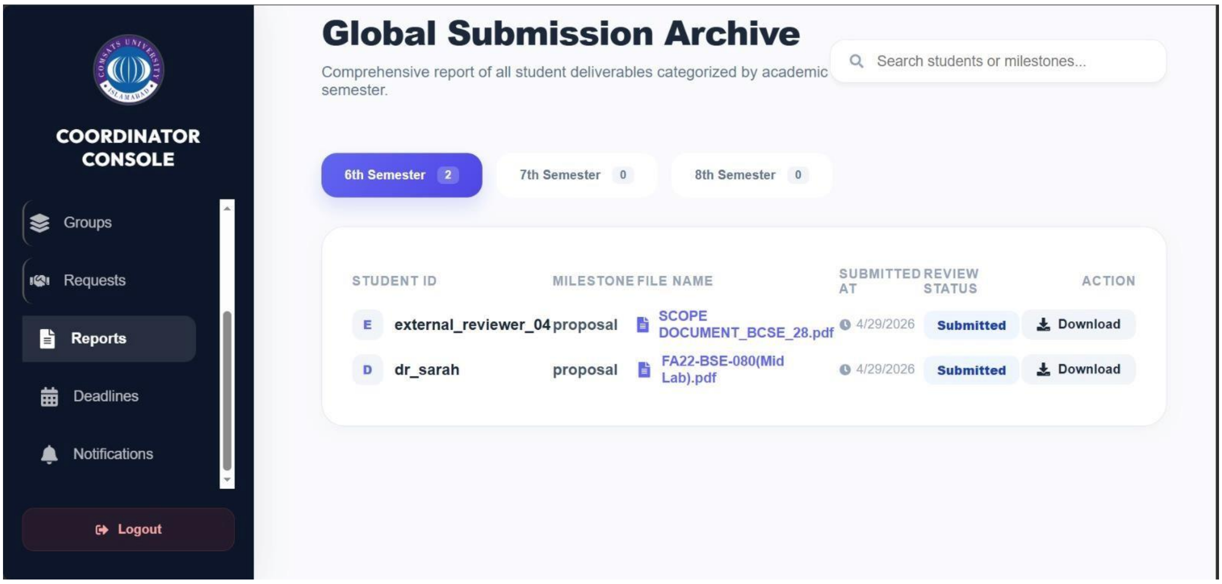

Figure 5-27: Global Submission Archive

Schedule Evaluation

The image shows a pop-up modal titled **"**Schedule Evaluation**"** within a coordinator dashboard. It is used to assign a panelist (internal or external) for evaluating an FYP (Final Year Project). The form includes fields to select the panelist type, with "External Panelist" currently chosen. Users can set a preferred date and time for the evaluation using input fields with calendar and clock icons. There is also a field to specify the venue, which is filled as "D2." At the bottom, two buttons are visible: "Cancel" and a highlighted "Send to External" for confirming the assignment. The blurred background shows a list of groups with "Assign Panelist" buttons, indicating this modal overlay the main interface.

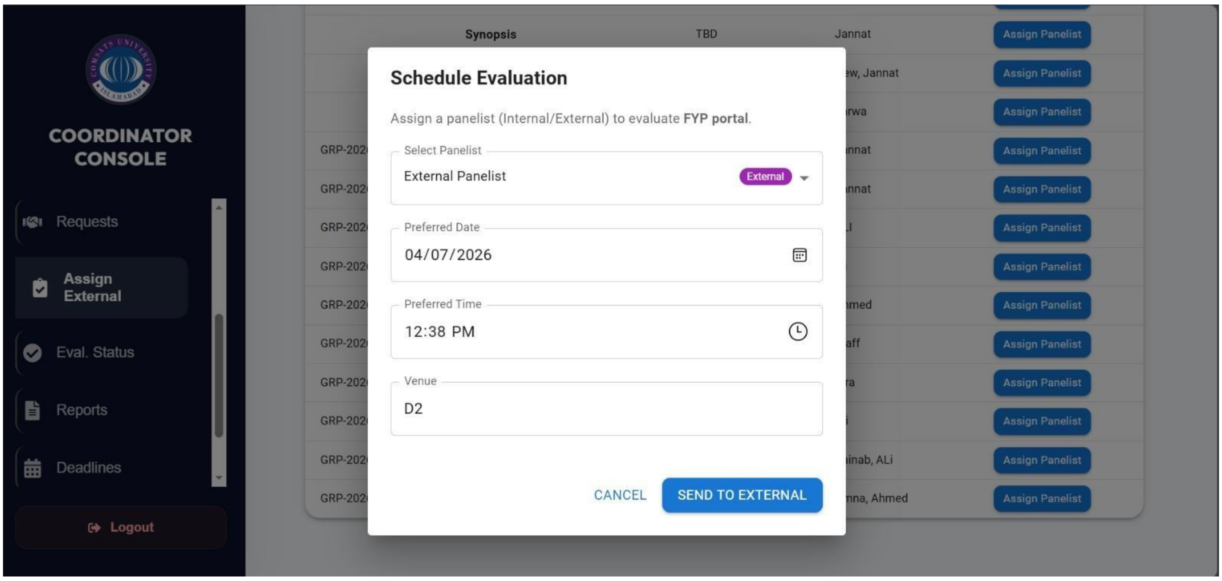

Figure 5-28: Schedule Evaluation

5.3.5 External Modules: Dashboard Screens

The External Module in FYP Genie allows external evaluators to securely log in to the system using assigned credentials. The profile screen enables them to view basic information and maintain their details. The dashboard provides access to assigned projects and related documents for evaluation. Evaluators can review submissions, assign marks, and provide feedback through a structured interface. It also ensures secure and limited access, focusing only on evaluation tasks.

Home Page

The image shows a clean and modern dashboard interface designed for an External Reviewer Panel. On the left side, a dark sidebar contains navigation options such as Home (selected), My Evaluations, New Invitations, and Profile, along with a logout button at the bottom and a university logo at the top. The main section greets the user with "Welcome, Dr. Ahmed Khan," followed by a brief description of the reviewer's role in evaluating student projects. Below this, three summary cards display key metrics: Assigned Projects, Evaluated, and Pending, each with an icon and placeholder values. A set of action buttons is provided underneath, including View Projects, Evaluate Projects, Submit Feedback, and Reports, making navigation quick and intuitive. Further down, there is a section for Recent Evaluation Requests, likely listing incoming tasks for review. On the right side, a "Reviewer Quick Tips" panel offers helpful guidance for evaluating projects effectively. The overall layout is user-friendly, visually balanced, and designed to streamline the evaluation workflow.

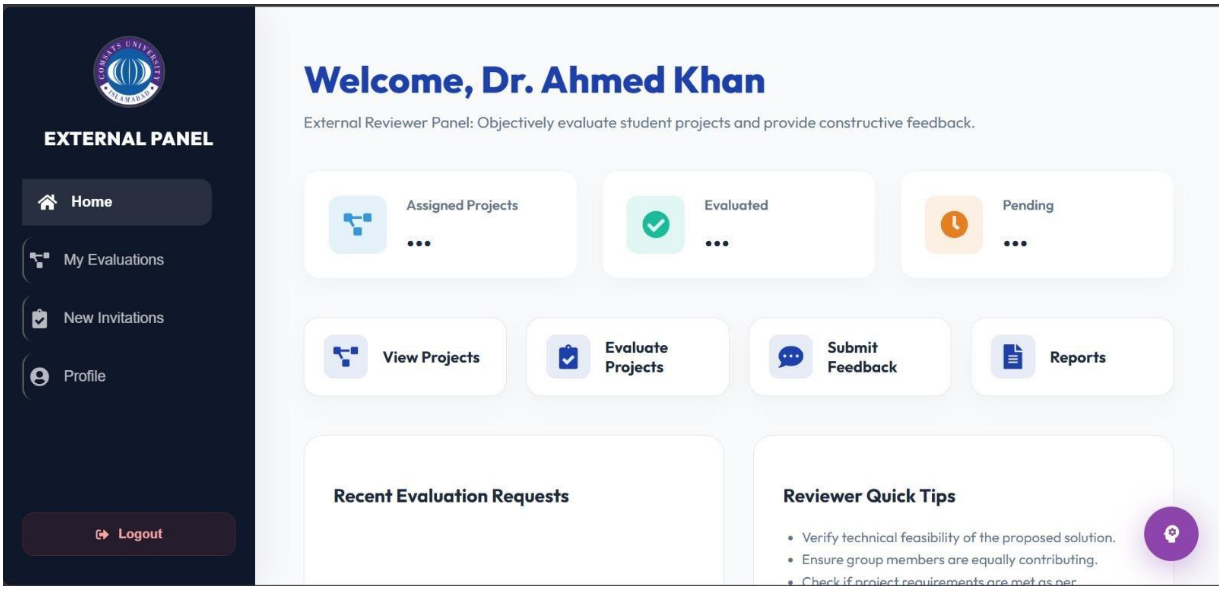

Figure 5-29: External Home Page

Evaluation Dashboard

The image shows a modern web-based Evaluation Dashboard designed for an external panel, likely used in an academic setting. On the left side, there is a dark vertical navigation bar with options such as Home, My Evaluations (currently selected), New Invitations, and Profile, along with a logout button at the bottom. At the top of this sidebar, a university logo is displayed, indicating an institutional platform. The main area features a clean, light-themed interface with the title "Evaluation Dashboard" prominently at the top. Two evaluation cards are visible: one marked as "Scheduled" with a date of 9/4/2026, and another marked as "Completed" with a date of 9/5/2026. Both cards belong to the Alpha Team under the supervision of Dr. Ali and are labeled as FYP Final projects with different sections. The scheduled card includes an active "Evaluate Project" button, while the completed one shows a disabled "Evaluated" status. Overall, the design is minimal, user-friendly, and clearly structured for managing project evaluations.

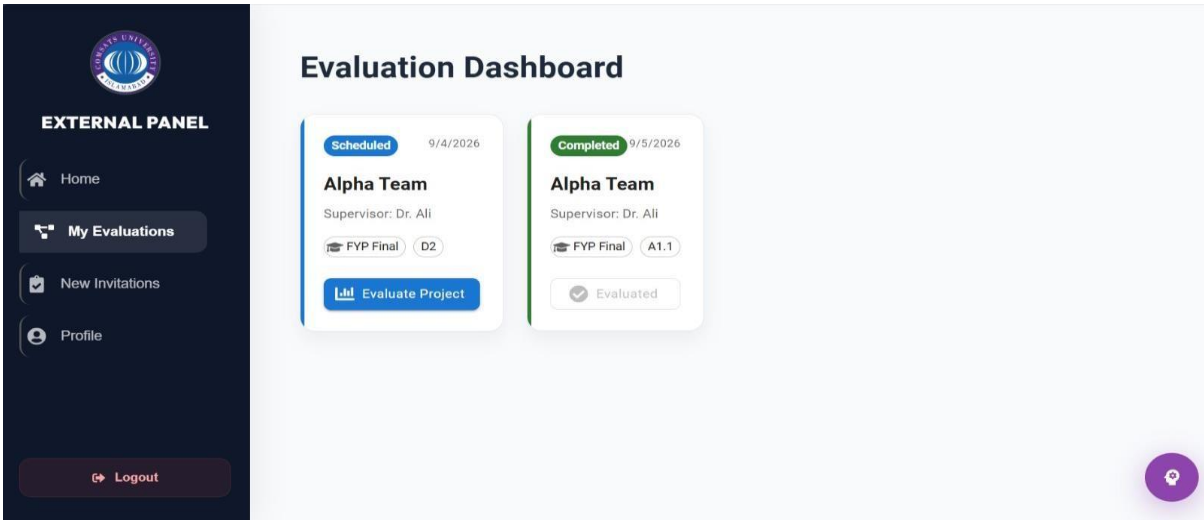

Figure 5-30: Evaluation Dashboard

Incoming Evaluation Request

It showcases the interaction between Students, Supervisors, and Administrators through a unified Web Interface, which acts as the primary gateway to the system's logic. Behind this interface, the backend processes manage critical workflows such as automated supervisor pairing, milestone tracking (D0, D1, D2), and digital rubric evaluations. Central to this architecture is the MongoDB database, which serves as a secure, non-relational repository for all project documentation and academic records. By decoupling the presentation layer from the database through a robust logic tier, the system ensures both high performance and data integrity. Overall, the diagram highlights a modern, scalable ecosystem designed to replace legacy manual processes with a streamlined, digital academic workflow.

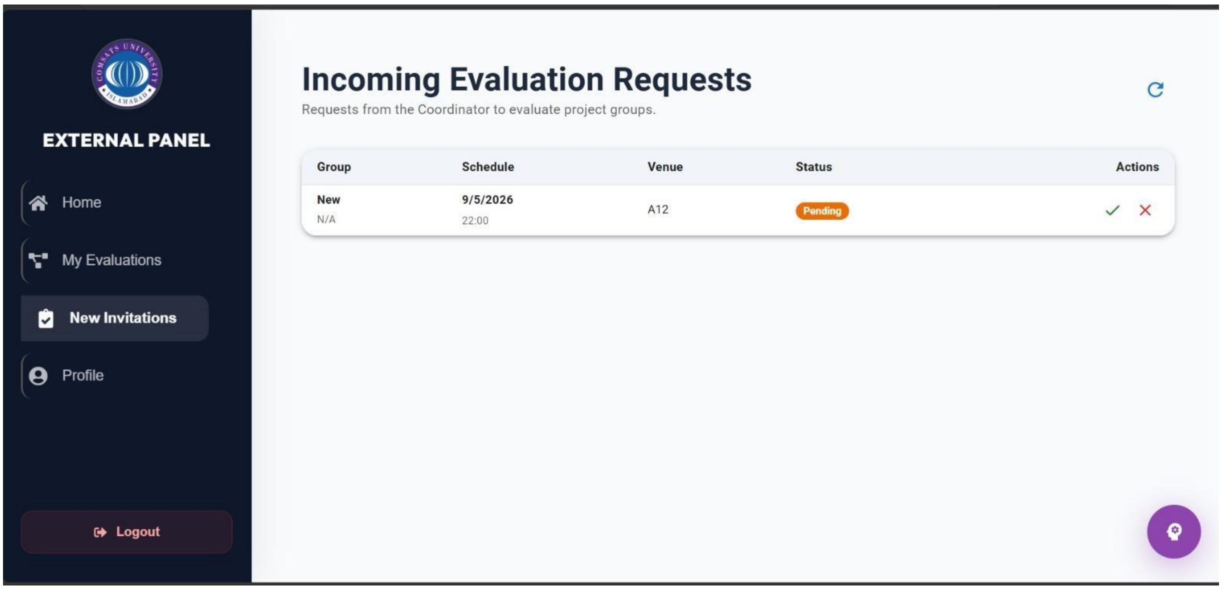

Figure 5-31: Incoming Evaluation Request

Chapter 6
Testing and Evaluation

6 Testing and Evaluation

This chapter presents the testing and evaluation of the FYP Genie system. Testing is a critical phase in the software development lifecycle that ensures the system functions correctly, meets requirements, and is free from major defects.

Different testing techniques are applied to validate system functionality, performance, and usability. The purpose of this chapter is to verify that all modules of the system operate as expected and fulfill the requirements.

6.1 Manual Testing

Manual testing is a software testing technique in which testers manually execute test cases without using any automated tools. In the FYP Genie system, manual testing was performed to verify that all functionalities work correctly according to the requirements.

In this approach, different user roles such as students, supervisors, and coordinators interacted with the system by performing real-world tasks like logging in, submitting proposals, uploading documents, and evaluating projects. The outputs were then compared with the expected results to identify any errors or inconsistencies.

Manual testing helped ensure that the system is user-friendly, reliable, and free from major defects before deployment.

Manual testing of FYP Genie was conducted by executing different test scenarios based on user roles. Each functionality was tested step-by-step, including login, proposal submission, milestone tracking, evaluation, and document management. The system responses were observed and recorded in testing tables, and any mismatches between expected and actual results were identified and corrected.

6.1.1 System Testing

System testing is the phase in which the complete FYP Genie system is tested as a fully integrated application. Instead of testing individual modules separately, this testing validates the system as a whole to ensure that all components work together correctly according to the requirements.

During system testing, real-world scenarios are executed to simulate how different users interact with the system. These include:

- User login and role-based dashboard access.
- Proposal submission and approval workflow.
- Automatic supervisor assignment.

- Milestone submission and tracking.
- Evaluation and result generation.
- Document upload and version control.
- Notification and alert system.

The goal is to ensure that all modules—authentication, proposal management, supervisor assignment, milestone tracking, evaluation, document management, and notifications—function correctly when integrated.

System testing also evaluates:

- **Performance:** Response time and system efficiency
- **Security:** Role-based access and data protection
- **Usability:** User interface consistency and ease of use
- **Error Handling:** Proper system response to invalid inputs

System testing confirms that the system is stable, reliable, and ready for deployment.

Table 6-1: System Testing

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
ST-01	User Login	Valid email & password	User redirected to dashboard	As expected,	Pass
ST-02	Proposal Submission	Valid proposal data	Proposal stored successfully	As expected,	Pass
ST-03	Supervisor Assignment	Approved proposal	Supervisor assigned automatically	As expected,	Pass
ST-04	Milestone Submission	Upload milestone	Status updated to Submitted	As expected,	Pass
ST-05	Evaluation Process	Submit marks	Marks stored & result generated	As expected,	Pass

6.1.2 Unit Testing

Unit testing was performed to verify that each individual module of the FYP Genie system works correctly according to the requirements. The Authentication module was tested with valid credentials, which successfully allowed user login, while invalid passwords correctly displayed an error message. The Proposal Module was tested by submitting a valid proposal, and the data was stored successfully in the database. The Milestone Module was checked to ensure that deadline status updates were processed accurately. The Evaluation Module was tested by entering marks, and the scores were stored correctly in the system. All unit test cases passed successfully, confirming the reliability and proper functionality of the individual system components.

Table 6-2: Unit Testing

Test Case ID	Module	Input	Expected Output	Status
UT-01	Authentication	Valid credentials	Login successful	Pass
UT-02	Authentication	Invalid password	Error message	Pass
UT-03	Proposal Module	Valid proposal	Stored in DB	Pass
UT-04	Milestone Module	Deadline check	Status updated correctly	Pass
UT-05	Evaluation Module	Marks input	Stored correctly	Pass

6.1.3 Functional Testing

Functional testing was conducted to ensure that the main features of the FYP Genie system operate correctly according to user requirements. The Login function was tested by entering valid credentials, and the user was successfully logged into the system. The Submit Proposal feature was verified by filling out the form and submitting it, resulting in the proposal being saved successfully. The Approve Proposal function was tested by allowing the supervisor to approve a proposal, after which the status changed to Approved. The Upload Document feature confirmed that files were stored successfully after submission. The Notification function was also tested by triggering system events, and notifications were generated correctly.

Table 6-3: Functional Testing

Test Case ID	Function	Steps	Expected Result	Status
FT-01	Login	Enter valid credentials	User logged in	Pass
FT-02	Submit Proposal	Fill form & submit	Proposal saved	Pass
FT-03	Approve Proposal	Supervisor approves	Status changes to Approved	Pass
FT-04	Upload Document	Upload file	File stored successfully	Pass
FT-05	Notification	Trigger event	Notification generated	Pass

6.1.4 Integration Testing

Integration testing was performed to verify that different modules of the FYP Genie system work together smoothly and exchange data correctly. The Login and Dashboard modules were tested by logging in successfully, after which the dashboard loaded without errors. The Proposal and Supervisor modules were tested by submitting a proposal, which was correctly sent to the assigned supervisor. The Milestone and Notification modules were checked by submitting a milestone, resulting in a notification being sent successfully. The Evaluation and Result modules were tested by submitting marks, which generated accurate results. Additionally, the Document and Database modules were tested by uploading a file, confirming that it was properly stored in the database.

Table 6-4: Integration Testing

Test Case ID	Modules Involved	Scenario	Expected Result	Status
IT-01	Login + Dashboard	User logs in	Dashboard loads correctly	Pass
IT-02	Proposal + Supervisor	Submit proposal	Sent to supervisor	Pass
IT-03	Milestone + Notification	Submit milestone	Notification sent	Pass
IT-04	Evaluation + Result	Submit marks	Result generated	Pass
IT-05	Document + Database	Upload file	Stored in DB	Pass

6.2 Automated Testing

Automated testing is used to validate system functionality using tools and scripts. It reduces manual effort and improves testing efficiency.

Tools Used:

- Postman (API testing)
- Jest / Mocha (backend testing)
- React Testing Library (frontend testing)

Automated testing was implemented to verify the functionality of the FYP Genie system using testing tools and scripts, which helped reduce manual effort and improve overall testing efficiency. Different tools such as Postman were used for API testing, Jest or Mocha for backend testing, and React Testing Library for frontend testing. The Login API was tested using Postman, which successfully returned a valid authentication token. The Proposal API was also tested and confirmed that data was stored correctly in the database. Backend authentication logic was validated using Jest, ensuring proper login functionality. The frontend

UI components were tested using React Testing Library to confirm correct rendering, while the Evaluation API was verified through Postman to ensure marks were stored successfully.

Table 6-5: Automated Testing

Test Case ID	**Tool Used**	**Test Scenario**	**Expected Result**	**Status**
AT-01	Postman	Login API	Returns token	Pass
AT-02	Postman	Proposal API	Data stored	Pass
AT-03	Jest	Auth Module	Validates login logic	Pass
AT-04	React Testing Library	UI Rendering	Component loads correctly	Pass
AT-05	Postman	Evaluation API	Marks stored	Pass

Chapter 7
Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The FYP Genie system has been successfully designed and implemented as a comprehensive web-based solution to manage the Final Year Project (FYP) lifecycle at COMSATS University Islamabad, Sahiwal Campus. The system addresses the limitations of traditional manual processes by providing an integrated platform for proposal submission, supervisor assignment, progress tracking, evaluation, and document management.

Through the use of modern web technologies such as the MERN stack, the system ensures a scalable, efficient, and user-friendly environment for all stakeholders, including students, supervisors, coordinators, and external evaluators. The adoption of a modular architecture and Agile-based development approach enabled the team to build the system incrementally while incorporating feedback at each stage.

The implementation of key features such as automated supervisor assignment, milestone tracking, role-based dashboards, digital evaluation, and document version control has significantly improved transparency, accuracy, and efficiency in managing FYP activities. The system also enhances communication among users through a notification mechanism, ensuring that deadlines and updates are properly managed.

Comprehensive testing and evaluation confirmed that the system meets all functional and non-functional requirements defined in the SRS. The application demonstrated reliable performance, secure data handling, and ease of use across different user roles.

Overall, FYP Genie provides a centralized and automated solution that simplifies academic project management, reduces administrative workload, minimizes errors, and improves the overall quality and monitoring of Final Year Projects.

7.2 Limitations of the System

Despite its effectiveness, the current version of FYP Genie has some limitations:

- The system is currently limited to web-based access and does not include a dedicated mobile application
- Real-time communication features such as chat or video conferencing are not fully implemented
- Advanced analytics and reporting features are limited
- Integration with plagiarism detection tools (e.g., Turnitin API) is not automated
- External evaluator access is basic and can be further enhanced

7.3 Future Work

The FYP Genie system can be further enhanced by incorporating the following features:

1 Mobile Application Development

Developing a mobile application (Android/iOS) will improve accessibility and allow users to interact with the system anytime and anywhere.

2 AI-Based Supervisor Recommendation

Integrating machine learning algorithms can improve supervisor assignment by analyzing historical data, project domains, and performance trends.

3 Advanced Analytics Dashboard

Adding data analytics features for coordinators can provide insights such as:

- Student performance trends
- Supervisor workload distribution
- Project success rates

4 Integration with External Tools

The system can be integrated with:

- Turnitin for plagiarism detection
- Email/SMS gateways for notifications
- Learning Management Systems (LMS)

5 Real-Time Communication System

Adding built-in chat or messaging functionality will improve collaboration between students and supervisors.

6 Cloud Deployment and Scalability

Deploying the system on cloud platforms (e.g., AWS, Azure) will enhance scalability, availability, and performance.

7 Enhanced Security Features

Future versions can include:

- Two-factor authentication (2FA)
- Data encryption at advanced levels
- Activity monitoring dashboards

8 Automated Report Generation

Generating automated reports for students and faculty (PDF format) will further reduce manual workload.

Chapter 8
References

8 References

[1] PMI, https://www.devex.com/organizations/project-management-institute-pmi-112703

[2] EDU, https://www.educause.edu

[3] IEEE, https://ewh.ieee.org/sb/kerala/gecb/ieee.html

[4] RG, https://www.researchgate.net/figure/Research-Gate-home-page-Retrieved-from-https-wwwresearchgatenet_fig7_337366737

[5] SJ, https://smithjones.co/about.html

[6] LMS, https://www.dcc.fc.up.pt/~zp/papers/SPeL_2015.pdf?utm_.

[7] Scrum Guides, https://www.scrum.org/resources/2017-scrum-guide-update-ken-schwaber-and-jeff-sutherland

[8] PB, https://spectrum.ieee.org/the-institutes-print-edition-is-now-inside-ieee-spectrum

[9] GFG, https://koulurunandakishorereddy.hashnode.dev/6-free-websites-to-learn-web-development

[10] Spring, https://www.glassdoor.co.in/Photos/Springer-Nature-Office-Photos IMG5378225.htm

[11] Scribe, https://www.scribd.com/document/609451297/EDUCAUSE-REVIEW-Digital-Transformation

[12] book, https://books.google.com/books?id=5rFvQgAACAAJ

[13] catalog, https://www.pearson.com/en-us/subject-catalog/p/software-engineering/P200000003297

[14] SA, https://www.oreilly.com/library/view/software-architecture-in/9780132943449/

[15] MERN, https://link.springer.com/book/10.1007/978-1-4842-4410-6

[16] DB, https://www.oreilly.com/library/view/mongodb-the-definitive/9781491954454/

[17] OWASP, https://owasp.org/www-project-top-ten/

[18] HED https://www.mheducation.com/highered/product/software-engineering-practitioner-s-approach-pressman-maxim/M9780078022128.html

[19] https://www.pearson.com/en-us/subject-catalog/p/software-engineering/P200000003295

[20] TEST, https://www.geeksforgeeks.org/software-testing-basics/